```
Last login: Wed Sep 30 17:18:42 on ttys000
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────────┐
│                 Welcome to  the Bocconi HPC system                          │
│                                                          $$$$$$$$$$$$       │
│                                                          $$$       $$$      │
│                                                          $h         8$      │
│                                  _                       $f         k$      │
│                                 (_)                      $h         8$      │
│                                 | |  _                   $$$       $$$      │
│                                 | | /'_`\                 $$$       $$$      │
│                                 | |( (_) )                $$$$$$$$$$$$       │
│                                 (_)`\___/'                                  │
│                                                          (Jupiter I)        │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
├── Information ──────────────────────────────────────────────────────────────┤
│ Support Requests  : hpc@unibocconi.it                                       │
│ Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC │
├── Software ─────────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                            │
│ Use 'lquota' to see the available resources in your home                    │
├── Cluster Usage ────────────────────────────────────────────────────────────┤
│ Running computational data processing tasks on the LOGIN node is forbidden. │
│ Please always launch computation using SLURM job submission,               │
│ so that the computation is executed on the COMPUTE nodes.                   │
└─────────────────────────────────────────────────────────────────────────────┘
(base) [3355245@slnode01 ~]$ sinfo
sacctmgr show assoc user=$USER format=account,partition,qos
module avail 2>&1 | head -50
quota -s; df -h $HOME
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
stud         up 1-00:10:00      2    mix gnode[01-02]
   Account  Partition                  QOS
---------- ---------- --------------------
   3355245               debug,normal,stud
   3355245       stud    debug,normal,stud
-------------------------- /usr/share/Modules/modulefiles --------------------------
dot  module-git  module-info  modules  null  use.own

------------------------------- /software/modules --------------------------------
cuda/12.1         MATLAB_R2025b                    sw/stata
cuda/12.2         miniconda3                       sw/statanow
cuda/12.3         mpi/intelmpi/mpi                 ucx/1.19.0-rh9
cuda/12.4         mpi/openmpi/5.0.9-rh9            ucx/1.19.0-rh9-cuda
cuda/12.8         mpi/openmpi/5.0.9-rh9-cuda
ior/3.3.0         mpi/openmpi/5.0.9-rh9-cuda-no-ucx
libs/hwloc-2.9.3  mpi/openmpi/5.0.9-rh9-no-ucx
libs/testlib      sw/miniconda3
Filesystem      Size  Used Avail Use% Mounted on
beegfs_nodev    112T   12T  101T  11% /mnt/beegfsstudents
(base) [3355245@slnode01 ~]$ scontrol show partition stud | grep -iE "maxnodes|maxtime|tres"
sacctmgr show qos stud format=name,maxtrespu%40,maxjobspu,maxwall
scontrol show node gnode01 | grep -iE "gres|cfgtres"
curl -sI https://huggingface.co | head -1
   MaxNodes=UNLIMITED MaxTime=1-00:10:00 MinNodes=0 LLN=NO MaxCPUsPerNode=UNLIMITED MaxCPUsPerSocket=UNLIMITED
   TRES=cpu=56,mem=1490000M,node=2,billing=120,gres/gpu=16
   TRESBillingWeights=CPU=1.0,GRES/gpu=4.0
      Name                                  MaxTRESPU MaxJobsPU     MaxWall
---------- ---------------------------------------- --------- -----------
      stud                                    cpu=8         1  1-00:00:00
   Gres=gpu:4g.40gb:4,gpu:3g.40gb:4
   CfgTRES=cpu=32,mem=990000M,billing=64,gres/gpu=8
   AllocTRES=cpu=16,mem=112G,gres/gpu=2
HTTP/2 200
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=1.5GB LIMIT=50GB
(base) [3355245@slnode01 ~]$ conda create -n inward python=3.11 -y
conda activate inward
pip install --no-cache-dir vllm transformers datasets accelerate peft nnsight pandas matplotlib
conda clean -a -y
lquota
2 channel Terms of Service accepted
Retrieving notices: done
Channels:
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done


==> WARNING: A newer version of conda exists. <==
    current version: 26.1.1
    latest version: 26.7.3

Please update conda by running

    $ conda update -n base -c defaults conda



## Package Plan ##

  environment location: /home/3355245/.conda/envs/inward

  added / updated specs:
    - python=3.11


The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    _openmp_mutex-5.1          |           52_gnu           7 KB
    ca-certificates-2026.8.13  |       h06a4308_0         107 KB
    ld_impl_linux-64-2.44      |       h9e0c5a2_3         725 KB
    libexpat-2.8.5             |       h7354ed3_1         129 KB
    libffi-3.4.8               |       h06d3fd0_3         137 KB
    libgcc-15.2.0              |       h69a1729_8         803 KB
    libgcc-ng-15.2.0           |       h166f726_8          28 KB
    libstdcxx-15.2.0           |       h39759b7_8         3.7 MB
    libzlib-1.3.2              |       h47b2149_0          60 KB
    ncurses-6.6                |       hfaaeb4e_0         893 KB
    openssl-3.5.8              |       h1b28b03_0         5.5 MB
    packaging-26.3             |  py311h06a4308_0         380 KB
    pip-26.2.1                 |     pyhc872135_0         1.1 MB
    pthread-stubs-0.3          |       h47b2149_2           7 KB
    python-3.11.16             |       h17756b0_0        29.7 MB
    setuptools-83.0.0          |  py311h06a4308_0         1.6 MB
    sqlite-3.53.4              |       h795bf6d_0         1.2 MB
    tzdata-2026c               |       he532380_0         117 KB
    wheel-0.47.0               |  py311h06a4308_0          75 KB
    xorg-libx11-1.8.13         |       h65de747_0         817 KB
    xorg-xorgproto-2025.1      |       h47b2149_0         555 KB
    xz-5.8.4                   |       h140a991_0         652 KB
    zlib-1.3.2                 |       h47b2149_0          92 KB
    ------------------------------------------------------------
                                           Total:        48.3 MB

The following NEW packages will be INSTALLED:

  _libgcc_mutex      pkgs/main/linux-64::_libgcc_mutex-0.1-main
  _openmp_mutex      pkgs/main/linux-64::_openmp_mutex-5.1-52_gnu
  bzip2              pkgs/main/linux-64::bzip2-1.0.8-h5eee18b_6
  ca-certificates    pkgs/main/linux-64::ca-certificates-2026.8.13-h06a4308_0
```

```
  ld_impl_linux-64   pkgs/main/linux-64::ld_impl_linux-64-2.44-h9e0c5a2_3
  libexpat           pkgs/main/linux-64::libexpat-2.8.5-h7354ed3_1
  libffi             pkgs/main/linux-64::libffi-3.4.8-h06d3fd0_3
  libgcc             pkgs/main/linux-64::libgcc-15.2.0-h69a1729_8
  libgcc-ng          pkgs/main/linux-64::libgcc-ng-15.2.0-h166f726_8
  libnsl             pkgs/main/linux-64::libnsl-2.0.0-h5eee18b_0
  libstdcxx          pkgs/main/linux-64::libstdcxx-15.2.0-h39759b7_8
  libuuid            pkgs/main/linux-64::libuuid-1.41.5-h5eee18b_0
  libxcb             pkgs/main/linux-64::libxcb-1.17.0-h9b100fa_0
  libzlib            pkgs/main/linux-64::libzlib-1.3.2-h47b2149_0
  ncurses            pkgs/main/linux-64::ncurses-6.6-hfaaeb4e_0
  openssl            pkgs/main/linux-64::openssl-3.5.8-h1b28b03_0
  packaging          pkgs/main/linux-64::packaging-26.3-py311h06a4308_0
  pip                pkgs/main/noarch::pip-26.2.1-pyhc872135_0
  pthread-stubs      pkgs/main/linux-64::pthread-stubs-0.3-h47b2149_2
  python             pkgs/main/linux-64::python-3.11.16-h17756b0_0
  readline           pkgs/main/linux-64::readline-8.3-hc2a1206_0
  setuptools         pkgs/main/linux-64::setuptools-83.0.0-py311h06a4308_0
  sqlite             pkgs/main/linux-64::sqlite-3.53.4-h795bf6d_0
  tk                 pkgs/main/linux-64::tk-8.6.15-h54e0aa7_0
  tzdata             pkgs/main/noarch::tzdata-2026c-he532380_0
  wheel              pkgs/main/linux-64::wheel-0.47.0-py311h06a4308_0
  xorg-libx11        pkgs/main/linux-64::xorg-libx11-1.8.13-h65de747_0
  xorg-libxau        pkgs/main/linux-64::xorg-libxau-1.0.12-h9b100fa_0
  xorg-libxdmcp      pkgs/main/linux-64::xorg-libxdmcp-1.1.5-h9b100fa_0
  xorg-xorgproto     pkgs/main/linux-64::xorg-xorgproto-2025.1-h47b2149_0
  xz                 pkgs/main/linux-64::xz-5.8.4-h140a991_0
  zlib               pkgs/main/linux-64::zlib-1.3.2-h47b2149_0


Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate inward
#
# To deactivate an active environment, use
#
#     $ conda deactivate


Channel "defaults" has the following notices:
  [info] -- Thu Aug  6 00:00:00 2026
  main-x (Anaconda's new authenticated channel) is now generally available, with thousands of additional packages from our secure supply chain. Get started at: https://anaconda.com/docs/getting-started/main-
x?utm_source=channel_notices

Collecting vllm
  Downloading vllm-0.30.0-cp38-abi3-manylinux_2_28_x86_64.whl.metadata (11 kB)
Collecting transformers                                                  Downloading transformers-5.17.0-py3-none-any.whl.metadata (32 kB)
Collecting datasets
  Downloading datasets-5.0.1-py3-none-any.whl.metadata (23 kB)
Collecting accelerate
  Downloading accelerate-1.15.0-py3-none-any.whl.metadata (19 kB)
Collecting peft
  Downloading peft-0.21.1-py3-none-any.whl.metadata (14 kB)
Collecting nnsight
  Downloading nnsight-0.7.0-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (17 kB)
Collecting pandas
  Downloading pandas-3.0.6-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (79 kB)
Collecting matplotlib
  Downloading matplotlib-3.11.2-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (80 kB)
Collecting regex (from vllm)
  Downloading regex-2026.9.29-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (40 kB)
Collecting cachetools (from vllm)
  Downloading cachetools-7.2.0-py3-none-any.whl.metadata (5.5 kB)
Collecting psutil (from vllm)
  Downloading psutil-7.2.2-cp36-abi3-manylinux2010_x86_64.manylinux_2_12_x86_64.manylinux_2_28_x86_64.whl.metadata (22 kB)
Collecting sentencepiece (from vllm)
  Downloading sentencepiece-0.2.2-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (33 kB)
Collecting numpy (from vllm)
  Downloading numpy-2.4.6-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Collecting requests>=2.26.0 (from vllm)
  Downloading requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting tqdm (from vllm)
  Downloading tqdm-4.70.1-py3-none-any.whl.metadata (57 kB)
Collecting blake3 (from vllm)
  Downloading blake3-1.0.10-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.9 kB)
Collecting py-cpuinfo (from vllm)
  Downloading py_cpuinfo-9.0.0-py3-none-any.whl.metadata (794 bytes)
Collecting huggingface_hub>=1.31.0 (from vllm)
  Downloading huggingface_hub-2.0.0-py3-none-any.whl.metadata (16 kB)
Collecting tokenizers>=0.21.1 (from vllm)
  Downloading tokenizers-0.23.2-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (9.8 kB)
Collecting safetensors>=0.6.2 (from vllm)
  Downloading safetensors-0.8.0-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.2 kB)
Collecting protobuf!=6.30.*,!=6.31.*,!=6.32.*,!=6.33.0.*,!=6.33.1.*,!=6.33.2.*,!=6.33.3.*,!=6.33.4.*,>=5.29.6 (from vllm)
  Downloading protobuf-7.36.2-cp310-abi3-manylinux2014_x86_64.whl.metadata (595 bytes)
Collecting fastapi<0.137.0,>=0.133.0 (from fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading fastapi-0.136.3-py3-none-any.whl.metadata (27 kB)
Collecting starlette>=1.0.1 (from vllm)
  Downloading starlette-1.7.0-py3-none-any.whl.metadata (6.6 kB)
Collecting aiohttp>=3.13.3 (from vllm)
  Downloading aiohttp-3.14.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (8.3 kB)
Collecting openai>=2.25.0 (from vllm)
  Downloading openai-3.22.1-py3-none-any.whl.metadata (44 kB)
Collecting pydantic>=2.12.0 (from vllm)
  Downloading pydantic-2.13.5-py3-none-any.whl.metadata (110 kB)
Collecting prometheus_client>=0.18.0 (from vllm)
  Downloading prometheus_client-0.26.0-py3-none-any.whl.metadata (2.1 kB)
Collecting pillow (from vllm)
  Downloading pillow-12.3.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (9.1 kB)
Collecting prometheus-fastapi-instrumentator>=8.0.0 (from vllm)
  Downloading prometheus_fastapi_instrumentator-8.1.0-py3-none-any.whl.metadata (13 kB)
Collecting tiktoken>=0.6.0 (from vllm)
  Downloading tiktoken-0.14.0-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (6.7 kB)
Collecting lm-format-enforcer==0.11.3 (from vllm)
  Downloading lm_format_enforcer-0.11.3-py3-none-any.whl.metadata (17 kB)
Collecting llguidance<1.8.0,>=1.7.0 (from vllm)
  Downloading llguidance-1.7.6-cp39-abi3-manylinux_2_31_x86_64.whl.metadata (10 kB)
Collecting outlines_core==0.2.14 (from vllm)
  Downloading outlines_core-0.2.14-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.6 kB)
Collecting lark==1.2.2 (from vllm)
  Downloading lark-1.2.2-py3-none-any.whl.metadata (1.8 kB)
Collecting xgrammar<1.0.0,>=0.2.1 (from vllm)
  Downloading xgrammar-0.2.8-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (10 kB)
Collecting typing_extensions>=4.10 (from vllm)
  Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting filelock>=3.16.1 (from vllm)
  Downloading filelock-4.0.7-py3-none-any.whl.metadata (2.0 kB)
Collecting partial-json-parser (from vllm)
  Downloading partial_json_parser-0.2.1.1.post7-py3-none-any.whl.metadata (6.1 kB)
Collecting jsonschema>=4.23.0 (from vllm)
  Downloading jsonschema-4.26.0-py3-none-any.whl.metadata (7.6 kB)
Collecting pyzmq>=25.0.0 (from vllm)
  Downloading pyzmq-27.2.0-cp311-cp311-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl.metadata (3.8 kB)
Collecting msgspec (from vllm)
  Downloading msgspec-0.22.0-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (6.0 kB)
Collecting mistral_common>=1.11.6 (from mistral_common[image]>=1.11.6->vllm)
  Downloading mistral_common-1.12.0-py3-none-any.whl.metadata (6.1 kB)
Collecting opencv-python-headless>=4.13.0 (from vllm)
  Downloading opencv_python_headless-5.0.0.93-cp37-abi3-manylinux_2_28_x86_64.whl.metadata (19 kB)
Collecting pyyaml (from vllm)
  Downloading pyyaml-6.0.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)
Collecting einops (from vllm)
  Downloading einops-0.8.2-py3-none-any.whl.metadata (13 kB)
Collecting compressed-tensors==0.17.0 (from vllm)
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  Downloading compressed_tensors-0.17.0-py3-none-any.whl.metadata (4.7 kB)
Collecting depyf==0.20.0 (from vllm)
  Downloading depyf-0.20.0-py3-none-any.whl.metadata (7.3 kB)
Collecting cloudpickle (from vllm)
  Downloading cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Collecting watchfiles (from vllm)
  Downloading watchfiles-1.3.0-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.9 kB)
Collecting python-json-logger (from vllm)
  Downloading python_json_logger-4.2.0-py3-none-any.whl.metadata (2.9 kB)
Collecting ninja (from vllm)
  Downloading ninja-1.13.2-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (5.4 kB)
Collecting pybase64 (from vllm)
  Downloading pybase64-1.5.0-cp311-cp311-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl.metadata (11 kB)
Collecting cbor2 (from vllm)
  Downloading cbor2-6.1.4-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (5.5 kB)
Collecting ijson (from vllm)
  Downloading ijson-3.5.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (23 kB)
Collecting setproctitle (from vllm)
  Downloading setproctitle-1.3.7-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (10 kB)
Collecting openai-harmony>=0.0.3 (from vllm)
  Downloading openai_harmony-0.0.8-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (8.0 kB)
Collecting anthropic>=0.71.0 (from vllm)
  Downloading anthropic-1.9.0-py3-none-any.whl.metadata (3.3 kB)
Collecting model-hosting-container-standards<1.0.0,>=0.1.14 (from vllm)
  Downloading model_hosting_container_standards-0.1.16-py3-none-any.whl.metadata (24 kB)
Collecting mcp<3.0.0,>=2.0.0 (from vllm)
  Downloading mcp-2.2.0-py3-none-any.whl.metadata (7.8 kB)
Collecting opentelemetry-sdk>=1.27.0 (from vllm)
  Downloading opentelemetry_sdk-1.45.0-py3-none-any.whl.metadata (1.6 kB)
Collecting opentelemetry-api>=1.27.0 (from vllm)
  Downloading opentelemetry_api-1.45.0-py3-none-any.whl.metadata (1.4 kB)
Collecting opentelemetry-exporter-otlp>=1.27.0 (from vllm)
  Downloading opentelemetry_exporter_otlp-1.45.0-py3-none-any.whl.metadata (2.4 kB)
Collecting opentelemetry-semantic-conventions-ai>=0.4.1 (from vllm)
  Downloading opentelemetry_semantic_conventions_ai-0.5.1-py3-none-any.whl.metadata (997 bytes)
Collecting numba==0.65.0 (from vllm)
  Downloading numba-0.65.0-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.9 kB)
Collecting torch==2.13.0 (from vllm)
  Downloading torch-2.13.0-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (38 kB)
Collecting torchaudio==2.11.0 (from vllm)
  Downloading torchaudio-2.11.0-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (6.9 kB)
Collecting torchvision==0.28.0 (from vllm)
  Downloading torchvision-0.28.0-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (5.6 kB)
Collecting torchcodec>=0.14 (from vllm)
  Downloading torchcodec-0.17.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (9.9 kB)
Collecting PyNvVideoCodec==2.0.4 (from vllm)
  Downloading pynvvideocodec-2.0.4-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (5.3 kB)
Collecting flashinfer-python==0.6.18.post1 (from vllm)
  Downloading flashinfer_python-0.6.18.post1-py3-none-any.whl.metadata (11 kB)
Collecting apache-tvm-ffi==0.1.11 (from vllm)
  Downloading apache_tvm_ffi-0.1.11-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (6.1 kB)
Collecting tilelang==0.1.12 (from vllm)
  Downloading tilelang-0.1.12-cp38-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (15 kB)
Collecting nvidia-cudnn-frontend>=1.19.1 (from vllm)
  Downloading nvidia_cudnn_frontend-1.30.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (17 kB)
Collecting nvtx==0.2.15 (from vllm)
  Downloading nvtx-0.2.15-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.8 kB)
Collecting fastsafetensors>=0.3.3 (from vllm)
  Downloading fastsafetensors-0.4.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (3.8 kB)
Collecting instanttensor>=0.1.9 (from vllm)
  Downloading instanttensor-0.2.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (26 kB)
Collecting nvidia-cutlass-dsl==4.7.1 (from nvidia-cutlass-dsl[cu13]==4.7.1->vllm)
  Downloading nvidia_cutlass_dsl-4.7.1-py3-none-any.whl.metadata (2.8 kB)
Collecting quack-kernels==0.6.5 (from vllm)
  Downloading quack_kernels-0.6.5-py3-none-any.whl.metadata (834 bytes)
Collecting tokenspeed-mla==0.1.8 (from vllm)
  Downloading tokenspeed_mla-0.1.8-py3-none-manylinux_2_28_x86_64.whl.metadata (10 kB)
Collecting humming-kernels==0.1.12 (from humming-kernels[cu13]==0.1.12->vllm)
  Downloading humming_kernels-0.1.12-py3-none-manylinux_2_28_x86_64.whl.metadata (4.0 kB)
Collecting loguru (from compressed-tensors==0.17.0->vllm)
  Downloading loguru-0.7.3-py3-none-any.whl.metadata (22 kB)
Collecting astor (from depyf==0.20.0->vllm)
  Downloading astor-0.8.1-py2.py3-none-any.whl.metadata (4.2 kB)
Collecting dill (from depyf==0.20.0->vllm)
  Downloading dill-0.4.1-py3-none-any.whl.metadata (10 kB)
Collecting click (from flashinfer-python==0.6.18.post1->vllm)
  Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting cuda-python>=12.0 (from flashinfer-python==0.6.18.post1->vllm)
  Downloading cuda_python-13.4.1-py3-none-any.whl.metadata (6.8 kB)
Collecting cuda-tile>=1.4.0 (from flashinfer-python==0.6.18.post1->vllm)
  Downloading cuda_tile-1.6.0-cp311-cp311-manylinux2014_x86_64.whl.metadata (7.4 kB)
Collecting nccl4py>=0.3.1 (from flashinfer-python==0.6.18.post1->vllm)
  Downloading nccl4py-0.6.0-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (3.2 kB)
Collecting nvidia-ml-py (from flashinfer-python==0.6.18.post1->vllm)
  Downloading nvidia_ml_py-13.615.71-py3-none-any.whl.metadata (9.7 kB)
Requirement already satisfied: packaging>=24.2 in /home/3355245/.conda/envs/inward/lib/python3.11/site-packages (from flashinfer-python==0.6.18.post1->vllm) (26.3)
Collecting tabulate (from flashinfer-python==0.6.18.post1->vllm)
  Downloading tabulate-0.10.0-py3-none-any.whl.metadata (40 kB)
Collecting triton (from humming-kernels==0.1.12->humming-kernels[cu13]==0.1.12->vllm)
  Downloading triton-3.8.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (978 bytes)
Collecting jinja2 (from humming-kernels==0.1.12->humming-kernels[cu13]==0.1.12->vllm)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting cuda-bindings (from humming-kernels==0.1.12->humming-kernels[cu13]==0.1.12->vllm)
  Downloading cuda_bindings-13.4.3-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)
Collecting nvidia-cuda-runtime (from humming-kernels[cu13]==0.1.12->vllm)
  Downloading nvidia_cuda_runtime-13.4.92-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cuda-cccl (from humming-kernels[cu13]==0.1.12->vllm)
  Downloading nvidia_cuda_cccl-13.3.4.3.1-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-nvcc (from humming-kernels[cu13]==0.1.12->vllm)
  Downloading nvidia_cuda_nvcc-13.4.92-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.9 kB)
Collecting nvidia-cuda-nvrtc (from humming-kernels[cu13]==0.1.12->vllm)
  Downloading nvidia_cuda_nvrtc-13.4.92-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl.metadata (1.8 kB)
Collecting interegular>=0.3.2 (from lm-format-enforcer==0.11.3->vllm)
  Downloading interegular-0.3.3-py37-none-any.whl.metadata (3.0 kB)
Collecting llvmlite<0.48,>=0.47.0dev0 (from numba==0.65.0->vllm)
  Downloading llvmlite-0.47.0-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (5.0 kB)
Collecting nvidia-cutlass-dsl-libs-base==4.7.1 (from nvidia-cutlass-dsl==4.7.1->nvidia-cutlass-dsl[cu13]==4.7.1->vllm)
  Downloading nvidia_cutlass_dsl_libs_base-4.7.1-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (2.9 kB)
Collecting nvidia-cutlass-dsl-libs-cu12==4.7.1 (from nvidia-cutlass-dsl==4.7.1->nvidia-cutlass-dsl[cu13]==4.7.1->vllm)
  Downloading nvidia_cutlass_dsl_libs_cu12-4.7.1-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (2.9 kB)
Collecting nvidia-cuda-nvdisasm<14,>=13.3 (from nvidia-cutlass-dsl-libs-base==4.7.1->nvidia-cutlass-dsl==4.7.1->nvidia-cutlass-dsl[cu13]==4.7.1->vllm)
  Downloading nvidia_cuda_nvdisasm-13.4.92-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.9 kB)
Collecting nvidia-cutlass-dsl-libs-core==4.7.1 (from nvidia-cutlass-dsl-libs-base==4.7.1->nvidia-cutlass-dsl==4.7.1->nvidia-cutlass-dsl[cu13]==4.7.1->vllm)
  Downloading nvidia_cutlass_dsl_libs_core-4.7.1-py3-none-any.whl.metadata (2.8 kB)
Collecting nvidia-cutlass-dsl-libs-cu13==4.7.1 (from nvidia-cutlass-dsl[cu13]==4.7.1->vllm)
  Downloading nvidia_cutlass_dsl_libs_cu13-4.7.1-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (2.9 kB)
Collecting torch-c-dlpack-ext (from quack-kernels==0.6.5->vllm)
  Downloading torch_c_dlpack_ext-0.1.5-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (14 kB)
Collecting ml-dtypes (from tilelang==0.1.12->vllm)
  Downloading ml_dtypes-0.6.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (8.8 kB)
Collecting z3-solver<4.15.5,>=4.13.0 (from tilelang==0.1.12->vllm)
  Downloading z3_solver-4.15.4.0-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (602 bytes)
Collecting tokenspeed-triton>=3.7.10.post20260531 (from tokenspeed-mla==0.1.8->vllm)
  Downloading tokenspeed_triton-3.8.10.post20260920-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (1.0 kB)
Requirement already satisfied: setuptools>=77.0.3 in /home/3355245/.conda/envs/inward/lib/python3.11/site-packages (from torch==2.13.0->vllm) (83.0.0)
Collecting sympy>=1.13.3 (from torch==2.13.0->vllm)
  Downloading sympy-1.14.0-py3-none-any.whl.metadata (12 kB)
Collecting networkx>=2.5.1 (from torch==2.13.0->vllm)
  Downloading networkx-3.6.1-py3-none-any.whl.metadata (6.8 kB)
Collecting fsspec>=0.8.5 (from torch==2.13.0->vllm)
  Downloading fsspec-2026.9.0-py3-none-any.whl.metadata (10 kB)
Collecting cuda-toolkit==13.0.3 (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
  Downloading cuda_toolkit-13.0.3.0-py2.py3-none-any.whl.metadata (17 kB)
Collecting nvidia-cudnn-cu13==9.20.0.48 (from torch==2.13.0->vllm)
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Collecting nvidia-cusparselt-cu13==0.8.1 (from torch==2.13.0->vllm)
  Downloading nvidia_cusparselt_cu13-0.8.1-py3-none-manylinux2014_x86_64.whl.metadata (12 kB)
Collecting nvidia-nccl-cu13==2.29.7 (from torch==2.13.0->vllm)
  Downloading nvidia_nccl_cu13-2.29.7-py3-none-manylinux_2_18_x86_64.whl.metadata (2.1 kB)
Collecting nvidia-nvshmem-cu13==3.4.5 (from torch==2.13.0->vllm)
  Downloading nvidia_nvshmem_cu13-3.4.5-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.1 kB)
```

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Collecting triton (from humming-kernels==0.1.12->humming-kernels[cu13]==0.1.12->vllm)
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Collecting nvidia-cublas==13.1.1.3.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
  Downloading nvidia_cublas-13.1.1.3-py3-none-manylinux_2_27_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cuda-nvrtc (from humming-kernels[cu13]==0.1.12->vllm)
  Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-runtime (from humming-kernels[cu13]==0.1.12->vllm)
  Downloading nvidia_cuda_runtime-13.0.96-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cufft==12.0.0.61.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
  Downloading nvidia_cufft-12.0.0.61-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-nvjitlink<14,>=13.0.88 (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
  Downloading nvidia_nvjitlink-13.4.92-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-cufile==1.15.1.6.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
  Downloading nvidia_cufile-1.15.1.6-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-cupti==13.0.85.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
  Downloading nvidia_cuda_cupti-13.0.85-py3-none-manylinux_2_25_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-curand==10.4.0.35.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
  Downloading nvidia_curand-10.4.0.35-py3-none-manylinux_2_27_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cusolver==12.0.4.66.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
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  Downloading nvidia_cusparse-12.6.3.3-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.8 kB)
Collecting nvidia-nvtx==13.0.85.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch==2.13.0->vllm)
  Downloading nvidia_nvtx-13.0.85-py3-none-manylinux1_x86_64.manylinux_2_5_x86_64.whl.metadata (1.8 kB)
Collecting cuda-pathfinder>=1.4.2 (from cuda-bindings->humming-kernels==0.1.12->humming-kernels[cu13]==0.1.12->vllm)
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Collecting typing-inspection>=0.4.2 (from fastapi<0.137.0,>=0.133.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
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Collecting annotated-doc>=0.0.2 (from fastapi<0.137.0,>=0.133.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
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Collecting fastapi-cli>=0.0.8 (from fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
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Collecting fastar>=0.9.0 (from fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading fastar-0.12.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
Collecting httpx<1.0.0,>=0.23.0 (from fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting python-multipart>=0.0.18 (from fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading python_multipart-0.0.32-py3-none-any.whl.metadata (2.1 kB)
Collecting email-validator>=2.0.0 (from fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading email_validator-2.3.0-py3-none-any.whl.metadata (26 kB)
Collecting uvicorn>=0.12.0 (from uvicorn[standard]>=0.12.0; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
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Collecting pydantic-settings>=2.0.0 (from fastapi[standard]<0.137.0,>=0.133.0->vllm)
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Collecting pydantic-extra-types>=2.0.0 (from fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading pydantic_extra_types-2.11.1-py3-none-any.whl.metadata (4.2 kB)
Collecting anyio (from httpx<1.0.0,>=0.23.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading anyio-4.15.1-py3-none-any.whl.metadata (4.7 kB)
Collecting certifi (from httpx<1.0.0,>=0.23.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Collecting httpcore==1.* (from httpx<1.0.0,>=0.23.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Collecting idna (from httpx<1.0.0,>=0.23.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading idna-3.20-py3-none-any.whl.metadata (7.2 kB)
Collecting h11>=0.16 (from httpcore==1.*->httpx<1.0.0,>=0.23.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting httpx2>=2.5.0 (from mcp<3.0.0,>=2.0.0->vllm)
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Collecting mcp-types==2.2.0 (from mcp<3.0.0,>=2.0.0->vllm)
  Downloading mcp_types-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting pyjwt>=2.10.1 (from pyjwt[crypto]>=2.10.1->mcp<3.0.0,>=2.0.0->vllm)
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Collecting sse-starlette>=3.0.0 (from mcp<3.0.0,>=2.0.0->vllm)
  Downloading sse_starlette-3.5.0-py3-none-any.whl.metadata (15 kB)
Collecting jmespath (from model-hosting-container-standards<1.0.0,>=0.1.14->vllm)
  Downloading jmespath-1.1.0-py3-none-any.whl.metadata (7.6 kB)
Collecting supervisor>=4.2.0 (from model-hosting-container-standards<1.0.0,>=0.1.14->vllm)
  Downloading supervisor-4.3.0-py2.py3-none-any.whl.metadata (87 kB)
Collecting huggingface_hub>=1.31.0 (from vllm)
  Downloading huggingface_hub-1.33.0-py3-none-any.whl.metadata (16 kB)
Collecting typer (from transformers)
  Downloading typer-0.27.2-py3-none-any.whl.metadata (16 kB)
Collecting hf-xet<2.0.0,>=1.6.0 (from huggingface_hub>=1.31.0->vllm)
  Downloading hf_xet-1.6.0-cp38-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (4.9 kB)
Collecting pyarrow>=21.0.0 (from datasets)
  Downloading pyarrow-25.0.1-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (3.0 kB)
Collecting xxhash (from datasets)
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Collecting multiprocess<0.70.20 (from datasets)
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Collecting fsspec>=0.8.5 (from torch==2.13.0->vllm)
  Downloading fsspec-2026.6.0-py3-none-any.whl.metadata (10 kB)
Collecting python-socketio[client] (from nnsight)
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Collecting toml (from nnsight)
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Collecting ipython (from nnsight)
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Collecting rich (from nnsight)
  Downloading rich-15.0.0-py3-none-any.whl.metadata (18 kB)
Collecting zstandard (from nnsight)
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Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.3-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.66.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (130 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (5.2 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Collecting aiohappyeyeballs>=2.5.0 (from aiohttp>=3.13.3->vllm)
  Downloading aiohappyeyeballs-2.7.1-py3-none-any.whl.metadata (5.9 kB)
Collecting aiosignal>=1.4.0 (from aiohttp>=3.13.3->vllm)
  Downloading aiosignal-1.4.0-py3-none-any.whl.metadata (3.7 kB)
Collecting attrs>=17.3.0 (from aiohttp>=3.13.3->vllm)
  Downloading attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting frozenlist>=1.1.1 (from aiohttp>=3.13.3->vllm)
  Downloading frozenlist-1.8.0-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (20 kB)
Collecting multidict<7.0,>=4.5 (from aiohttp>=3.13.3->vllm)
  Downloading multidict-6.9.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (5.3 kB)
Collecting propcache>=0.2.0 (from aiohttp>=3.13.3->vllm)
  Downloading propcache-0.5.4-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (25 kB)
Collecting yarl<2.0,>=1.17.0 (from aiohttp>=3.13.3->vllm)
  Downloading yarl-1.25.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (111 kB)
Collecting docstring-parser<1,>=0.15 (from anthropic>=0.71.0->vllm)
  Downloading docstring_parser-0.18.0-py3-none-any.whl.metadata (3.5 kB)
Collecting jiter<1,>=0.4.0 (from anthropic>=0.71.0->vllm)
  Downloading jiter-0.17.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (5.2 kB)
Collecting sniffio<2,>=1 (from anthropic>=0.71.0->vllm)
  Downloading sniffio-1.3.1-py3-none-any.whl.metadata (3.9 kB)
Collecting httpcore2==2.13.1 (from httpx2>=2.5.0->mcp<3.0.0,>=2.0.0->vllm)
  Downloading httpcore2-2.13.1-py3-none-any.whl.metadata (26 kB)
Collecting truststore>=0.10 (from httpx2>=2.5.0->mcp<3.0.0,>=2.0.0->vllm)
  Downloading truststore-0.10.4-py3-none-any.whl.metadata (4.4 kB)
Collecting annotated-types>=0.6.0 (from pydantic>=2.12.0->vllm)
  Downloading annotated_types-0.8.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.5 (from pydantic>=2.12.0->vllm)
  Downloading pydantic_core-2.46.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.6 kB)
Collecting cuda-core~=1.2.0 (from cuda-python>=12.0->flashinfer-python==0.6.18.post1->vllm)
  Downloading cuda_core-1.2.1-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (3.3 kB)
Collecting dnspython>=2.0.0 (from email-validator>=2.0.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
Collecting rich-toolkit>=0.14.8 (from fastapi-cli>=0.0.8->fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading rich_toolkit-0.20.5-py3-none-any.whl.metadata (3.0 kB)
Collecting fastapi-cloud-cli>=0.1.1 (from fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading fastapi_cloud_cli-0.26.0-py3-none-any.whl.metadata (3.1 kB)
Collecting rignore>=0.5.1 (from fastapi-cloud-cli>=0.1.1->fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
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  Downloading rignore-0.8.1-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.2 kB)
Collecting sentry-sdk>=2.20.0 (from fastapi-cloud-cli>=0.1.1->fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading sentry_sdk-2.71.0-py3-none-any.whl.metadata (11 kB)
Collecting detect-installer>=0.1.0 (from fastapi-cloud-cli>=0.1.1->fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading detect_installer-0.2.1-py3-none-any.whl.metadata (1.2 kB)
Collecting agent-detector>=1.1.0 (from fastapi-cloud-cli>=0.1.1->fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading agent_detector-2.0.0-py3-none-any.whl.metadata (6.6 kB)
Collecting MarkupSafe>=2.0 (from jinja2->humming-kernels==0.1.12->humming-kernels[cu13]==0.1.12->vllm)
  Downloading markupsafe-3.0.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Collecting jsonschema-specifications>=2023.03.6 (from jsonschema>=4.23.0->vllm)
  Downloading jsonschema_specifications-2025.9.1-py3-none-any.whl.metadata (2.9 kB)
Collecting referencing>=0.28.4 (from jsonschema>=4.23.0->vllm)
  Downloading referencing-0.37.0-py3-none-any.whl.metadata (2.8 kB)
Collecting rpds-py>=0.25.0 (from jsonschema>=4.23.0->vllm)
  Downloading rpds_py-2026.6.3-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
Collecting numpy (from vllm)
  Downloading numpy-2.3.5-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (62 kB)
Collecting opentelemetry-exporter-otlp-proto-grpc==1.45.0 (from opentelemetry-exporter-otlp>=1.27.0->vllm)
  Downloading opentelemetry_exporter_otlp_proto_grpc-1.45.0-py3-none-any.whl.metadata (2.6 kB)
Collecting opentelemetry-exporter-otlp-proto-http==1.45.0 (from opentelemetry-exporter-otlp>=1.27.0->vllm)
  Downloading opentelemetry_exporter_otlp_proto_http-1.45.0-py3-none-any.whl.metadata (3.1 kB)
Collecting googleapis-common-protos~=1.57 (from opentelemetry-exporter-otlp-proto-grpc==1.45.0->opentelemetry-exporter-otlp>=1.27.0->vllm)
  Downloading googleapis_common_protos-1.75.5-py3-none-any.whl.metadata (8.5 kB)
Collecting grpcio<2.0.0,>=1.63.2 (from opentelemetry-exporter-otlp-proto-grpc==1.45.0->opentelemetry-exporter-otlp>=1.27.0->vllm)
  Downloading grpcio-1.84.0-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (3.8 kB)
Collecting opentelemetry-exporter-otlp-common==0.66b0 (from opentelemetry-exporter-otlp-proto-grpc==1.45.0->opentelemetry-exporter-otlp>=1.27.0->vllm)
  Downloading opentelemetry_exporter_otlp_common-0.66b0-py3-none-any.whl.metadata (1.9 kB)
Collecting opentelemetry-exporter-otlp-proto-common==1.45.0 (from opentelemetry-exporter-otlp-proto-grpc==1.45.0->opentelemetry-exporter-otlp>=1.27.0->vllm)
  Downloading opentelemetry_exporter_otlp_proto_common-1.45.0-py3-none-any.whl.metadata (1.8 kB)
Collecting opentelemetry-proto==1.45.0 (from opentelemetry-exporter-otlp-proto-grpc==1.45.0->opentelemetry-exporter-otlp>=1.27.0->vllm)
  Downloading opentelemetry_proto-1.45.0-py3-none-any.whl.metadata (2.3 kB)
Collecting opentelemetry-exporter-http-transport==0.66b0 (from opentelemetry-exporter-http-transport[urllib3]==0.66b0->opentelemetry-exporter-otlp-proto-http==1.45.0->opentelemetry-exporter-otlp>=1.27.0->vll
m)
  Downloading opentelemetry_exporter_http_transport-0.66b0-py3-none-any.whl.metadata (2.3 kB)
Collecting urllib3>=1.26 (from opentelemetry-exporter-http-transport[urllib3]==0.66b0->opentelemetry-exporter-otlp-proto-http==1.45.0->opentelemetry-exporter-otlp>=1.27.0->vllm)
  Downloading urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)
Collecting opentelemetry-semantic-conventions==0.66b0 (from opentelemetry-sdk>=1.27.0->vllm)
  Downloading opentelemetry_semantic_conventions-0.66b0-py3-none-any.whl.metadata (2.4 kB)
Collecting pycountry>=23 (from pydantic-extra-types[pycountry]>=2.10.5->mistral_common>=1.11.6->mistral_common[image]>=1.11.6->vllm)
  Downloading pycountry-26.2.16-py3-none-any.whl.metadata (12 kB)
Collecting python-dotenv>=0.21.0 (from pydantic-settings>=2.0.0->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading python_dotenv-1.2.3-py3-none-any.whl.metadata (29 kB)
Collecting cryptography>=3.4.0 (from pyjwt[crypto]>=2.10.1->mcp<3.0.0,>=2.0.0->vllm)
  Downloading cryptography-50.0.2-cp311-abi3-manylinux_2_34_x86_64.whl.metadata (4.4 kB)
Collecting cffi>=2.0.0 (from cryptography>=3.4.0->pyjwt[crypto]>=2.10.1->mcp<3.0.0,>=2.0.0->vllm)
  Downloading cffi-2.1.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.5 kB)
Collecting pycparser (from cffi>=2.0.0->cryptography>=3.4.0->pyjwt[crypto]>=2.10.1->mcp<3.0.0,>=2.0.0->vllm)
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting charset_normalizer<4,>=2 (from requests>=2.26.0->vllm)
  Downloading charset_normalizer-3.5.2-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (46 kB)
Collecting markdown-it-py>=2.2.0 (from rich->nnsight)
  Downloading markdown_it_py-4.2.0-py3-none-any.whl.metadata (7.4 kB)
Collecting pygments<3.0.0,>=2.13.0 (from rich->nnsight)
  Downloading pygments-2.21.0-py3-none-any.whl.metadata (2.5 kB)
Collecting mdurl~=0.1 (from markdown-it-py>=2.2.0->rich->nnsight)
  Downloading mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Collecting mpmath<1.4,>=1.1.0 (from sympy>=1.13.3->torch==2.13.0->vllm)
  Downloading mpmath-1.3.0-py3-none-any.whl.metadata (8.6 kB)
Collecting shellingham>=1.3.0 (from typer->transformers)
  Downloading shellingham-1.5.4-py2.py3-none-any.whl.metadata (3.5 kB)
Collecting httptools>=0.8.0 (from uvicorn[standard]>=0.12.0; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading httptools-0.8.0-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (3.5 kB)
Collecting uvloop>=0.15.1 (from uvicorn[standard]>=0.12.0; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading uvloop-0.22.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (4.9 kB)
Collecting websockets>=13.0 (from uvicorn[standard]>=0.12.0; extra == "standard"->fastapi[standard]<0.137.0,>=0.133.0->vllm)
  Downloading websockets-17.1-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (6.3 kB)
Collecting ipython-pygments-lexers>=1.0.0 (from ipython->nnsight)
  Downloading ipython_pygments_lexers-1.1.1-py3-none-any.whl.metadata (1.1 kB)
Collecting jedi>=0.18.2 (from ipython->nnsight)
  Downloading jedi-0.20.0-py2.py3-none-any.whl.metadata (23 kB)
Collecting matplotlib-inline>=0.1.6 (from ipython->nnsight)
  Downloading matplotlib_inline-0.2.2-py3-none-any.whl.metadata (2.4 kB)
Collecting pexpect>4.6 (from ipython->nnsight)
  Downloading pexpect-4.9.0-py2.py3-none-any.whl.metadata (2.5 kB)
Collecting prompt_toolkit<3.1.0,>=3.0.41 (from ipython->nnsight)
  Downloading prompt_toolkit-3.0.53-py3-none-any.whl.metadata (6.4 kB)
Collecting stack_data>=0.6.0 (from ipython->nnsight)
  Downloading stack_data-0.6.3-py3-none-any.whl.metadata (18 kB)
Collecting traitlets>=5.13.0 (from ipython->nnsight)
  Downloading traitlets-5.16.1-py3-none-any.whl.metadata (10 kB)
Collecting wcwidth>=0.1.4 (from prompt_toolkit<3.1.0,>=3.0.41->ipython->nnsight)
  Downloading wcwidth-0.9.1-cp310-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (23 kB)
Collecting parso<0.9.0,>=0.8.6 (from jedi>=0.18.2->ipython->nnsight)
  Downloading parso-0.8.7-py2.py3-none-any.whl.metadata (8.2 kB)
Collecting ptyprocess>=0.5 (from pexpect>4.6->ipython->nnsight)
  Downloading ptyprocess-0.7.0-py2.py3-none-any.whl.metadata (1.3 kB)
Collecting executing>=1.2.0 (from stack_data>=0.6.0->ipython->nnsight)
  Downloading executing-2.2.1-py2.py3-none-any.whl.metadata (8.9 kB)
Collecting asttokens>=2.1.0 (from stack_data>=0.6.0->ipython->nnsight)
  Downloading asttokens-3.0.2-py3-none-any.whl.metadata (5.7 kB)
Collecting pure-eval (from stack_data>=0.6.0->ipython->nnsight)
  Downloading pure_eval-0.2.4-py3-none-any.whl.metadata (6.4 kB)
Collecting nvidia-nvvm (from nvidia-cuda-nvcc->humming-kernels[cu13]==0.1.12->vllm)
  Downloading nvidia_nvvm-13.4.92-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cuda-crt (from nvidia-cuda-nvcc->humming-kernels[cu13]==0.1.12->vllm)
  Downloading nvidia_cuda_crt-13.4.92-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (1.7 kB)
Collecting bidict>=0.21.0 (from python-socketio[client]->nnsight)
  Downloading bidict-0.24.1-py3-none-any.whl.metadata (8.7 kB)
Collecting python-engineio>=4.13.2 (from python-socketio[client]->nnsight)
  Downloading python_engineio-4.14.0-py3-none-any.whl.metadata (2.5 kB)
Collecting websocket-client>=0.54.0 (from python-socketio[client]->nnsight)
  Downloading websocket_client-1.9.2-py3-none-any.whl.metadata (8.3 kB)
Collecting simple-websocket>=0.10.0 (from python-engineio>=4.13.2->python-socketio[client]->nnsight)
  Downloading simple_websocket-1.1.0-py3-none-any.whl.metadata (1.5 kB)
Collecting wsproto (from simple-websocket>=0.10.0->python-engineio>=4.13.2->python-socketio[client]->nnsight)
  Downloading wsproto-1.3.2-py3-none-any.whl.metadata (5.2 kB)
Downloading vllm-0.30.0-cp38-abi3-manylinux_2_28_x86_64.whl (314.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 314.9/314.9 MB 115.5 MB/s  0:00:02
Downloading apache_tvm_ffi-0.1.11-cp311-cp311-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (2.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.8/2.8 MB 130.4 MB/s  0:00:00
Downloading compressed_tensors-0.17.0-py3-none-any.whl (211 kB)
Downloading depyf-0.20.0-py3-none-any.whl (39 kB)
Downloading flashinfer_python-0.6.18.post1-py3-none-any.whl (18.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 18.3/18.3 MB 117.6 MB/s  0:00:00
Downloading humming_kernels-0.1.12-py3-none-manylinux_2_28_x86_64.whl (322 kB)
Downloading lark-1.2.2-py3-none-any.whl (111 kB)
Downloading lm_format_enforcer-0.11.3-py3-none-any.whl (45 kB)
Downloading numba-0.65.0-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (3.7 MB)
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Downloading nvidia_cutlass_dsl-4.7.1-py3-none-any.whl (10 kB)
Downloading nvidia_cutlass_dsl_libs_base-4.7.1-cp311-cp311-manylinux_2_28_x86_64.whl (2.8 MB)
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Downloading nvidia_cutlass_dsl_libs_core-4.7.1-py3-none-any.whl (1.2 MB)
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Downloading nvidia_cutlass_dsl_libs_cu12-4.7.1-cp311-cp311-manylinux_2_28_x86_64.whl (88.7 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 88.7/88.7 MB 60.6 MB/s  0:00:01
Downloading nvidia_cutlass_dsl_libs_cu13-4.7.1-cp311-cp311-manylinux_2_28_x86_64.whl (88.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 88.3/88.3 MB 106.2 MB/s  0:00:00
Downloading nvtx-0.2.15-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (808 kB)
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Downloading outlines_core-0.2.14-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.3 MB)
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Downloading pynvvideocodec-2.0.4-cp311-cp311-manylinux_2_28_x86_64.whl (43.2 MB)
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Downloading quack_kernels-0.6.5-py3-none-any.whl (729 kB)
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Downloading tilelang-0.1.12-cp38-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (50.5 MB)
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Downloading tokenspeed_mla-0.1.8-py3-none-manylinux_2_28_x86_64.whl (755 kB)
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   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 755.8/755.8 kB 330.2 MB/s  0:00:00
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Downloading xxhash-4.0.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (261 kB)
Downloading zstandard-0.25.0-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (5.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.6/5.6 MB 108.2 MB/s  0:00:00
Installing collected packages: z3-solver, torchaudio, supervisor, py-cpuinfo, pure-eval, ptyprocess, nvtx, nvidia-ml-py, nvidia-cusparselt-cu13, mpmath, cuda-toolkit, zstandard, xxhash, websockets, websocket-client, wcwidth, uvloop, urllib3, typing_extensions, trusststore, triton, traitlets, tqdm, torchcodec, toml, tokenspeed-triton, tabulate, sympy, sniffio, six, shellingham, setproctitle, sentencepiece, safetensors, rpds-py, rignore, regex, pyzmq, pyyaml, python-multipart, python-json-logger, python-dotenv, pyparsing, PyNvVideoCodec, pyjwt, pygments, pycparser, pycountry, pybase64, pyarrow, psutil, protobuf, propcache, prometheus_client, pillow, pexpect, partial-json-parser, parso, outlines_core, nvidia-nvvm, nvidia-nvtx, nvidia-nvshmem-cu13, nvidia-nvjitlink, nvidia-nccl-cu13, nvidia-curand, nvidia-cufile, nvidia-cuda-runtime, nvidia-cuda-nvrtc, nvidia-cuda-nvdisasm, nvidia-cuda-cupti, nvidia-cuda-crt, nvidia-cuda-cccl, numpy, ninja, networkx, multidict, msgspec, mdurl, MarkupSafe, loguru, llvmlite, llguidance, lark, kiwisolver, jmespath, jiter, interegular, ijson, idna, httptools, hf-xet, h11, fsspec, frozenlist, fonttools, filelock, fastar, executing, einops, docstring-parser, dnspython, dill, detect-installer, cycler, cuda-pathfinder, cloudpickle, click, charset_normalizer, certifi, cbor2, cachetools, attrs, asttokens, astor, annotated-types, annotated-doc, aiohappyeyeballs, agent-detector, yarl, wsproto, uvicorn, typing-inspection, stack_data, sentry-sdk, requests, referencing, python-dateutil, pydantic-core, prompt_toolkit, opentelemetry-proto, opentelemetry-api, opencv-python-headless, nvidia-cusparse, nvidia-cufft, nvidia-cuda-nvcc, nvidia-cublas, numba, multiprocess, ml-dtypes, matplotlib-inline, markdown-it-py, jinja2, jedi, ipython-pygments-lexers, httpcore2, httpcore, grpcio, googleapis-common-protos, email-validator, depyf, cuda-tile, cuda-core, cuda-bindings, contourpy, cffi, blake3, bidict, apache-tvm-ffi, anyio, aiosignal, watchfiles, tiktoken, starlette, simple-websocket, rich, pydantic, pandas, opentelemetry-semantic-conventions, opentelemetry-exporter-otlp-proto-common, opentelemetry-exporter-http-transport, nvidia-cusolver, nvidia-cudnn-cu13, nccl4py, matplotlib, jsonschema-specifications, ipython, httpx2, httpx, cuda-python, cryptography, aiohttp, typer, sse-starlette, rich-toolkit, python-engineio, pydantic-settings, pydantic-extra-types, prometheus-fastapi-instrumentator, opentelemetry-sdk, openai-harmony, openai, nvidia-cutlass-dsl-libs-core, mcp-types, lm-format-enforcer, jsonschema, huggingface_hub, fastapi, anthropic, torch, tokenizers, python-socketio, opentelemetry-semantic-conventions-ai, opentelemetry-exporter-otlp-common, nvidia-cutlass-dsl-libs-base, model-hosting-container-standards, mcp, fastsafetensors, fastapi-cloud-cli, fastapi-cli, datasets, transformers, torchvision, torch-c-dlpack-ext, opentelemetry-exporter-otlp-proto-http, opentelemetry-exporter-otlp-proto-grpc, nvidia-cutlass-dsl-libs-cu13, nvidia-cutlass-dsl-libs-cu12, mistral_common, instanttensor, humming-kernels, accelerate, xgrammar, tilelang, peft, opentelemetry-exporter-otlp, nvidia-cutlass-dsl, nnsight, compressed-tensors, tokenspeed-mla, quack-kernels, nvidia-cudnn-frontend, flashinfer-python, vllm
Successfully installed MarkupSafe-3.0.3 PyNvVideoCodec-2.0.4 accelerate-1.15.0 agent-detector-2.0.0 aiohappyeyeballs-2.7.1 aiohttp-3.14.3 aiosignal-1.4.0 annotated-doc-0.0.5 annotated-types-0.8.0 anthropic-1.9.0 anyio-4.15.1 apache-tvm-ffi-0.1.11 astor-0.8.1 asttokens-3.0.2 attrs-26.1.0 bidict-0.24.1 blake3-1.0.10 cachetools-7.2.0 cbor2-6.1.4 certifi-2026.7.22 cffi-2.1.1 charset_normalizer-3.5.2 click-8.5.0 cloudpickle-3.1.2 compressed-tensors-0.17.0 contourpy-1.3.3 cryptography-50.0.2 cuda-bindings-13.4.3 cuda-core-1.2.1 cuda-pathfinder-1.8.2 cuda-python-13.4.1 cuda-tile-1.6.0 cuda-toolkit-13.0.3.0 cycler-0.12.1 datasets-5.0.1 depyf-0.20.0 detect-installer-0.2.1 dill-0.4.1 dnspython-2.8.0 docstring-parser-0.18.0 einops-0.8.2 email-validator-2.3.0 executing-2.2.1 fastapi-0.136.3 fastapi-cli-0.0.32 fastapi-cloud-cli-0.26.0 fastar-0.12.0 fastsafetensors-0.4.0 filelock-4.0.7 flashinfer-python-0.6.18.post1 fonttools-4.66.1 frozenlist-1.8.0 fsspec-2026.6.0 googleapis-common-protos-1.75.5 grpcio-1.84.0 h11-0.16.0 hf-xet-1.6.0 httpcore-1.0.9 httpcore2-2.13.1 httptools-0.8.0 httpx-0.28.1 httpx2-2.13.1 huggingface_hub-1.33.0 humming-kernels-0.1.12 idna-3.20 ijson-3.5.1 instanttensor-0.2.0 interegular-0.3.3 ipython-9.17.1 ipython-pygments-lexers-1.1.1 jedi-0.20.0 jinja2-3.1.6 jiter-0.17.0 jmespath-1.1.0 jsonschema-4.26.0 jsonschema-specifications-2025.9.1 kiwisolver-1.5.1 lark-1.2.2 llguidance-1.7.6 llvmlite-0.47.0 lm-format-enforcer-0.11.3 loguru-0.7.3 markdown-it-py-4.2.0 matplotlib-3.11.2 matplotlib-inline-0.2.2 mcp-2.2.0 mcp-types-2.2.0 mdurl-0.1.2 mistral_common-1.12.0 ml-dtypes-0.6.0 model-hosting-container-standards-0.1.16 mpmath-1.3.0 msgspec-0.22.0 multidict-6.9.1 multiprocess-0.70.19 nccl4py-0.6.0 networkx-3.6.1 ninja-1.13.2 nnsight-0.7.0 numba-0.65.0 numpy-2.3.5 nvidia-cublas-13.1.1.3 nvidia-cuda-cccl-13.3.4.3.1 nvidia-cuda-crt-13.4.92 nvidia-cuda-cupti-13.0.85 nvidia-cuda-nvcc-13.4.92 nvidia-cuda-nvdisasm-13.4.92 nvidia-cuda-nvrtc-13.0.88 nvidia-cuda-runtime-13.0.96 nvidia-cudnn-cu13-9.20.0.48 nvidia-cudnn-frontend-1.30.0 nvidia-cufft-12.0.0.61 nvidia-cufile-1.15.1.6 nvidia-curand-10.4.0.35 nvidia-cusolver-12.0.4.66 nvidia-cusparse-12.6.3.3 nvidia-cusparselt-cu13-0.8.1 nvidia-cutlass-dsl-4.7.1 nvidia-cutlass-dsl-libs-base-4.7.1 nvidia-cutlass-dsl-libs-core-4.7.1 nvidia-cutlass-dsl-libs-cu12-4.7.1 nvidia-cutlass-dsl-libs-cu13-4.7.1 nvidia-ml-py-13.615.71 nvidia-nccl-cu13-2.29.7 nvidia-nvjitlink-13.4.92 nvidia-nvshmem-cu13-3.4.5 nvidia-nvtx-13.0.85 nvidia-nvvm-13.4.92 nvtx-0.2.15 openai-3.22.1 openai-harmony-0.0.8 opencv-python-headless-5.0.0.93 opentelemetry-api-1.45.0 opentelemetry-exporter-http-transport-0.66b0 opentelemetry-exporter-otlp-1.45.0 opentelemetry-exporter-otlp-common-0.66b0 opentelemetry-exporter-otlp-proto-common-1.45.0 opentelemetry-exporter-otlp-proto-grpc-1.45.0 opentelemetry-exporter-otlp-proto-http-1.45.0 opentelemetry-proto-1.45.0 opentelemetry-sdk-1.45.0 opentelemetry-semantic-conventions-0.66b0 opentelemetry-semantic-conventions-ai-0.5.1 outlines_core-0.2.14 pandas-3.0.6 parso-0.8.7 partial-json-parser-0.2.1.1.post7 peft-0.21.1 pexpect-4.9.0 pillow-12.3.0 prometheus-fastapi-instrumentator-8.1.0 prometheus_client-0.26.0 prompt_toolkit-3.0.53 propcache-0.5.4 protobuf-7.36.2 psutil-7.2.2 ptyprocess-0.7.0 pure-eval-0.2.4 py-cpuinfo-9.0.0 pyarrow-25.0.1 pybase64-1.5.0 pycountry-26.2.16 pycparser-3.0 pydantic-2.13.5 pydantic-core-2.46.5 pydantic-extra-types-2.11.1 pydantic-settings-2.15.0 pygments-2.21.0 pyjwt-2.15.1 pyparsing-3.3.3 python-dateutil-2.9.0.post0 python-dotenv-1.2.3 python-engineio-4.14.0 python-json-logger-4.2.0 python-multipart-0.0.32 python-socketio-5.17.0 pyyaml-6.0.3 pyzmq-27.2.0 quack-kernels-0.6.5 referencing-0.37.0 regex-2026.9.29 requests-2.34.2 rich-15.0.0 rich-toolkit-0.20.5 rignore-0.8.1 rpds-py-2026.6.3 safetensors-0.8.0 sentencepiece-0.2.2 sentry-sdk-2.71.0 setproctitle-1.3.7 shellingham-1.5.4 simple-websocket-1.1.0 six-1.17.0 sniffio-1.3.1 sse-starlette-3.5.0 stack_data-0.6.3 starlette-1.7.0 supervisor-4.3.0 sympy-1.14.0 tabulate-0.10.0 tiktoken-0.14.0 tilelang-0.1.12 tokenizers-0.23.2 tokenspeed-mla-0.1.8 tokenspeed-triton-3.8.10.post20260920 toml-0.10.2 torch-2.13.0 torch-c-dlpack-ext-0.1.5 torchaudio-2.11.0 torchcodec-0.17.0 torchvision-0.28.0 tqdm-4.70.1 traitlets-5.16.1 transformers-5.17.0 triton-3.7.1 trusststore-0.10.4 typer-0.27.2 typing-inspection-0.4.4 typing_extensions-4.16.0 urllib3-2.8.0 uvicorn-0.54.0 uvloop-0.22.1 vllm-0.30.0 watchfiles-1.3.0 wcwidth-0.9.1 websocket-client-1.9.2 websockets-17.1 wsproto-1.3.2 xgrammar-0.2.8 xxhash-4.0.1 yarl-1.25.1 z3-solver-4.15.4.0 zstandard-0.25.0
Will remove 45 (113.4 MB) tarball(s).
Will remove 1 index cache(s).
Will remove 5 (3.1 MB) package(s).
There are no tempfile(s) to remove.
There are no logfile(s) to remove.
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=11GB LIMIT=50GB
(inward) [3355245@slnode01 ~]$ mkdir -p ~/inward/{models,data,results,logs}
cd ~/inward
git clone --depth 1 https://github.com/ofirpress/self-ask.git
cp self-ask/datasets/compositional_celebrities.json data/

nohup bash -c 'for m in Qwen3-0.6B Qwen3-1.7B; do hf download Qwen/$m --local-dir models/$m; done' > logs/download.log 2>&1 &
Cloning into 'self-ask'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 9 (delta 0), reused 1 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), 281.46 KiB | 2.63 MiB/s, done.
[1] 461292
(inward) [3355245@slnode01 inward]$ tail -3 logs/download.log
du -sh models/*
nohup: ignoring input
bash: line 1: hf: command not found
bash: line 1: hf: command not found
[1]+  Exit 127                nohup bash -c 'for m in Qwen3-0.6B Qwen3-1.7B; do hf download Qwen/$m --local-dir models/$m; done' > logs/download.log 2>&1
du: cannot access 'models/*': No such file or directory
(inward) [3355245@slnode01 inward]$ nohup python -c "
from huggingface_hub import snapshot_download
for m in ['Qwen3-0.6B', 'Qwen3-1.7B']:
    snapshot_download(f'Qwen/{m}', local_dir=f'models/{m}')
    print('done', m, flush=True)
" > logs/download.log 2>&1 &
[1] 461402
(inward) [3355245@slnode01 inward]$ lquota
USED=11GB LIMIT=50GB
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
(inward) [3355245@slnode01 inward]$ tail -3 logs/download.log
du -sh models/*
done Qwen3-0.6B
Fetching 12 files: 100%|██████████| 12/12 [00:59<00:00,  4.97s/it]
done Qwen3-1.7B
[1]+  Done                    nohup python -c "
from huggingface_hub import snapshot_download
for m in ['Qwen3-0.6B', 'Qwen3-1.7B']:
    snapshot_download(f'Qwen/{m}', local_dir=f'models/{m}')
    print('done', m, flush=True)
" > logs/download.log 2>&1
1.5G    models/Qwen3-0.6B
3.8G    models/Qwen3-1.7B
(inward) [3355245@slnode01 inward]$ cat > gpu_test.sh << 'EOF'
#!/bin/bash
#SBATCH -J gputest
#SBATCH -p stud
#SBATCH --qos=stud
#SBATCH --gres=gpu:4g.40gb:1
#SBATCH --cpus-per-task=8
#SBATCH --mem=64G
#SBATCH --time=00:15:00
#SBATCH -o logs/%x-%j.out

eval "$(conda shell.bash hook)"
conda activate inward
export HF_HUB_OFFLINE=1

nvidia-smi
python -c "import torch; print('CUDA ok:', torch.cuda.is_available(), torch.cuda.get_device_name(0))"
python -c "
from vllm import LLM, SamplingParams
llm = LLM('$HOME/inward/models/Qwen3-0.6B', gpu_memory_utilization=0.8, max_model_len=2048, enforce_eager=True)
print('OUTPUT:', llm.generate(['The capital of France is'], SamplingParams(max_tokens=8))[0].outputs[0].text)
"
EOF

sbatch gpu_test.sh
squeue -u $USER
Submitted batch job 677541
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677541      stud  gputest  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ squeue -p stud                # stud 分区里所有人的任务，R 是在跑，PD 是在排队
squeue -p stud -t PD | wc -l   # 排队中的任务数（结果减 1，因为第一行是表头）
squeue --start -j 677541       # SLURM 估计你的任务什么时候开始
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677541      stud  gputest  3355245 PD       0:00      1 (Resources)
            677443      stud gate-smo  3343816 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            677538      stud    sim22  3195720 PD       0:00      1 (Priority)
            677499      stud robust-4  3161355 PD       0:00      1 (Dependency)
            677105      stud sae135-L  3343855 PD       0:00      1 (Dependency)
            677508      stud euclidea  3246955  R      44:52      1 gnode01
            677309      stud cotd-tra  3343816  R    3:42:17      1 gnode01
            677513      stud evolve-l  3242106  R      40:05      1 gnode02
            677536      stud pca-refi  3160552  R       3:39      1 gnode01
            677491      stud robust-4  3161355  R      57:31      1 gnode01
            677104      stud sae135-L  3343855  R    7:30:39      1 gnode01
            676454      stud  kg_tune  3310986  R   21:25:17      1 gnode02
```

```
           677306      stud vllm-qwe  3221337  R    3:45:22      1 gnode02
6
            JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
           677541      stud  gputest  3355245 PD 2026-09-30T20:38:15      1 gnode01             (Resources)
(inward) [3355245@slnode01 inward]$ scancel 677541
sed -i 's/gpu:4g.40gb:1/gpu:1/' gpu_test.sh
sbatch gpu_test.sh
squeue -u $USER
Submitted batch job 677576
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
           677576      stud  gputest  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
           677576      stud  gputest  3355245  R       0:33      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -u $USER
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
           677576      stud  gputest  3355245  R       2:03      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -u $USER
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ sacct -j 677576 --format=JobID,State,Elapsed
JobID           State    Elapsed
------------ ---------- ----------
677576        COMPLETED   00:02:54
677576.batch  COMPLETED   00:02:54
(inward) [3355245@slnode01 inward]$ cat logs/gputest-677576.out | tail -30
Loading safetensors checkpoint shards: 100% Completed | 1/1 [00:03<00:00,  3.10s/it]
(EngineCore pid=1605545)
(EngineCore pid=1605545) INFO 09-30 18:47:35 [default_loader.py:430] Loading weights took 3.15 seconds
(EngineCore pid=1605545) INFO 09-30 18:47:36 [model_runner.py:428] Model loading took 1.12 GiB memory and 11.703850 seconds
(EngineCore pid=1605545) INFO 09-30 18:47:36 [topk_topp_sampler.py:78] Using FlashInfer for top-p & top-k sampling.
(EngineCore pid=1605545) INFO 09-30 18:47:36 [utils.py:320] Using LBNHC KV cache layout.
(EngineCore pid=1605545) INFO 09-30 18:47:39 [gpu_worker.py:640] Available KV cache memory: 29.78 GiB
(EngineCore pid=1605545) INFO 09-30 18:47:39 [kv_cache_utils.py:2395] GPU KV cache size: 278,816 tokens, Maximum concurrency for 2,048 tokens per request: 136.14x
(EngineCore pid=1605545) INFO 09-30 18:47:39 [kernel_warmup.py:172] JIT kernel warmup starting.
(EngineCore pid=1605545) INFO 09-30 18:47:39 [kernel_warmup.py:185] JIT kernel warmup finished in 0.00s.
(EngineCore pid=1605545) INFO 09-30 18:47:49 [gpu_worker.py:888] Free memory on device (38.97/39.25 GiB) on startup. Desired GPU memory utilization is (0.8, 31.4 GiB). Actual usage is 1.42 GiB for consumed m
emory (weights + non-torch), 0.2 GiB for peak activation, and 0.0 GiB for CUDAGraph memory. Replace gpu_memory_utilization config with `--kv-cache-memory=31819662234` (29.63 GiB) to fit into requested memory
, or `--kv-cache-memory=39948577280` (37.21 GiB) to fully utilize gpu memory. Current kv cache memory in use is 29.78 GiB.
(EngineCore pid=1605545) INFO 09-30 18:47:57 [jit_monitor.py:85] Kernel JIT monitor activated; monitored JIT compilations during inference will use mode=warn.
(EngineCore pid=1605545) INFO 09-30 18:47:57 [torch_utils.py:287] Reducing Torch threads from 8 to 1 for serving. Set OMP_NUM_THREADS in the external environment to override.
(EngineCore pid=1605545) INFO 09-30 18:47:57 [core.py:379] init engine (profile, create kv cache, warmup model) took 21.02 s
(EngineCore pid=1605545) INFO 09-30 18:47:59 [kernel.py:416] Final IR op priority after setting platform defaults: IrOpPriorityConfig(rms_norm=['vllm_c', 'native'], fused_add_rms_norm=['vllm_c', 'native'], g
elu_and_mul_sparse=['triton', 'native'])
INFO 09-30 18:48:01 [hf.py:642] Detected the chat template content format to be 'string'. You can set `--chat-template-content-format` to override this.
Rendering prompts: 100%|██████████| 1/1 [00:00<00:00,  1.40it/s]
Processed prompts: 100%|██████████| 1/1 [00:00<00:00,  8.36it/s, est. speed input: 43.61 toks/s, output: 69.77 toks/s]
OUTPUT:  Paris, and the President of France is
INFO 09-30 18:48:02 [utils.py:632] [shutdown] Process manager: send sigterm to process EngineCore
(EngineCore pid=1605545) INFO 09-30 18:48:02 [core.py:1342] [shutdown] EngineCore: trigger received signal=SIGTERM
(EngineCore pid=1605545) INFO 09-30 18:48:02 [core.py:1493] [shutdown] EngineCore: start mode=abort timeout=0s
(EngineCore pid=1605545) INFO 09-30 18:48:02 [core.py:1524] [shutdown] EngineCore: request processing complete; starting resource teardown
(EngineCore pid=1605545) INFO 09-30 18:48:02 [core.py:1355] [shutdown] EngineCore: exiting busy loop
WARNING 09-30 18:48:03 [core_client.py:791] [shutdown] MPClient: engine core exited unexpectedly; starting cleanup
INFO 09-30 18:48:03 [core_client.py:744] [shutdown] MPClient: start timeout=default
INFO 09-30 18:48:03 [core_client.py:746] [shutdown] MPClient: stopping engine manager
INFO 09-30 18:48:03 [core_client.py:748] [shutdown] MPClient: engine manager stopped
INFO 09-30 18:48:03 [core_client.py:749] [shutdown] MPClient: cleaning up background resources
INFO 09-30 18:48:03 [core_client.py:751] [shutdown] MPClient: complete
(inward) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=16.5GB LIMIT=50GB
(inward) [3355245@slnode01 inward]$ cd ~/inward
ls exp1_gap.py run_exp1.sh
sed -i 's/gpu:4g.40gb:1/gpu:1/' run_exp1.sh
sbatch run_exp1.sh Qwen3-0.6B Qwen3-1.7B
squeue -u $USER
exp1_gap.py  run_exp1.sh
Submitted batch job 677584
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
           677584      stud     exp1  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
           677584      stud     exp1  3355245 PD       0:00      1 (Resources)
(inward) [3355245@slnode01 inward]$ squeue -p stud
squeue --start -j 677584
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
           677584      stud     exp1  3355245 PD       0:00      1 (Resources)
           677499      stud robust-4  3161355 PD       0:00      1 (Dependency)
           677105      stud sae135-L  3343855 PD       0:00      1 (Dependency)
           677582      stud latdefus  3310909 PD       0:00      1 (Priority)
           677583      stud latdefus  3310909 PD       0:00      1 (Dependency)
           677575      stud cotd-tra  3343816  R       4:31      1 gnode01
           677570      stud    sim21  3195720  R      17:04      1 gnode01
           677580      stud medsam3-  3160552  R       8:36      1 gnode02
           677491      stud robust-4  3161355  R    1:36:07      1 gnode01
           677104      stud sae135-L  3343855  R    8:09:15      1 gnode01
           676454      stud  kg_tune  3310986  R   22:03:53      1 gnode02
           677306      stud vllm-qwe  3221337  R    4:23:58      1 gnode02
            JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
           677584      stud     exp1  3355245 PD 2026-09-30T20:45:15      1 gnode02             (Resources)
(inward) [3355245@slnode01 inward]$ squeue -p stud
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
           677499      stud robust-4  3161355 PD       0:00      1 (Resources)
           677105      stud sae135-L  3343855 PD       0:00      1 (Dependency)
           677582      stud latdefus  3310909 PD       0:00      1 (Priority)
           677583      stud latdefus  3310909 PD       0:00      1 (Dependency)
           677584      stud     exp1  3355245  R       0:14      1 gnode01
           677575      stud cotd-tra  3343816  R       4:57      1 gnode01
           677570      stud    sim21  3195720  R      17:30      1 gnode01
           677580      stud medsam3-  3160552  R       9:02      1 gnode02
           677104      stud sae135-L  3343855  R    8:09:41      1 gnode01
           676454      stud  kg_tune  3310986  R   22:04:19      1 gnode02
           677306      stud vllm-qwe  3221337  R    4:24:24      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue --start -k 677584
squeue: invalid option -- 'k'
Try "squeue --help" for more information
(inward) [3355245@slnode01 inward]$ squeue --start -j 677584
            JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
            JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
           677584      stud     exp1  3355245  R       2:02      1 gnode01
(inward) [3355245@slnode01 inward]$ Read from remote host slogin.hpc.unibocconi.it: Can't assign requested address
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────┐
                Welcome to  the Bocconi HPC system
                                                      $$$$$$$$$$$$
                                                      $$$       $$$
                                                      $h         8$
                                  _                   $f         k$
                                 (_)                  $h         8$
                                 | |   _              $$$       $$$
                                 | | /'_`\            $$$$$$$$$$$$
                                 | |( (_) )           $$$$$$$$$$$$
                                 (_)`\___/'
                                                      (Jupiter I)

 2x Dell PowerEdge R750XA:
 2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM

 2x Dell PowerEdge R750XA:
 2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM

├── Information ──────────────────────────────────────────────────────────┤
 Support Requests  : hpc@unibocconi.it
 Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC
├── Software ─────────────────────────────────────────────────────────────┤
 Use 'module avail' to see the installed software
 Use 'lquota' to see the available resources in your home
├── Cluster Usage ────────────────────────────────────────────────────────┤
```

```
╭──────────────────────────────────────────────────────────────────────────────╮
│ Running computational data processing tasks on the LOGIN node is forbidden.  │
│ Please always launch computation using SLURM job submission,                 │
│ so that the computation is executed on the COMPUTE nodes.                    │
╰──────────────────────────────────────────────────────────────────────────────╯
Last login: Wed Sep 30 17:20:45 2026 from 10.5.171.188
(base) [3355245@slnode01 ~]$ cd ~/inward
conda activate inward
squeue -u $USER
tail -20 logs/exp1-677584.out
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
Rendering prompts: 100%|██████████| 21800/21800 [00:07<00:00, 2802.02it/s]
Processed prompts: 100%|██████████| 21800/21800 [00:18<00:00, 1172.26it/s, est. speed input: 76106.49 toks/s, output: 7745.97 toks/s]
generation took 283s
{
  "n": 8688,
  "acc_q1": 0.35681399631675875,
  "acc_q2": 0.3819060773480663,
  "acc_comp": 0.05785948434622466,
  "acc_ctx": 0.2224907918968924,
  "n_both_hops_correct": 1358,
  "gap": 0.8144329896907216,
  "gap_with_context": 0.46686303387334316
}

GAP: of 1358 items where both hops are known, 81.4% fail when composed; 46.7% still fail with hop 1 given in context.
INFO 09-30 19:08:40 [utils.py:632] [shutdown] Process manager: send sigterm to process EngineCore
(EngineCore pid=2558987) INFO 09-30 19:08:40 [core.py:1342] [shutdown] EngineCore: trigger received signal=SIGTERM
(EngineCore pid=2558987) INFO 09-30 19:08:40 [core.py:1493] [shutdown] EngineCore: start mode=abort timeout=0s
(EngineCore pid=2558987) INFO 09-30 19:08:40 [core.py:1524] [shutdown] EngineCore: request processing complete; starting resource teardown
(EngineCore pid=2558987) INFO 09-30 19:08:40 [core.py:1355] [shutdown] EngineCore: exiting busy loop
(inward) [3355245@slnode01 inward]$ sequeue -U $USER
-bash: sequeue: command not found
(inward) [3355245@slnode01 inward]$ sequeue -u $USER
-bash: sequeue: command not found
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ grep "GAP" logs/exp1-677584.out
GAP: of 182 items where both hops are known, 54.9% fail when composed; 67.0% still fail with hop 1 given in context.
GAP: of 1358 items where both hops are known, 81.4% fail when composed; 46.7% still fail with hop 1 given in context.
(inward) [3355245@slnode01 inward]$ tree
-bash: tree: command not found
(inward) [3355245@slnode01 inward]$ python - <<'EOF'
import json, random
random.seed(0)
rows = [json.loads(l) for l in open('results/exp1_Qwen3-1.7B/items.jsonl')]
both = [r for r in rows if r['q1'] and r['q2']]
fail = [r for r in both if not r['comp']]
print("=== 两跳都对、组合答错的例子 ===")
for r in random.sample(fail, 10):
    print(r['category'], '|', r['question'], '| 标准:', r['gold'], '| 组合:', repr(r['pred_comp']), '| 给提示:', repr(r['pred_ctx']))
print("\n=== 被判为单跳答错的例子（检查判分是否太严）===")
wrong = [r for r in rows if not r['q2']]
for r in random.sample(wrong, 10):
    print(r['category'], '| 回答:', repr(r['pred_q2']))
s = json.load(open('results/exp1_Qwen3-1.7B/summary.json'))
print("\n=== 分类别 ===")
for c, v in s['by_category'].items():
    print(f"{c:30s} 两跳都对={v['n_both_hops_correct']:4d}  缺口={v.get('gap', float('nan')):.2f}")
EOF
=== 两跳都对、组合答错的例子 ===
birthplace_rus_common_name | What is the Russian name of the birthplace of Saddam Hussein? | 标准: ['Ирак'] | 组合: 'Saddam Hussein\'s birthplace in Russia is known as "Saddam Hussein\'s birthplace" or "Sadd
am\'s birthplace," but the' | 给提示: "Saddam Hussein's birthplace in Russia is not applicable as it is not a Russian name. The answer is not applicable."
birthplace_spa_common_name | What is the Spanish name of the birthplace of Lionel Messi? | 标准: ['Argentina'] | 组合: 'El Poble de la Barca' | 给提示: 'Marruecos'
birthplace_callingcode | What is the calling code of the birthplace of Virat Kohli? | 标准: ['+91'] | 组合: 'India' | 给提示: '91'
birthplace_currency_short | What is the currency abbreviation in the birthplace of Bashar al-Assad? | 标准: ['SYP'] | 组合: 'Jordan' | 给提示: 'Syria'
birthplace_tld | What is the top-level domain of the birthplace of Lee Chong Wei? | 标准: ['.my'] | 组合: 'Malaysia' | 给提示: 'com.au'
birthplace_spa_common_name | What is the Spanish name of the birthplace of Cung Le? | 标准: ['Vietnam'] | 组合: 'Nueva York' | 给提示: 'Vietnam'
birthplace_rus_common_name | What is the Russian name of the birthplace of Maria Sharapova? | 标准: ['Россия'] | 组合: 'Moscow' | 给提示: 'Russia'
birthplace_est_common_name | What is the Estonian name of the birthplace of Thich Quang Duc? | 标准: ['Vietnam'] | 组合: 'Tõstmevõimurii' | 给提示: 'Vietnam'
birthplace_spa_common_name | What is the Spanish name of the birthplace of Pep Guardiola? | 标准: ['España'] | 组合: 'El Paseo de Gracia' | 给提示: 'Spain'
birthplace_rus_common_name | What is the Russian name of the birthplace of Lali Espósito? | 标准: ['Аргентина'] | 组合: 'St. Petersburg' | 给提示: 'Argentina'

=== 被判为单跳答错的例子（检查判分是否太严）===
birthyear_nobelLiterature | 回答: 'William Faulkner'
birthplace_est_common_name | 回答: 'Morocco in Estonian is "Moori"'
birthyear_masterchamp | 回答: 'Jack Nicklaus'
birthplace_currency_short | 回答: 'J$'
birthplace_rounded_lat | 回答: '25° S'
birthplace_currency_short | 回答: 'LTL'
birthplace_ccn3 | 回答: 'PERU'
birthyear_nobelLiterature | 回答: 'William S. Burroughs'
birthplace_jpn_common_name | 回答: 'Hong Kong Island'
birthyear_masterchamp | 回答: 'Jerry Wofino'

=== 分类别 ===
birthdate_uspresident          两跳都对=   1  缺口=1.00
birthplace_callingcode         两跳都对= 174  缺口=0.98
birthplace_capital             两跳都对= 192  缺口=0.61
birthplace_ccn3                两跳都对=   0  缺口=nan
birthplace_currency            两跳都对= 168  缺口=0.57
birthplace_currency_short      两跳都对= 151  缺口=0.98
birthplace_currency_symbol     两跳都对=  45  缺口=0.76
birthplace_est_common_name     两跳都对=  60  缺口=0.97
birthplace_jpn_common_name     两跳都对=  46  缺口=1.00
birthplace_rounded_lat         两跳都对=  48  缺口=1.00
birthplace_rounded_lng         两跳都对=  13  缺口=0.85
birthplace_rus_common_name     两跳都对= 125  缺口=0.99
birthplace_spa_common_name     两跳都对= 163  缺口=0.93
birthplace_tld                 两跳都对= 157  缺口=0.56
birthplace_urd_common_name     两跳都对=  11  缺口=1.00
birthyear_masterchamp          两跳都对=   4  缺口=0.75
birthyear_nobelLiterature      两跳都对=   0  缺口=nan
(inward) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=16.5GB LIMIT=50GB
(inward) [3355245@slnode01 inward]$ cd ~/inward
sbatch run_exp1_v2.sh Qwen3-0.6B Qwen3-1.7B
squeue -u $USER
Submitted batch job 677608
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD       0:00      1 (Resources)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD       0:00      1 (Resources)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD       0:00      1 (Resources)
(inward) [3355245@slnode01 inward]$ squeue -p stud
squeue --start -j 677584
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            677105      stud sae135-L  3343855 PD       0:00      1 (Dependency)
            677582      stud latdefus  3310909 PD       0:00      1 (Priority)
            677583      stud latdefus  3310909 PD       0:00      1 (Dependency)
            677575      stud cotd-tra  3343816  R      55:27      1 gnode01
            677570      stud    sim21  3195720  R    1:08:00      1 gnode01
            677580      stud medsam3-  3160552  R      59:32      1 gnode02
            677499      stud robust-4  3161355  R      41:19      1 gnode01
            677104      stud sae135-L  3343855  R    9:00:11      1 gnode01
            676454      stud  kg_tune  3310986  R   22:54:49      1 gnode02
            677306      stud vllm-qwe  3221337  R    5:14:54      1 gnode02
slurm_load_jobs error: Invalid job id specified
(inward) [3355245@slnode01 inward]$ squeue --start -j 677608
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD 2026-09-30T20:45:15      1 gnode02             (Resources)
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD       0:00      1 (Resources)
```

```
            677105      stud sae135-L  3343855 PD       0:00      1 (Dependency)
            677582      stud latdefus  3310909 PD       0:00      1 (Priority)
            677583      stud latdefus  3310909 PD       0:00      1 (Dependency)
            677575      stud cotd-tra  3343816  R      56:26      1 gnode01
            677570      stud    sim21  3195720  R    1:08:59      1 gnode01
            677580      stud medsam3-  3160552  R    1:00:31      1 gnode02
            677499      stud robust-4  3161355  R      42:18      1 gnode01
            677104      stud sae135-L  3343855  R    9:01:10      1 gnode01
            676454      stud  kg_tune  3310986  R   22:55:48      1 gnode02
            677306      stud vllm-qwe  3221337  R    5:15:53      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue --start -j 677608
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD 2026-09-30T20:45:15      1 gnode02             (Resources)
(inward) [3355245@slnode01 inward]$ Read from remote host slogin.hpc.unibocconi.it: Can't assign requested address
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────────┐
│                 Welcome to  the Bocconi HPC system                          │
│                                                          $$$$$$$$$$$$       │
│                                                          $$$       $$$      │
│                                                          $h         8$      │
│                                  _                       $f         k$      │
│                                 (_)                      $h         8$      │
│                                 | |  _                   $$$       $$$      │
│                                 | | /'_`\                $$$$$$$$$$$$       │
│                                 | |( (_) )               $$$$$$$$$$$$       │
│                                 (_)`\___/'                                  │
│                                                          (Jupiter I)        │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
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│ Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC │
├── Software ─────────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                            │
│ Use 'lquota' to see the available resources in your home                    │
├── Cluster Usage ────────────────────────────────────────────────────────────┤
│ Running computational data processing tasks on the LOGIN node is forbidden. │
│ Please always launch computation using SLURM job submission,                │
│ so that the computation is executed on the COMPUTE nodes.                   │
└─────────────────────────────────────────────────────────────────────────────┘
Last login: Wed Sep 30 19:10:58 2026 from 10.5.153.81
(base) [3355245@slnode01 ~]$ cd ~/inwards
-bash: cd: /home/3355245/inwards: No such file or directory
(base) [3355245@slnode01 ~]$ cd ~/inward
(base) [3355245@slnode01 inward]$ cd ~/inward
squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            677608      stud   exp1v2  3355245 PD       0:00      1 (Resources)
(base) [3355245@slnode01 inward]$ cd ~/inward
ls results/ logs/
history > commands_history.txt
logs/:
download.log  exp1-677584.out  gputest-677541.out  gputest-677576.out

results/:
exp1_Qwen3-0.6B  exp1_Qwen3-1.7B
(base) [3355245@slnode01 inward]$ Connection to slogin.hpc.unibocconi.it closed by remote host.
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────────┐
│                 Welcome to  the Bocconi HPC system                          │
│                                                          $$$$$$$$$$$$       │
│                                                          $$$       $$$      │
│                                                          $h         8$      │
│                                  _                       $f         k$      │
│                                 (_)                      $h         8$      │
│                                 | |  _                   $$$       $$$      │
│                                 | | /'_`\                $$$$$$$$$$$$       │
│                                 | |( (_) )               $$$$$$$$$$$$       │
│                                 (_)`\___/'                                  │
│                                                          (Jupiter I)        │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
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├── Software ─────────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                            │
│ Use 'lquota' to see the available resources in your home                    │
├── Cluster Usage ────────────────────────────────────────────────────────────┤
│ Running computational data processing tasks on the LOGIN node is forbidden. │
│ Please always launch computation using SLURM job submission,                │
│ so that the computation is executed on the COMPUTE nodes.                   │
└─────────────────────────────────────────────────────────────────────────────┘
Last login: Wed Sep 30 20:03:43 2026 from 10.5.221.149
(base) [3355245@slnode01 ~]$ cd ~/inward
conda activate inward
sacct -j 677608 --format=JobID,State,Elapsed
grep -A20 "category" logs/exp1v2-677608.out
grep "GAP" logs/exp1v2-677608.out
JobID             State    Elapsed
------------ ---------- ----------
677608        COMPLETED   00:09:21
677608.batch  COMPLETED   00:09:21
category                       both  gap_orig gap_clear  gap_ctx  stop@1
birthplace_callingcode            7      1.00      1.00     1.00    0.86
birthplace_capital               29      0.45      0.38     0.62    0.17
birthplace_currency              27      0.33      0.37     0.93    0.19
birthplace_currency_short        34      0.68      0.74     0.74    0.15
birthplace_currency_symbol        7      0.43      0.43     0.43    0.14
birthplace_est_common_name        9      0.56      0.67     0.56    0.00
birthplace_rounded_lat            8      0.88      0.62     0.88    0.25
birthplace_rounded_lng            2      1.00      1.00     1.00    1.00
birthplace_spa_common_name       13      0.85      0.46     0.31    0.15
birthplace_tld                   42      0.21      0.19     0.26    0.00

GAP (2 categories with >= 30 usable items): of 76 items where both hops are known, 43.4% fail when composed (original wording: 42.1%); 6.6% stop at the intermediate answer; 47.4% still fail with hop 1 given
in context.
INFO 09-30 20:41:14 [utils.py:632] [shutdown] Process manager: send sigterm to process EngineCore
(EngineCore pid=1607879) INFO 09-30 20:41:14 [core.py:1342] [shutdown] EngineCore: trigger received signal=SIGTERM
(EngineCore pid=1607879) INFO 09-30 20:41:14 [core.py:1493] [shutdown] EngineCore: start mode=abort timeout=0s
(EngineCore pid=1607879) INFO 09-30 20:41:14 [core.py:1524] [shutdown] EngineCore: request processing complete; starting resource teardown
(EngineCore pid=1607879) INFO 09-30 20:41:14 [core.py:1355] [shutdown] EngineCore: exiting busy loop
===== Qwen3-1.7B =====
8688 items, 28347 unique prompts
INFO 09-30 20:42:55 [api_utils.py:286] non-default args: {'max_model_len': 1024, 'gpu_memory_utilization': 0.85, 'disable_log_stats': True, 'model': 'models/Qwen3-1.7B'}
--
category                       both  gap_orig gap_clear  gap_ctx  stop@1
birthdate_uspresident             1      1.00      1.00     0.00    0.00
birthplace_callingcode          181      0.98      0.86     0.90    0.53
birthplace_capital              198      0.60      0.72     0.78    0.15
birthplace_currency             174      0.57      0.57     0.37    0.09
birthplace_currency_short       154      0.98      0.95     0.53    0.62
birthplace_currency_symbol       46      0.76      0.78     0.61    0.11
birthplace_est_common_name       57      0.96      1.00     0.12    0.00
birthplace_jpn_common_name       43      1.00      1.00     1.00    0.28
birthplace_rounded_lat           57      1.00      1.00     1.00    0.60
birthplace_rounded_lng           11      0.91      1.00     1.00    0.45
```

```
birthplace_rus_common_name       126      1.00      0.98     1.00    0.30
birthplace_spa_common_name       172      0.92      0.59     0.25    0.05
birthplace_tld                   158      0.56      0.54     0.49    0.37
birthplace_urd_common_name        11      1.00      0.91     0.73    0.18
birthyear_masterchamp              4      0.75      0.75     0.75    0.00

GAP (11 categories with >= 30 usable items): of 1366 items where both hops are known, 76.6% fail when composed (original wording: 81.2%); 28.7% stop at the intermediate answer; 61.9% still fail with hop 1 gi
ven in context.
INFO 09-30 20:46:02 [utils.py:632] [shutdown] Process manager: send sigterm to process EngineCore
(EngineCore pid=1608100) INFO 09-30 20:46:02 [core.py:1342] [shutdown] EngineCore: trigger received signal=SIGTERM
(EngineCore pid=1608100) INFO 09-30 20:46:02 [core.py:1493] [shutdown] EngineCore: start mode=abort timeout=0s
GAP (2 categories with >= 30 usable items): of 76 items where both hops are known, 43.4% fail when composed (original wording: 42.1%); 6.6% stop at the intermediate answer; 47.4% still fail with hop 1 given
in context.
GAP (11 categories with >= 30 usable items): of 1366 items where both hops are known, 76.6% fail when composed (original wording: 81.2%); 28.7% stop at the intermediate answer; 61.9% still fail with hop 1 gi
ven in context.
(inward) [3355245@slnode01 inward]$ Connection to slogin.hpc.unibocconi.it closed by remote host.
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
Permission denied, please try again.
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────┐
│                 Welcome to  the Bocconi HPC system                      │
│                                                        $$$$$$$$$$$$     │
│                                                        $$$       $$$    │
│                                                        $h         8$    │
│                                  _                     $f         k$    │
│                                 (_)                    $h         8$    │
│                                 | |  _.                $$$       $$$    │
│                                 | | /'_`\              $$$       $$$    │
│                                 | |( (_) )             $$$$$$$$$$$$     │
│                                 (_)`\___/'                              │
│                                                        (Jupiter I)      │
│                                                                         │
│  2x Dell PowerEdge R750XA:                                              │
│  2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM
│                                                                         │
│  2x Dell PowerEdge R750XA:                                              │
│  2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM
│                                                                         │
├── Information ──────────────────────────────────────────────────────────┤
│  Support Requests  : hpc@unibocconi.it                                  │
│  Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC
├── Software ─────────────────────────────────────────────────────────────┤
│  Use 'module avail' to see the installed software                       │
│  Use 'lquota' to see the available resources in your home               │
├── Cluster Usage ────────────────────────────────────────────────────────┤
│  Running computational data processing tasks on the LOGIN node is forbidden.
│  Please always launch computation using SLURM job submission,           │
│  so that the computation is executed on the COMPUTE nodes.              │
└─────────────────────────────────────────────────────────────────────────┘
Last failed login: Thu Oct  1 10:32:30 CEST 2026 from 10.5.165.50 on ssh:notty
There was 1 failed login attempt since the last successful login.
Last login: Thu Oct  1 10:22:35 2026 from 10.5.165.50
(base) [3355245@slnode01 ~]$ cd ~/inward
python - <<'EOF'
import json, random
random.seed(1)
rows = [json.loads(l) for l in open('results/exp1v2_Qwen3-1.7B/items.jsonl')]
both = [r for r in rows if r['q1'] and r['q2']]
print("给提示后复述中间答案的比例:", round(sum(r['ctx_echo'] for r in both)/len(both), 3))
for cat in ['birthplace_callingcode', 'birthplace_jpn_common_name', 'birthplace_rounded_lat']:
    bad = [r for r in both if r['category'] == cat and not r['ctx']]
    print(f"\n=== {cat} ===")
    for r in random.sample(bad, min(4, len(bad))):
        print('标准:', r['gold'], '| 单问第二跳:', repr(r['pred_q2']), '| 清楚版:', repr(r['pred_clear']), '| 给提示:', repr(r['pred_ctx']))
EOF
给提示后复述中间答案的比例: 0.521

=== birthplace_callingcode ===
标准: ['+1'] | 单问第二跳: '1-800-267-3777' | 清楚版: 'Canada' | 给提示: 'Canada'
标准: ['+41'] | 单问第二跳: '+41' | 清楚版: '+49' | 给提示: 'Switzerland'
标准: ['+43'] | 单问第二跳: '43' | 清楚版: 'Switzerland' | 给提示: 'Austria'
标准: ['+20'] | 单问第二跳: '+20' | 清楚版: 'Egyptian Egypt' | 给提示: 'Egypt'

=== birthplace_jpn_common_name ===
标准: ['デンマーク'] | 单问第二跳: 'Denmark in Japanese is called "デンマーク" (Denmaruke).' | 清楚版: 'Denmark' | 给提示: 'Denmark'
标准: ['ポルトガル'] | 单问第二跳: 'Portugal in Japanese is "ポルトガル" (Portugals).' | 清楚版: 'Japan' | 给提示: 'Portugal'
标准: ['ポーランド'] | 单问第二跳: 'Poland in Japanese is called "ポーランド" (Pōrando).' | 清楚版: 'Poland' | 给提示: 'Poland'
标准: ['ポルトガル'] | 单问第二跳: 'Portugal in Japanese is "ポルトガル" (Portugals).' | 清楚版: 'India' | 给提示: 'Portugal'

=== birthplace_rounded_lat ===
标准: [60] | 单问第二跳: 'Russia is located in the northern hemisphere, so its latitude is rounded down to 60 degrees north.' | 清楚版: 'Russia' | 给提示: 'Russia'
标准: [56] | 单问第二跳: '56° N' | 清楚版: 'Šiaurės šaly' | 给提示: 'Lithuania'
标准: [47] | 单问第二跳: "Switzerland's latitude is approximately 47.3 degrees north." | 清楚版: 'Geneva' | 给提示: 'Switzerland'
标准: [52] | 单问第二跳: '52° N' | 清楚版: 'Nuenen' | 给提示: 'Netherlands'
(base) [3355245@slnode01 inward]$ nohup python -c "
from huggingface_hub import snapshot_download
snapshot_download('Qwen/Qwen3-8B', local_dir='models/Qwen3-8B')
print('done 8B', flush=True)
" > logs/download_8b.log 2>&1 &
[1] 570820
(base) [3355245@slnode01 inward]$ du -sh ~/inward/models/Qwen3-8B
tail -2 ~/inward/logs/download_8b.log
du: cannot access '/home/3355245/inward/models/Qwen3-8B': No such file or directory
[1]+  Exit 1                  nohup python -c "
from huggingface_hub import snapshot_download
snapshot_download('Qwen/Qwen3-8B', local_dir='models/Qwen3-8B')
print('done 8B', flush=True)
" > logs/download_8b.log 2>&1
    from huggingface_hub import snapshot_download
ModuleNotFoundError: No module named 'huggingface_hub'
(base) [3355245@slnode01 inward]$ conda activate inward
nohup python -c "
from huggingface_hub import snapshot_download
snapshot_download('Qwen/Qwen3-8B', local_dir='models/Qwen3-8B')
print('done 8B', flush=True)
" > logs/download_8b.log 2>&1 &
[1] 571340
(inward) [3355245@slnode01 inward]$ du -sh ~/inward/models/Qwen3-8B
tail -2 ~/inward/logs/download_8b.log
286M    /home/3355245/inward/models/Qwen3-8B
Fetching 15 files:   0%|          | 0/15 [00:00<?, ?it/s]Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Fetching 15 files:  40%|████      | 6/15 [00:15<00:17,  1.97s/it](inward) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=16.5GB LIMIT=50GB
(inward) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=33GB LIMIT=50GB
[1]+  Done                    nohup python -c "
from huggingface_hub import snapshot_download
snapshot_download('Qwen/Qwen3-8B', local_dir='models/Qwen3-8B')
print('done 8B', flush=True)
" > logs/download_8b.log 2>&1
(inward) [3355245@slnode01 inward]$ du -sh ~/inward/models/Qwen3-8B
tail -2 ~/inward/logs/download_8b.log
16G     /home/3355245/inward/models/Qwen3-8B
Fetching 15 files: 100%|██████████| 15/15 [02:21<00:00,  9.45s/it]
done 8B
(inward) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=33GB LIMIT=50GB
(inward) [3355245@slnode01 inward]$ sbatch run_exp1_v2.sh Qwen3-8B
Submitted batch job 678097
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678097      stud   exp1v2  3355245 PD       0:00      1 (Resources)
(inward) [3355245@slnode01 inward]$ squeue --start -j stud
squeue: error: Invalid job id: stud
(inward) [3355245@slnode01 inward]$ squeue -stud
squeue: error: Invalid job_step id: tud.??
```

```
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678097      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678063      stud routeb-m  3195720 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Dependency)
            677810      stud rsrun-bi  3257706  R    8:20:09      1 gnode01
            678030      stud cotd-tra  3343816  R      31:35      1 gnode01
            677580      stud medsam3-  3160552  R   16:04:37      1 gnode02
            678003      stud robust-4  3161355  R    2:00:51      1 gnode01
            677105      stud sae135-L  3343855  R    3:15:39      1 gnode01
            677648      stud  kg_tune  3310986  R   14:08:54      1 gnode02
            677306      stud vllm-qwe  3221337  R   20:19:59      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678097      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678063      stud routeb-m  3195720 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Dependency)
            677810      stud rsrun-bi  3257706  R    8:22:20      1 gnode01
            678030      stud cotd-tra  3343816  R      33:46      1 gnode01
            677580      stud medsam3-  3160552  R   16:06:48      1 gnode02
            678003      stud robust-4  3161355  R    2:03:02      1 gnode01
            677105      stud sae135-L  3343855  R    3:17:50      1 gnode01
            677648      stud  kg_tune  3310986  R   14:11:05      1 gnode02
            677306      stud vllm-qwe  3221337  R   20:22:10      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -u $USER
scontrol update JobId=任务号 TimeLimit=01:00:00
squeue --start -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678097      stud   exp1v2  3355245 PD       0:00      1 (Resources)
Invalid job id specified for job 任务号
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678097      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
(inward) [3355245@slnode01 inward]$ scontrol update JobId=678097 TimeLimit=01:00:00
squeue --start -u $USER
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678097      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (None)
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678097      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678063      stud routeb-m  3195720 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Dependency)
            677810      stud rsrun-bi  3257706  R    8:24:36      1 gnode01
            678030      stud cotd-tra  3343816  R      36:02      1 gnode01
            677580      stud medsam3-  3160552  R   16:09:04      1 gnode02
            678003      stud robust-4  3161355  R    2:05:18      1 gnode01
            677105      stud sae135-L  3343855  R    3:20:06      1 gnode01
            677648      stud  kg_tune  3310986  R   14:13:21      1 gnode02
            677306      stud vllm-qwe  3221337  R   20:24:26      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678097      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678063      stud routeb-m  3195720 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Dependency)
            677810      stud rsrun-bi  3257706  R    8:28:28      1 gnode01
            678030      stud cotd-tra  3343816  R      39:54      1 gnode01
            677580      stud medsam3-  3160552  R   16:12:56      1 gnode02
            678003      stud robust-4  3161355  R    2:09:10      1 gnode01
            677105      stud sae135-L  3343855  R    3:23:58      1 gnode01
            677648      stud  kg_tune  3310986  R   14:17:13      1 gnode02
            677306      stud vllm-qwe  3221337  R   20:28:18      1 gnode02
(inward) [3355245@slnode01 inward]$ squene -p stud
-bash: squene: command not found
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678063      stud routeb-m  3195720 PD       0:00      1 (Resources)
            678105      stud cotd-tra  3343816 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Dependency)
            678097      stud   exp1v2  3355245  R       4:32      1 gnode01
            677810      stud rsrun-bi  3257706  R    8:35:39      1 gnode01
            677580      stud medsam3-  3160552  R   16:20:07      1 gnode02
            678003      stud robust-4  3161355  R    2:16:21      1 gnode01
            677105      stud sae135-L  3343855  R    3:31:09      1 gnode01
            677648      stud  kg_tune  3310986  R   14:24:24      1 gnode02
            677306      stud vllm-qwe  3221337  R   20:35:29      1 gnode02
(inward) [3355245@slnode01 inward]$ lquta
-bash: lquta: command not found
(inward) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=33GB LIMIT=50GB
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678105      stud cotd-tra  3343816 PD       0:00      1 (Resources)
            678004      stud robust-4  3161355 PD       0:00      1 (Dependency)
            677810      stud rsrun-bi  3257706  R    8:42:03      1 gnode01
            678063      stud routeb-m  3195720  R       3:55      1 gnode01
            677580      stud medsam3-  3160552  R   16:26:31      1 gnode02
            678003      stud robust-4  3161355  R    2:22:45      1 gnode01
            677105      stud sae135-L  3343855  R    3:37:33      1 gnode01
            677648      stud  kg_tune  3310986  R   14:30:48      1 gnode02
            677306      stud vllm-qwe  3221337  R   20:41:53      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ sacct -j 678097 --format=JobID,State,Elapsed
grep -A20 "category" logs/exp1v2-678097.out
JobID             State    Elapsed
------------ ---------- ----------
678097        COMPLETED   00:07:00
678097.batch  COMPLETED   00:07:00
category                        both  gap_orig gap_clear  gap_ctx  stop@1
birthdate_uspresident             83      0.83      0.83     0.86    0.00
birthplace_callingcode           323      0.21      0.25     0.03    0.00
birthplace_capital               325      0.54      0.67     0.07    0.42
birthplace_ccn3                   46      0.98      0.93     0.78    0.20
birthplace_currency              254      0.28      0.32     0.07    0.04
birthplace_currency_short        306      0.22      0.33     0.03    0.01
birthplace_currency_symbol       157      0.27      0.33     0.24    0.01
birthplace_est_common_name       132      0.99      0.86     0.84    0.03
birthplace_jpn_common_name       135      0.99      0.88     0.94    0.49
birthplace_rounded_lat           107      0.77      0.76     0.50    0.00
birthplace_rounded_lng            83      0.78      0.69     0.45    0.00
birthplace_rus_common_name       200      0.98      0.67     0.91    0.38
birthplace_spa_common_name       312      0.97      0.18     0.12    0.08
birthplace_tld                   322      0.31      0.17     0.02    0.02
birthplace_urd_common_name       113      0.99      0.71     0.28    0.08
birthyear_masterchamp             56      0.59      0.59     0.84    0.00
birthyear_nobelLiterature         59      0.98      0.98     0.98    0.00

GAP (17 categories with >= 30 usable items): of 3013 items where both hops are known, 47.4% fail when composed (original wording: 58.1%); 11.5% stop at the intermediate answer; 29.9% still fail with hop 1 gi
ven in context.
INFO 10-01 11:13:02 [utils.py:632] [shutdown] Process manager: send sigterm to process EngineCore
(inward) [3355245@slnode01 inward]$ scp -r 3355245@slogin.hpc.unibocconi.it:~/inward/results ~/Desktop/Deepseek-猜想实验/
ssh: Could not resolve hostname slogin.hpc.unibocconi.it: Name or service not known
scp: Connection closed
(inward) [3355245@slnode01 inward]$ scp -r 3355245@slogin.hpc.unibocconi.it:~/inward/results ~/Desktop/Deepseek-猜想实验/
ssh: Could not resolve hostname slogin.hpc.unibocconi.it: Name or service not known
scp: Connection closed
(inward) [3355245@slnode01 inward]$ cd ~/inward
rm -rf models/Qwen3-0.6B models/Qwen3-1.7B models/Qwen3-8B
lquota
nohup python -c "
from huggingface_hub import snapshot_download
snapshot_download('Qwen/Qwen3-14B', local_dir='models/Qwen3-14B')
print('done 14B', flush=True)
" > logs/download_14b.log 2>&1 &
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=33GB LIMIT=50GB
[1] 577209
(inward) [3355245@slnode01 inward]$ du -sh ~/inward/models/Qwen3-14B
tail -2 ~/inward/logs/download_14b.log
1.7M    /home/3355245/inward/models/Qwen3-14B
Fetching 18 files:   0%|          | 0/18 [00:00<?, ?it/s]Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
```

```
Fetching 18 files:  33%|███       | du -sh ~/inward/models/Qwen3-14Bward) [3355245@slnode01 inward]$ du -sh ~/inward/models/Qwen3-14B
tail -2 ~/inward/logs/download_14b.log
28G     /home/3355245/inward/models/Qwen3-14B
[1]+  Done                    nohup python -c "
from huggingface_hub import snapshot_download
snapshot_download('Qwen/Qwen3-14B', local_dir='models/Qwen3-14B')
print('done 14B', flush=True)
" > logs/download_14b.log 2>&1
Fetching 18 files: 100%|██████████| 18/18 [03:57<00:00, 13.19s/it]
done 14B
(inward) [3355245@slnode01 inward]$ sed -i 's/04:00:00/01:00:00/' run_exp1_v2.sh
sbatch run_exp1_v2.sh Qwen3-14B
squeue -u $USER
Submitted batch job 678144
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678138      stud cotd-tra  3343816 PD       0:00      1 (Dependency)
            678004      stud robust-4  3161355 PD       0:00      1 (Dependency)
            677810      stud rsrun-bi  3257706  R    8:53:11      1 gnode01
            678063      stud routeb-m  3195720  R      15:03      1 gnode01
            678105      stud cotd-tra  3343816  R       4:49      1 gnode01
            677580      stud medsam3-  3160552  R   16:37:39      1 gnode02
            678003      stud robust-4  3161355  R    2:33:53      1 gnode01
            677105      stud sae135-L  3343855  R    3:48:41      1 gnode01
            677648      stud  kg_tune  3310986  R   14:41:56      1 gnode02
            677306      stud vllm-qwe  3221337  R   20:53:01      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678138      stud cotd-tra  3343816 PD       0:00      1 (Dependency)
            678146      stud fall_ver  3258901 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Dependency)
            677810      stud rsrun-bi  3257706  R    8:57:53      1 gnode01
            678063      stud routeb-m  3195720  R      19:45      1 gnode01
            678105      stud cotd-tra  3343816  R       9:31      1 gnode01
            677580      stud medsam3-  3160552  R   16:42:21      1 gnode02
            678003      stud robust-4  3161355  R    2:38:35      1 gnode01
            677105      stud sae135-L  3343855  R    3:53:23      1 gnode01
            677648      stud  kg_tune  3310986  R   14:46:38      1 gnode02
            677306      stud vllm-qwe  3221337  R   20:57:43      1 gnode02
(inward) [3355245@slnode01 inward]$ lquota
USED=40.5GB LIMIT=50GB
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
            678168      stud    sim26  3195720 PD 2026-10-01T15:36:00      1 gnode02             (Priority)
            678146      stud fall_ver  3258901 PD 2026-10-01T17:23:22      1 gnode01             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678004      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678138      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$ squeue --start -j 678144
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678168      stud    sim26  3195720 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            678138      stud cotd-tra  3343816 PD       0:00      1 (Dependency)
            678146      stud fall_ver  3258901 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Priority)
            678174      stud gb-engin  3163450  R       0:26      1 gnode01
            677810      stud rsrun-bi  3257706  R    9:31:17      1 gnode01
            678175      stud routeb-m  3195720  R      11:30      1 gnode01
            678105      stud cotd-tra  3343816  R      42:55      1 gnode01
            677580      stud medsam3-  3160552  R   17:15:45      1 gnode02
            677105      stud sae135-L  3343855  R    4:26:47      1 gnode01
            677648      stud  kg_tune  3310986  R   15:20:02      1 gnode02
            677306      stud vllm-qwe  3221337  R   21:31:07      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
            678146      stud fall_ver  3258901 PD 2026-10-01T15:36:00      1 gnode02             (Priority)
            678004      stud robust-4  3161355 PD 2026-10-01T16:36:00      1 gnode02             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678138      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
            678168      stud    sim26  3195720 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
(inward) [3355245@slnode01 inward]$ squeue - stud
squeue: error: Unrecognized option: -
Usage: squeue [-A account] [--clusters names] [-i seconds] [--job jobid]
              [-n name] [-o format] [--only-job-state] [-p partitions]
              [--qos qos] [--reservation reservation] [--sort fields] [--start]
              [--step step_id] [-t states] [-u user_name] [--usage]
              [-L licenses] [-w nodes] [--federation] [--local] [--sibling]
              [--expand-patterns] [--json=data_parser] [--yaml=data_parser]
              [-ahjlrsv]
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678168      stud    sim26  3195720 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            678138      stud cotd-tra  3343816 PD       0:00      1 (Dependency)
            678146      stud fall_ver  3258901 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Priority)
            678174      stud gb-engin  3163450  R       8:16      1 gnode01
            677810      stud rsrun-bi  3257706  R    9:39:07      1 gnode01
            678175      stud routeb-m  3195720  R      19:20      1 gnode01
            678105      stud cotd-tra  3343816  R      50:45      1 gnode01
            677580      stud medsam3-  3160552  R   17:23:35      1 gnode02
            677105      stud sae135-L  3343855  R    4:34:37      1 gnode01
            677648      stud  kg_tune  3310986  R   15:27:52      1 gnode02
            677306      stud vllm-qwe  3221337  R   21:38:57      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678168      stud    sim26  3195720 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            678138      stud cotd-tra  3343816 PD       0:00      1 (Dependency)
            678146      stud fall_ver  3258901 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Priority)
            678174      stud gb-engin  3163450  R      15:46      1 gnode01
            677810      stud rsrun-bi  3257706  R    9:46:37      1 gnode01
            678175      stud routeb-m  3195720  R      26:50      1 gnode01
            678105      stud cotd-tra  3343816  R      58:15      1 gnode01
            677580      stud medsam3-  3160552  R   17:31:05      1 gnode02
            677105      stud sae135-L  3343855  R    4:42:07      1 gnode01
            677648      stud  kg_tune  3310986  R   15:35:22      1 gnode02
            677306      stud vllm-qwe  3221337  R   21:46:27      1 gnode02
(inward) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=40.5GB LIMIT=50GB
(inward) [3355245@slnode01 inward]$ squeue -p stud
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
            678168      stud    sim26  3195720 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            678138      stud cotd-tra  3343816 PD       0:00      1 (Dependency)
            678146      stud fall_ver  3258901 PD       0:00      1 (Priority)
            678004      stud robust-4  3161355 PD       0:00      1 (Priority)
            678216      stud robust-4  3161355 PD       0:00      1 (Dependency)
            678174      stud gb-engin  3163450  R      28:13      1 gnode01
            677810      stud rsrun-bi  3257706  R    9:59:04      1 gnode01
            678175      stud routeb-m  3195720  R      39:17      1 gnode01
            678105      stud cotd-tra  3343816  R    1:10:42      1 gnode01
            677580      stud medsam3-  3160552  R   17:43:32      1 gnode02
            677105      stud sae135-L  3343855  R    4:54:34      1 gnode01
            677648      stud  kg_tune  3310986  R   15:47:49      1 gnode02
```

```
           677306      stud vllm-qwe  3221337  R   21:58:54      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
            678146      stud fall_ver  3258901 PD 2026-10-01T15:36:00      1 gnode02             (Priority)
            678004      stud robust-4  3161355 PD 2026-10-01T16:36:00      1 gnode02             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678138      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
            678168      stud    sim26  3195720 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$ squeue --start -j 678144
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
(inward) [3355245@slnode01 inward]$ squeue --start -j 678144
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD       0:00      1 (Resources)
(inward) [3355245@slnode01 inward]$ scontrol show job 678144
JobId=678144 JobName=exp1v2
   UserId=3355245(1553) GroupId=grp_hpcclusterstudentsbocconiusers(1009) MCS_label=N/A
   Priority=8470 Nice=0 Account=3355245 QOS=stud
   JobState=PENDING Reason=Resources Dependency=(null)
   Requeue=1 Restarts=0 BatchFlag=1 Reboot=0 ExitCode=0:0
   RunTime=00:00:00 TimeLimit=01:00:00 TimeMin=N/A
   SubmitTime=2026-10-01T11:27:58 EligibleTime=2026-10-01T11:27:58
   AccrueTime=2026-10-01T11:27:58
   StartTime=2026-10-01T14:35:10 EndTime=2026-10-01T15:35:10 Deadline=N/A
   SuspendTime=None SecsPreSuspend=0 LastSchedEval=2026-10-01T13:11:33 Scheduler=Backfill:*
   Partition=stud AllocNode:Sid=slnode01:570479
   ReqNodeList=(null) ExcNodeList=(null)
   NodeList= SchedNodeList=gnode02
   NumNodes=1 NumCPUs=8 NumTasks=1 CPUs/Task=8 ReqB:S:C:T=0:0:*:*
   ReqTRES=cpu=8,mem=64G,node=1,billing=12,gres/gpu=1
   AllocTRES=(null)
   Socks/Node=* NtasksPerN:B:S:C=0:0:*:* CoreSpec=*
   MinCPUsNode=8 MinMemoryNode=64G MinTmpDiskNode=0
   Features=(null) DelayBoot=00:00:00
   OverSubscribe=OK Contiguous=0 Licenses=(null) LicensesAlloc=(null) Network=(null)
   Command=/mnt/beegfsstudents/home/3355245/inward/run_exp1_v2.sh
   WorkDir=/mnt/beegfsstudents/home/3355245/inward
   StdErr=/mnt/beegfsstudents/home/3355245/inward/logs/exp1v2-678144.out
   StdIn=/dev/null
   StdOut=/mnt/beegfsstudents/home/3355245/inward/logs/exp1v2-678144.out
   TresPerNode=gres/gpu:1
   TresPerTask=cpu=8


(inward) [3355245@slnode01 inward]$ sinfo -p stud
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
stud         up 1-00:10:00      1   mix- gnode02
stud         up 1-00:10:00      1  alloc gnode01
(inward) [3355245@slnode01 inward]$ claude
-bash: claude: command not found
(inward) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=40.5GB LIMIT=50GB
(inward) [3355245@slnode01 inward]$ squeue --start
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
            678146      stud fall_ver  3258901 PD 2026-10-01T15:36:00      1 gnode02             (Priority)
            678004      stud robust-4  3161355 PD 2026-10-01T16:36:00      1 gnode02             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678138      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
            678168      stud    sim26  3195720 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$ squeue --start
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
            678168      stud    sim26  3195720 PD 2026-10-01T15:36:00      1 gnode02             (Priority)
            678146      stud fall_ver  3258901 PD 2026-10-01T19:36:00      1 gnode02             (Priority)
            678004      stud robust-4  3161355 PD 2026-10-01T19:55:05      1 gnode01             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678274      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$ squeue --start
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
            678168      stud    sim26  3195720 PD 2026-10-01T15:36:00      1 gnode02             (Priority)
            678146      stud fall_ver  3258901 PD 2026-10-01T19:36:00      1 gnode02             (Priority)
            678004      stud robust-4  3161355 PD 2026-10-01T19:55:05      1 gnode01             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678306      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678144      stud   exp1v2  3355245 PD 2026-10-01T14:35:10      1 gnode02             (Resources)
            678168      stud    sim26  3195720 PD 2026-10-01T15:36:00      1 gnode02             (Priority)
            678146      stud fall_ver  3258901 PD 2026-10-01T19:36:00      1 gnode02             (Priority)
            678004      stud robust-4  3161355 PD 2026-10-01T19:55:05      1 gnode01             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678306      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$    #SBATCH --mail-type=END,FAIL
   #SBATCH --mail-user=weiyu.chen2@studbocconi.it
(inward) [3355245@slnode01 inward]$ sqeue -u $USER
-bash: sqeue: command not found
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678144      stud   exp1v2  3355245  R       5:35      1 gnode02
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ sacct -j 678144 --format=JobID,State,ExitCode,Elapsed,MaxRSS
tail -50 logs/exp1v2-678144.out
JobID             State ExitCode    Elapsed     MaxRSS
------------ ---------- -------- ---------- ----------
678144        COMPLETED      0:0   00:09:20
678144.batch  COMPLETED      0:0   00:09:20  32986464K
(EngineCore pid=1621687) INFO 10-01 14:40:01 [decorators.py:719] saved AOT compiled function to /home/3355245/.cache/vllm/torch_compile_cache/torch_aot_compile/d54980d30cb20e815da8ef4e06abdb42ee591182b40eece
229544f984c9f316e/rank_0_0/model
(EngineCore pid=1621687) INFO 10-01 14:40:01 [monitor.py:53] torch.compile took 57.32 s in total
(EngineCore pid=1621687) INFO 10-01 14:40:03 [monitor.py:81] Initial profiling/warmup run took 2.05 s
Capturing CUDA graphs (PIECEWISE): 100%|██████████| 51/51 [00:05<00:00,  8.64it/s]
Capturing CUDA graphs (FULL): 100%|██████████| 2/2 [00:00<00:00, 12.89it/s]
(EngineCore pid=1621687) INFO 10-01 14:40:13 [model_runner.py:1066] Graph capturing finished in 7 secs, took 0.45 GiB
(EngineCore pid=1621687) INFO 10-01 14:40:13 [gpu_worker.py:640] Available KV cache memory: 3.49 GiB
(EngineCore pid=1621687) INFO 10-01 14:40:13 [gpu_worker.py:655] CUDA graph memory profiling is enabled (default since v0.21.0). The current --gpu-memory-utilization=0.8500 is equivalent to --gpu-memory-util
ization=0.8369 without CUDA graph memory profiling. To maintain the same effective KV cache size as before, increase --gpu-memory-utilization to 0.8631. To disable, set VLLM_MEMORY_PROFILER_ESTIMATE_CUDAGRAP
HS=0.
(EngineCore pid=1621687) INFO 10-01 14:40:13 [kv_cache_utils.py:2395] GPU KV cache size: 22,832 tokens, Maximum concurrency for 1,024 tokens per request: 22.30x
(EngineCore pid=1621687) INFO 10-01 14:40:13 [kernel_warmup.py:172] JIT kernel warmup starting.
(EngineCore pid=1621687) INFO 10-01 14:40:13 [kernel_warmup.py:185] JIT kernel warmup finished in 0.00s.
Capturing CUDA graphs (PIECEWISE): 100%|██████████| 51/51 [00:05<00:00,  8.80it/s]
Capturing CUDA graphs (FULL): 100%|██████████| 35/35 [00:02<00:00, 15.00it/s]
```

```
(EngineCore pid=1621687) INFO 10-01 14:40:36 [model_runner.py:1066] Graph capturing finished in 9 secs, took 0.27 GiB
(EngineCore pid=1621687) INFO 10-01 14:40:36 [gpu_worker.py:825] CUDA graph pool memory: 0.27 GiB (actual), 0.51 GiB (estimated), difference: 0.24 GiB (90.6%).
(EngineCore pid=1621687) INFO 10-01 14:40:36 [gpu_worker.py:888] Free memory on device (38.89/39.25 GiB) on startup. Desired GPU memory utilization is (0.85, 33.36 GiB). Actual usage is 27.77 GiB for consume
d memory (weights + non-torch), 2.11 GiB for peak activation, and 0.27 GiB for CUDAGraph memory. Replace gpu_memory_utilization config with `--kv-cache-memory=3295966516` (3.07 GiB) to fit into requested mem
ory, or `--kv-cache-memory=9229320704` (8.6 GiB) to fully utilize gpu memory. Current kv cache memory in use is 3.49 GiB.
(EngineCore pid=1621687) INFO 10-01 14:40:43 [jit_monitor.py:85] Kernel JIT monitor activated; monitored JIT compilations during inference will use mode=warn.
(EngineCore pid=1621687) INFO 10-01 14:40:44 [torch_utils.py:287] Reducing Torch threads from 8 to 1 for serving. Set OMP_NUM_THREADS in the external environment to override.
(EngineCore pid=1621687) INFO 10-01 14:40:44 [core.py:372] init engine (profile, create kv cache, warmup model) took 118.27 s (compilation: 57.32 s)
(EngineCore pid=1621687) INFO 10-01 14:40:46 [kernel.py:416] Final IR op priority after setting platform defaults: IrOpPriorityConfig(rms_norm=['native'], fused_add_rms_norm=['native'], gelu_and_mul_sparse=[
'triton', 'native'])
INFO 10-01 14:40:48 [hf.py:642] Detected the chat template content format to be 'string'. You can set `--chat-template-content-format` to override this.
Rendering prompts: 100%|██████████| 28347/28347 [00:09<00:00, 2978.39it/s]
Processed prompts: 100%|██████████| 28347/28347 [03:39<00:00, 129.23it/s, est. speed input: 8069.65 toks/s, output: 595.51 toks/s]
generation took 546s

category                        both  gap_orig gap_clear  gap_ctx  stop@1
birthdate_uspresident            310      0.91      0.91     0.10    0.00
birthplace_callingcode           349      0.13      0.13     0.00    0.00
birthplace_capital               355      0.30      0.34     0.01    0.06
birthplace_ccn3                  144      0.81      0.80     0.51    0.00
birthplace_currency              261      0.22      0.32     0.06    0.04
birthplace_currency_short        347      0.20      0.22     0.02    0.00
birthplace_currency_symbol       275      0.31      0.33     0.15    0.00
birthplace_est_common_name       242      0.92      0.59     0.22    0.15
birthplace_jpn_common_name       277      0.99      0.62     0.07    0.45
birthplace_rounded_lat           197      0.71      0.64     0.42    0.00
birthplace_rounded_lng           119      0.82      0.76     0.33    0.00
birthplace_rus_common_name       331      0.89      0.33     0.03    0.13
birthplace_spa_common_name       332      0.87      0.15     0.03    0.02
birthplace_tld                   354      0.22      0.19     0.01    0.00
birthplace_urd_common_name       140      0.96      0.63     0.19    0.13
birthyear_masterchamp            117      0.91      0.91     0.74    0.00
birthyear_nobelLiterature        193      0.96      0.96     0.95    0.01

GAP (17 categories with >= 30 usable items): of 4343 items where both hops are known, 45.0% fail when composed (original wording: 59.5%); 6.1% stop at the intermediate answer; 15.7% still fail with hop 1 giv
en in context.
INFO 10-01 14:44:39 [utils.py:632] [shutdown] Process manager: send sigterm to process EngineCore
(EngineCore pid=1621687) INFO 10-01 14:44:39 [core.py:1342] [shutdown] EngineCore: trigger received signal=SIGTERM
(EngineCore pid=1621687) INFO 10-01 14:44:39 [core.py:1493] [shutdown] EngineCore: start mode=abort timeout=0s
(EngineCore pid=1621687) INFO 10-01 14:44:39 [core.py:1524] [shutdown] EngineCore: request processing complete; starting resource teardown
(EngineCore pid=1621687) INFO 10-01 14:44:39 [core.py:1355] [shutdown] EngineCore: exiting busy loop
(inward) [3355245@slnode01 inward]$ cd ~/inward
ls -la
echo "===== run script ====="
cat run_exp1_v2.sh
echo "===== python files ====="
ls *.py
echo "===== model name ====="
grep -rn -E "model|MODEL" run_exp1_v2.sh *.py | head -20
echo "===== result files ====="
ls -la results/ outputs/ out/ 2>/dev/null
find . -maxdepth 2 \( -name "*.jsonl" -o -name "*.json" -o -name "*.csv" -o -name "*.parquet" \) -newer run_exp1_v2.sh | head -20
total 32
drwxr-xr-x  7 3355245 grp_hpcclusterstudentsbocconiusers   11 Oct  1 11:27 .
drwx------ 12 3355245 grp_hpcclusterstudentsbocconiusers   15 Sep 30 18:39 ..
-rw-r--r--  1 3355245 grp_hpcclusterstudentsbocconiusers 7323 Sep 30 20:06 commands_history.txt
drwxr-xr-x  2 3355245 grp_hpcclusterstudentsbocconiusers    1 Sep 30 18:14 data
-rw-r--r--  1 3355245 grp_hpcclusterstudentsbocconiusers 7545 Sep 30 18:56 exp1_gap.py
-rw-r--r--  1 3355245 grp_hpcclusterstudentsbocconiusers 9842 Sep 30 19:43 exp1_gap_v2.py
-rw-r--r--  1 3355245 grp_hpcclusterstudentsbocconiusers  647 Sep 30 18:45 gpu_test.sh
drwxr-xr-x  2 3355245 grp_hpcclusterstudentsbocconiusers    9 Oct  1 14:35 logs
drwxr-xr-x  3 3355245 grp_hpcclusterstudentsbocconiusers    1 Oct  1 11:20 models
drwxr-xr-x  8 3355245 grp_hpcclusterstudentsbocconiusers    6 Oct  1 14:44 results
-rw-r--r--  1 3355245 grp_hpcclusterstudentsbocconiusers  515 Sep 30 18:56 run_exp1.sh
-rw-r--r--  1 3355245 grp_hpcclusterstudentsbocconiusers  525 Oct  1 11:27 run_exp1_v2.sh
drwxr-xr-x  4 3355245 grp_hpcclusterstudentsbocconiusers    5 Sep 30 18:14 self-ask
===== run script =====
#!/bin/bash
#SBATCH -J exp1v2
#SBATCH -p stud
#SBATCH --qos=stud
#SBATCH --gres=gpu:1
#SBATCH --cpus-per-task=8
#SBATCH --mem=64G
#SBATCH --time=01:00:00
#SBATCH -o logs/%x-%j.out

# Usage: sbatch run_exp1_v2.sh Qwen3-0.6B Qwen3-1.7B
# Runs experiment 1 on each listed model, one after another, inside a single job.

eval "$(conda shell.bash hook)"
conda activate inward
export HF_HUB_OFFLINE=1
cd ~/inward

for m in "$@"; do
    echo "===== $m ====="
    python exp1_gap_v2.py --model models/$m --out results/exp1v2_$m
done
===== python files =====
exp1_gap.py  exp1_gap_v2.py
===== model name =====
run_exp1_v2.sh:12:# Runs experiment 1 on each listed model, one after another, inside a single job.
run_exp1_v2.sh:21:    python exp1_gap_v2.py --model models/$m --out results/exp1v2_$m
exp1_gap.py:4:For every 2-hop question in Compositional Celebrities we ask the model:
exp1_gap.py:10:Gap = share of items where the model gets BOTH hops right on their own
exp1_gap.py:14:  python exp1_gap.py --model models/Qwen3-0.6B --out results/exp1_Qwen3-0.6B
exp1_gap.py:94:    ap.add_argument("--model", default=None)
exp1_gap.py:113:        # fake model: answers each hop correctly, composed questions correctly 60% of the time
exp1_gap.py:125:        tok = AutoTokenizer.from_pretrained(args.model)
exp1_gap.py:126:        llm = LLM(args.model, gpu_memory_utilization=0.85, max_model_len=1024, seed=args.seed)
exp1_gap.py:166:    summary = {"model": args.model or "dry-run", "overall": summarize(rows), "by_category": {}}
exp1_gap_v2.py:4:For every 2-hop question in Compositional Celebrities we ask the model:
exp1_gap_v2.py:10:Gap = share of items where the model gets BOTH hops right on their own
exp1_gap_v2.py:14:  python exp1_gap.py --model models/Qwen3-0.6B --out results/exp1_Qwen3-0.6B
exp1_gap_v2.py:104:    ap.add_argument("--model", default=None)
exp1_gap_v2.py:123:        # fake model: answers each hop correctly, composed questions correctly 60% of the time
exp1_gap_v2.py:136:        tok = AutoTokenizer.from_pretrained(args.model)
exp1_gap_v2.py:137:        llm = LLM(args.model, gpu_memory_utilization=0.85, max_model_len=1024, seed=args.seed)
exp1_gap_v2.py:194:    summary = {"model": args.model or "dry-run", "min_both_per_category": MIN_BOTH,
===== result files =====
results/:
total 4
drwxr-xr-x 8 3355245 grp_hpcclusterstudentsbocconiusers  6 Oct  1 14:44 .
drwxr-xr-x 7 3355245 grp_hpcclusterstudentsbocconiusers 11 Oct  1 11:27 ..
drwxr-xr-x 2 3355245 grp_hpcclusterstudentsbocconiusers  2 Sep 30 19:03 exp1_Qwen3-0.6B
drwxr-xr-x 2 3355245 grp_hpcclusterstudentsbocconiusers  2 Sep 30 19:08 exp1_Qwen3-1.7B
drwxr-xr-x 2 3355245 grp_hpcclusterstudentsbocconiusers  2 Sep 30 20:41 exp1v2_Qwen3-0.6B
drwxr-xr-x 2 3355245 grp_hpcclusterstudentsbocconiusers  2 Oct  1 14:44 exp1v2_Qwen3-14B
drwxr-xr-x 2 3355245 grp_hpcclusterstudentsbocconiusers  2 Sep 30 20:46 exp1v2_Qwen3-1.7B
drwxr-xr-x 2 3355245 grp_hpcclusterstudentsbocconiusers  2 Oct  1 11:13 exp1v2_Qwen3-8B
(inward) [3355245@slnode01 inward]$ cd ~/inward
ls -la results/exp1v2_Qwen3-14B/
echo "===== row sample ====="
for f in results/exp1v2_Qwen3-14B/*; do echo "--- $f"; head -c 1500 "$f"; echo; done
echo "===== scaling ====="
for m in 0.6B 1.7B 8B 14B; do echo "--- Qwen3-$m"; grep -o '"overall"[^}]*}' results/exp1v2_Qwen3-$m/*.json 2>/dev/null | head -c 600; echo; done
echo "===== script ====="
cat exp1_gap_v2.py
echo "===== disk ====="
lquota
du -sh models/* ~/.cache/* 2>/dev/null
total 3490
drwxr-xr-x 2 3355245 grp_hpcclusterstudentsbocconiusers       2 Oct  1 14:44 .
drwxr-xr-x 8 3355245 grp_hpcclusterstudentsbocconiusers       6 Oct  1 14:44 ..
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 3562684 Oct  1 14:44 items.jsonl
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers    8774 Oct  1 14:44 summary.json
===== row sample =====
--- results/exp1v2_Qwen3-14B/items.jsonl
{"category": "birthdate_uspresident", "person": "Maggie Smith", "question": "Who was the President of the United States when Maggie Smith was born?", "gold": ["Franklin D. Roosevelt"], "pred_q1": "July 28, 1
934", "pred_q2": "Franklin D. Roosevelt", "pred_comp": "Dwight D. Eisenhower", "pred_clear": "Dwight D. Eisenhower", "pred_ctx": "Franklin D. Roosevelt", "q1": false, "q2": true, "comp": false, "clear": fals
```

e, "ctx": true, "stop_hop1": false, "ctx_echo": false}
{"category": "birthdate_uspresident", "person": "Jane Goodall", "question": "Who was the President of the United States when Jane Goodall was born?", "gold": ["Franklin D. Roosevelt"], "pred_q1": "April 3, 1934", "pred_q2": "Franklin D. Roosevelt", "pred_comp": "Dwight D. Eisenhower", "pred_clear": "Dwight D. Eisenhower", "pred_ctx": "Franklin D. Roosevelt", "q1": true, "q2": true, "comp": false, "clear": false, "ctx": true, "stop_hop1": false, "ctx_echo": false}
{"category": "birthdate_uspresident", "person": "Charles Manson", "question": "Who was the President of the United States when Charles Manson was born?", "gold": ["Franklin D. Roosevelt"], "pred_q1": "November 12, 1930", "pred_q2": "Franklin D. Roosevelt", "pred_comp": "Lyndon B. Johnson", "pred_clear": "Lyndon B. Johnson", "pred_ctx": "Franklin D. Roosevelt", "q1": false, "q2": true, "comp": false, "clear": false, "ctx": true, "stop_hop1": false, "ctx_echo": false}
{"category": "birthdate_uspresident", "person": "Carl Sagan", "question": "Who was the Presi
--- results/exp1v2_Qwen3-14B/summary.json

```
{
  "model": "models/Qwen3-14B",
  "min_both_per_category": 30,
  "categories_in_headline": [
    "birthdate_uspresident",
    "birthplace_callingcode",
    "birthplace_capital",
    "birthplace_ccn3",
    "birthplace_currency",
    "birthplace_currency_short",
    "birthplace_currency_symbol",
    "birthplace_est_common_name",
    "birthplace_jpn_common_name",
    "birthplace_rounded_lat",
    "birthplace_rounded_lng",
    "birthplace_rus_common_name",
    "birthplace_spa_common_name",
    "birthplace_tld",
    "birthplace_urd_common_name",
    "birthyear_masterchamp",
    "birthyear_nobelLiterature"
  ],
  "headline": {
    "n": 8688,
    "acc_q1": 0.7467771639042358,
    "acc_q2": 0.6726519337016574,
    "acc_comp": 0.2460865561694291,
    "acc_ctx": 0.6151012891344383,
    "n_both_hops_correct": 4343,
    "gap_original": 0.5952106838590836,
    "gap_clear": 0.45014966612940366,
    "gap_context": 0.1574948192493668,
    "stopped_at_hop1": 0.06078747409624683,
    "context_echo": 0.03684089339166475
  },
  "all_items": {
    "n": 8688,
    "acc_q1": 0.7467771639042358,
    "acc_q2": 0.6726519337016574,
    "acc_comp": 0.2460865561694291,
    "acc_ctx": 0.6151012891344383,
    "n_both_hops_correct": 4343,
    "gap_original": 0.5952106838590836,
    "gap_clear": 0.45014966612940366,
    "gap_context": 0.1574948192493668,
    "stopped_at_hop1": 0.06078747409624683,
    "context_echo": 0.03684089339166475
  },
  "by_category": {
    "birthdate_uspresident": {
      "n": 708,
```


===== scaling =====
--- Qwen3-0.6B

--- Qwen3-1.7B

--- Qwen3-8B

--- Qwen3-14B

===== script =====
```
"""
Experiment 1: measure the knowledge–utilization (compositionality) gap.

For every 2-hop question in Compositional Celebrities we ask the model:
  q1   – first hop alone           ("What is the birthplace of Rumi?")
  q2   – second hop alone          ("What is the capital of Afghanistan?")
  comp – the composed question     ("What is the capital of the birthplace of Rumi?")
  ctx  – composed question with the first-hop fact given in context

Gap = share of items where the model gets BOTH hops right on their own
      but still gets the composed question wrong.

Usage:
  python exp1_gap.py --model models/Qwen3-0.6B --out results/exp1_Qwen3-0.6B
  python exp1_gap.py --dry-run          # logic test without a GPU
"""
import argparse
import json
import os
import random
import re
import time
import unicodedata
from collections import defaultdict

SYSTEM = "Answer the question with only the answer itself: a name, a number, or a short phrase. No explanation."


# ---------------------------------------------------------------- scoring
def norm(s):
    s = unicodedata.normalize("NFKC", str(s)).lower()
    s = re.sub(r"<think>.*?</think>", " ", s, flags=re.S)
    s = s.replace("+", " ")
    s = re.sub(r"[^\w\s]", " ", s)          # drop punctuation, keep all scripts
    s = re.sub(r"\b(the|a|an)\b", " ", s)
    s = re.sub(r"\b0+(\d)", r"\1", s)        # 004 -> 4
    return re.sub(r"\s+", " ", s).strip()


def is_correct(pred, aliases):
    p = norm(pred)
    for a in aliases:
        raw = unicodedata.normalize("NFKC", str(a)).strip()
        a = norm(a)
        if not a:                                       # pure symbols, e.g. currency sign "؋" or "$"
            if raw and raw in unicodedata.normalize("NFKC", str(pred)):
                return True
            continue
        if re.fullmatch(r"\d+", a):                     # numbers: whole-token match
            if re.search(rf"(?<!\d){a}(?!\d)", p):
                return True
            continue
        if a in p:                                      # exact phrase inside answer
            return True
        toks = [t for t in a.split() if len(t) > 1]     # "Franklin D. Roosevelt" vs "Franklin Roosevelt"
        if len(toks) >= 2 and all(re.search(rf"\b{re.escape(t)}\b", p) for t in toks):
            return True
    return False


# ---------------------------------------------------------------- prompts
def build_items(data_path, per_category, seed):
    data = json.load(open(data_path))["data"]
    by_cat = defaultdict(list)
    for e in data:
        if any(str(e[k][0]).strip() == "" for k in ("A1", "A2", "Answer")):
            continue                            # a few items in the dataset have an empty gold answer
        by_cat[e["category"]].append(e)
    rng = random.Random(seed)
    items = []
```

```
    for cat in sorted(by_cat):
        pool = by_cat[cat]
        if per_category and len(pool) > per_category:
            pool = rng.sample(pool, per_category)
        items.extend(pool)
    return items


def clear_question(e):
    """'the birthplace of X' usually means a city in English, but the gold answer is about the country.
    Make the composed question say 'country' explicitly, like the single-hop Q1 does."""
    p = e["person"]
    return e["Question"].replace(f"the birthplace of {p}", f"the country where {p} was born")


def ctx_question(e):
    """Composed question with the first-hop fact stated up front."""
    verb = "on" if e["category"] == "birthdate_uspresident" else "in"
    return (f"Background fact: {e['person']} was born {verb} {e['A1'][0]}.\n"
            f"Using this fact, answer the question: {clear_question(e)}")


def to_prompt(tok, question):
    msgs = [{"role": "system", "content": SYSTEM}, {"role": "user", "content": question}]
    if tok is None:  # dry run
        return question
    return tok.apply_chat_template(msgs, tokenize=False, add_generation_prompt=True,
                                   enable_thinking=False)


# ---------------------------------------------------------------------- main
def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--model", default=None)
    ap.add_argument("--data", default="data/compositional_celebrities.json")
    ap.add_argument("--out", default="results/exp1_dry")
    ap.add_argument("--per-category", type=int, default=0, help="0 = use all items")
    ap.add_argument("--seed", type=int, default=0)
    ap.add_argument("--max-tokens", type=int, default=32)
    ap.add_argument("--dry-run", action="store_true")
    args = ap.parse_args()

    items = build_items(args.data, args.per_category, args.seed)
    questions = {}
    for e in items:
        for q in (e["Q1"], e["Q2"], e["Question"], clear_question(e), ctx_question(e)):
            questions.setdefault(q, None)
    uniq = list(questions)
    print(f"{len(items)} items, {len(uniq)} unique prompts", flush=True)

    t0 = time.time()
    if args.dry_run:
        # fake model: answers each hop correctly, composed questions correctly 60% of the time
        rng = random.Random(1)
        gold = {}
        for e in items:
            gold[e["Q1"]] = e["A1"][0]
            gold[e["Q2"]] = e["A2"][0]
            gold[e["Question"]] = e["Answer"][0] if rng.random() < 0.6 else "Unknown"
            gold[clear_question(e)] = e["Answer"][0] if rng.random() < 0.7 else e["A1"][0]
            gold[ctx_question(e)] = e["Answer"][0] if rng.random() < 0.9 else "Unknown"
        outs = [str(gold[q]) for q in uniq]
    else:
        from vllm import LLM, SamplingParams
        from transformers import AutoTokenizer
        tok = AutoTokenizer.from_pretrained(args.model)
        llm = LLM(args.model, gpu_memory_utilization=0.85, max_model_len=1024, seed=args.seed)
        prompts = [to_prompt(tok, q) for q in uniq]
        res = llm.generate(prompts, SamplingParams(temperature=0.0, max_tokens=args.max_tokens))
        outs = [r.outputs[0].text.strip() for r in res]
    answers = dict(zip(uniq, outs))
    print(f"generation took {time.time() - t0:.0f}s", flush=True)

    os.makedirs(args.out, exist_ok=True)
    rows = []
    with open(os.path.join(args.out, "items.jsonl"), "w") as f:
        for e in items:
            r = {
                "category": e["category"], "person": e["person"], "question": e["Question"],
                "gold": e["Answer"],
                "pred_q1": answers[e["Q1"]], "pred_q2": answers[e["Q2"]],
                "pred_comp": answers[e["Question"]], "pred_clear": answers[clear_question(e)],
                "pred_ctx": answers[ctx_question(e)],
            }
            r["q1"] = is_correct(r["pred_q1"], e["A1"])
            r["q2"] = is_correct(r["pred_q2"], e["A2"])
            r["comp"] = is_correct(r["pred_comp"], e["Answer"])        # original dataset wording
            r["clear"] = is_correct(r["pred_clear"], e["Answer"])      # unambiguous wording
            r["ctx"] = is_correct(r["pred_ctx"], e["Answer"])
            # "stopped halfway": gave the intermediate answer (hop 1) instead of the final one
            r["stop_hop1"] = (not r["clear"]) and is_correct(r["pred_clear"], e["A1"])
            r["ctx_echo"] = (not r["ctx"]) and is_correct(r["pred_ctx"], e["A1"])
            rows.append(r)
            f.write(json.dumps(r, ensure_ascii=False) + "\n")

    def summarize(rs):
        n = len(rs)
        both = [r for r in rs if r["q1"] and r["q2"]]
        s = {
            "n": n,
            "acc_q1": sum(r["q1"] for r in rs) / n,
            "acc_q2": sum(r["q2"] for r in rs) / n,
            "acc_comp": sum(r["comp"] for r in rs) / n,
            "acc_ctx": sum(r["ctx"] for r in rs) / n,
            "n_both_hops_correct": len(both),
        }
        if both:
            k = len(both)
            s["gap_original"] = 1 - sum(r["comp"] for r in both) / k   # knows both hops, fails to combine
            s["gap_clear"] = 1 - sum(r["clear"] for r in both) / k     # same, unambiguous wording (headline)
            s["gap_context"] = 1 - sum(r["ctx"] for r in both) / k     # still fails with hop-1 fact given
            s["stopped_at_hop1"] = sum(r["stop_hop1"] for r in both) / k
            s["context_echo"] = sum(r["ctx_echo"] for r in both) / k
        return s

    by_cat = defaultdict(list)
    for r in rows:
        by_cat[r["category"]].append(r)
    cats = {c: summarize(rs) for c, rs in sorted(by_cat.items())}
    # headline: only categories where enough items have both hops right (small cells are too noisy)
    MIN_BOTH = 30
    kept = [c for c, s in cats.items() if s["n_both_hops_correct"] >= MIN_BOTH]
    headline = summarize([r for c in kept for r in by_cat[c]]) if kept else {}
    summary = {"model": args.model or "dry-run", "min_both_per_category": MIN_BOTH,
               "categories_in_headline": kept, "headline": headline,
               "all_items": summarize(rows), "by_category": cats}
    json.dump(summary, open(os.path.join(args.out, "summary.json"), "w"), indent=2, ensure_ascii=False)

    print(f"\n{'category':30s} {'both':>5s} {'gap_orig':>9s} {'gap_clear':>9s} {'gap_ctx':>8s} {'stop@1':>7s}")
    for c, s in cats.items():
        if s["n_both_hops_correct"]:
            print(f"{c:30s} {s['n_both_hops_correct']:5d} {s['gap_original']:9.2f} {s['gap_clear']:9.2f} "
                  f"{s['gap_context']:8.2f} {s['stopped_at_hop1']:7.2f}")
    h = headline
    if h:
        print(f"\nGAP ({len(kept)} categories with >= {MIN_BOTH} usable items): "
              f"of {h['n_both_hops_correct']} items where both hops are known, "
              f"{h['gap_clear']:.1%} fail when composed (original wording: {h['gap_original']:.1%}); "
              f"{h['stopped_at_hop1']:.1%} stop at the intermediate answer; "
              f"{h['gap_context']:.1%} still fail with hop 1 given in context.")
```

```
if __name__ == "__main__":
    main()
===== disk =====
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=40.5GB LIMIT=50GB
28G     models/Qwen3-14B
2.0K    /home/3355245/.cache/conda
1.5K    /home/3355245/.cache/conda-anaconda-tos
5.7M    /home/3355245/.cache/flashinfer
2.0M    /home/3355245/.cache/huggingface
239M    /home/3355245/.cache/vllm
(inward) [3355245@slnode01 inward]$ cd ~/inward
cat > run_exp2.sh << 'EOF'
#!/bin/bash
#SBATCH -J lens14b
#SBATCH -p stud
#SBATCH --qos=stud
#SBATCH --gres=gpu:1
#SBATCH --cpus-per-task=8
#SBATCH --mem=64G
#SBATCH --time=01:30:00
#SBATCH -o logs/%x-%j.out
#SBATCH --mail-type=END,FAIL
#SBATCH --mail-user=weiyu.chen2@studbocconi.it

eval "$(conda shell.bash hook)"
conda activate inward
export HF_HUB_OFFLINE=1
cd ~/inward
python exp2_lens.py --model models/Qwen3-14B \
    --items results/exp1v2_Qwen3-14B/items.jsonl \
    --out results/lens_Qwen3-14B
EOF
sbatch run_exp2.sh
squeue -u $USER
Submitted batch job 678361
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678361      stud  lens14b  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678361      stud  lens14b  3355245 PD 2026-10-01T15:55:05      1 gnode01             (Resources)
            678004      stud robust-4  3161355 PD 2026-10-01T17:26:00      1 gnode01             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678351      stud     isv2  3195720 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
            678357      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$ python -c "import json;[print(m, {k:round(v,3) for k,v in json.load(open(f'results/exp1v2_Qwen3-{m}/summary.json'))['headline'].items() if k.startswith(('gap','acc','n_bot
h'))}) for m in ['0.6B','1.7B','8B','14B']]"
0.6B {'acc_q1': 0.143, 'acc_q2': 0.422, 'acc_comp': 0.179, 'acc_ctx': 0.226, 'n_both_hops_correct': 76, 'gap_original': 0.421, 'gap_clear': 0.434, 'gap_context': 0.474}
1.7B {'acc_q1': 0.451, 'acc_q2': 0.548, 'acc_comp': 0.073, 'acc_ctx': 0.238, 'n_both_hops_correct': 1366, 'gap_original': 0.812, 'gap_clear': 0.766, 'gap_context': 0.619}
8B {'acc_q1': 0.653, 'acc_q2': 0.545, 'acc_comp': 0.194, 'acc_ctx': 0.385, 'n_both_hops_correct': 3013, 'gap_original': 0.581, 'gap_clear': 0.474, 'gap_context': 0.299}
14B {'acc_q1': 0.747, 'acc_q2': 0.673, 'acc_comp': 0.246, 'acc_ctx': 0.615, 'n_both_hops_correct': 4343, 'gap_original': 0.595, 'gap_clear': 0.45, 'gap_context': 0.157}
(inward) [3355245@slnode01 inward]$ python -c "
import json
rows=[json.loads(l) for l in open('results/exp1v2_Qwen3-14B/items.jsonl')]
for cat in ['birthplace_jpn_common_name','birthyear_nobelLiterature','birthyear_masterchamp']:
    bad=[r for r in rows if r['category']==cat and r['q1'] and r['q2'] and not r['clear']][:6]
    print('=====',cat)
    for r in bad: print('  gold:',r['gold'][:2],'| clear:',r['pred_clear'],'| ctx:',r['pred_ctx'])
"
===== birthplace_jpn_common_name
  gold: ['アフガニスタン'] | clear: Afghanistan | ctx: アフガニスタン
  gold: ['アフガニスタン'] | clear: Afghanistan | ctx: アフガニスタン
  gold: ['アルバニア'] | clear: Albania | ctx: アルバニア
  gold: ['アンゴラ'] | clear: Angola | ctx: アンゴラ
  gold: ['アルゼンチン'] | clear: キューバ | ctx: アルゼンチン
  gold: ['アルメニア'] | clear: アゼルバイジャン | ctx: アルメニア
===== birthyear_nobelLiterature
  gold: ["Eugene O'Neill"] | clear: Gabriel García Márquez | ctx: Winston Churchill
  gold: ["Eugene O'Neill"] | clear: John Steinbeck | ctx: No one won the Nobel Prize in Literature in 1936.
  gold: ["Eugene O'Neill"] | clear: Ernest Hemingway | ctx: Winston Churchill
  gold: ["Eugene O'Neill"] | clear: Ernest Hemingway | ctx: No one won the Nobel Prize in Literature in 1936.
  gold: ["Eugene O'Neill"] | clear: Samuel Beckett | ctx: Winston Churchill
  gold: ["Eugene O'Neill"] | clear: John Steinbeck | ctx: Winston Churchill
===== birthyear_masterchamp
  gold: ['Horton Smith'] | clear: Jack Nicklaus | ctx: Sam Snead
  gold: ['Horton Smith'] | clear: Gary Player | ctx: Sam Snead
  gold: ['Horton Smith'] | clear: Jack Nicklaus | ctx: Sam Snead
  gold: ['Horton Smith'] | clear: Sam Snead | ctx: Sam Snead
  gold: ['Horton Smith'] | clear: Sam Snead | ctx: Sam Snead
  gold: ['Horton Smith'] | clear: Gary Player | ctx: Sam Snead
(inward) [3355245@slnode01 inward]$ python -c "
import json
def load(m): return [json.loads(l) for l in open(f'results/exp1v2_Qwen3-{m}/items.jsonl')]
r14=load('14B')
for cat in ['birthyear_nobelLiterature','birthyear_masterchamp']:
    bad=[r for r in r14 if r['category']==cat and r['q1'] and r['q2'] and not r['clear']][:4]
    print('=====',cat)
    for r in bad: print('  q1:',r['pred_q1'],'| q2:',r['pred_q2'],'| gold:',r['gold'][0])
print('===== fair scaling: birthplace categories only, no translation categories')
for m in ['1.7B','8B','14B']:
    rs=[r for r in load(m) if r['category'].startswith('birthplace') and 'common_name' not in r['category'] and r['q1'] and r['q2']]
    k=len(rs); print(m,'n_both',k,'gap_clear',round(1-sum(r['clear'] for r in rs)/k,3),'gap_ctx',round(1-sum(r['ctx'] for r in rs)/k,3))
"
===== birthyear_nobelLiterature
  q1: 1936 | q2: Eugene O'Neill | gold: Eugene O'Neill
  q1: 1936 | q2: Eugene O'Neill | gold: Eugene O'Neill
  q1: 1936 | q2: Eugene O'Neill | gold: Eugene O'Neill
  q1: 1936 | q2: Eugene O'Neill | gold: Eugene O'Neill
===== birthyear_masterchamp
  q1: 1934 | q2: Horton Smith | gold: Horton Smith
  q1: 1934 | q2: Horton Smith | gold: Horton Smith
  q1: 1934 | q2: Horton Smith | gold: Horton Smith
  q1: 1934 | q2: Horton Smith | gold: Horton Smith
===== fair scaling: birthplace categories only, no translation categories
1.7B n_both 979 gap_clear 0.749 gap_ctx 0.651
8B n_both 1923 gap_clear 0.399 gap_ctx 0.122
14B n_both 2401 gap_clear 0.341 gap_ctx 0.11
(inward) [3355245@slnode01 inward]$ scancel 678361
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ grep -c cat_type ~/inward/exp2_lens.py
cd ~/inward && sbatch run_exp2.sh && squeue -u $USER
2
Submitted batch job 678387
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678387      stud  lens14b  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ squeue --start -j 678387
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678387      stud  lens14b  3355245 PD 2026-10-01T18:44:44      1 gnode02             (Resources)
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678387      stud  lens14b  3355245 PD 2026-10-01T18:44:44      1 gnode02             (Resources)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678351      stud     isv2  3195720 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
            678357      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ sacct -j 678387 --format=JobID,State,ExitCode,Elapsed
tail -60 logs/lens14b-678387.out
JobID           State ExitCode    Elapsed
------------ ---------- -------- ----------
678387        COMPLETED      0:0   00:04:49
678387.batch  COMPLETED      0:0   00:04:49
```

```
================================================================================================
CLEAN categories: birth country -> ordinary 2nd hop (MAIN RESULT)

[clean] composed, model FAILS -> bridge rank  (n=300)   bridge ever in top-10: 85%   median best layer: 35/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk   63011  61977  58824  78328  73281  74608   1620     56      3     16
  % top-10       0%     0%     0%     0%     0%     0%     1%    21%    68%    29%

[clean] composed, model SUCCEEDS -> bridge rank  (n=300)   bridge ever in top-10: 92%   median best layer: 34/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk   67948  62542  59654  65538  59006  60632    540     41      1     10
  % top-10       0%     0%     0%     0%     0%     0%     2%    30%    82%    50%

[clean] hop-1 alone (control) -> bridge rank  (n=600)   bridge ever in top-10: 100%   median best layer: 33/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk   65120  62805  57802  72443  61569  71023    396      1      0      0
  % top-10       0%     0%     0%     0%     0%     0%     2%    78%    99%   100%

[clean] composed, model FAILS -> final answer rank  (n=300)   bridge ever in top-10: 94%   median best layer: 39/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk  140470 122160 126134 119002 135262 118300 133217  52740    656      1
  % top-10       0%     0%     0%     0%     0%     0%     0%     3%    23%    90%

[clean] composed, model SUCCEEDS -> final answer rank  (n=300)   bridge ever in top-10: 99%   median best layer: 39/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk  117876 104655 115542 118052 126294 119594  87830   4512      8      0
  % top-10       0%     0%     0%     0%     0%     0%     0%    12%    52%    99%

================================================================================================
TRANSLATION categories: country name in another language (failures often = right country in English; bridge trivially present)

[trans] composed, model FAILS -> bridge rank  (n=100)   bridge ever in top-10: 97%   median best layer: 33/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk   61351  59430  62521  65784  84814  81590  11084      6      0      1
  % top-10       0%     0%     0%     0%     0%     0%     0%    54%    92%    83%

[trans] composed, model SUCCEEDS -> bridge rank  (n=100)   bridge ever in top-10: 100%   median best layer: 32/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk   67530  69804  55169  68806  61416  69228   3077      0      0      1
  % top-10       0%     0%     0%     0%     0%     0%     0%    81%    98%    87%

[trans] hop-1 alone (control) -> bridge rank  (n=200)   bridge ever in top-10: 100%   median best layer: 33/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk   65333  62441  58642  67015  68982  71422    540      1      0      0
  % top-10       0%     0%     0%     0%     0%     0%     1%    78%    98%   100%

[trans] composed, model FAILS -> final answer rank  (n=100)   bridge ever in top-10: 85%   median best layer: 40/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk  107238  79527  84476  76539 105856  86679  59044  30773    744      2
  % top-10       0%     0%     0%     0%     0%     0%     0%     3%    15%    82%

[trans] composed, model SUCCEEDS -> final answer rank  (n=100)   bridge ever in top-10: 100%   median best layer: 39/40
  layer          4      8     12     16     20     24     28     32     36     40
  median rk   88440  65808  82774  89106  97758  86153  32491   5630     53      0
  % top-10       0%     0%     0%     0%     0%     0%     0%    23%    36%   100%

How to read: if the FAIL group's bridge still reaches the top-10 in middle layers (close to the
SUCCESS group), the hop-1 fact is retrieved but not used. If it never surfaces in the FAIL
group, hop 1 is not resolved internally on the composed prompt.
(inward) [3355245@slnode01 inward]$ cd ~ && tar czf inward_results.tgz inward/results inward/logs inward/data inward/*.py inward/*.sh inward/commands_history.txt && ls -lh inward_results.tgz
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 2.6M Oct  1 15:27 inward_results.tgz
(inward) [3355245@slnode01 ~]$ cd ~/inward
sed -e 's/lens14b/patch14b/' -e 's#exp2_lens.py#exp3_backpatch.py#' -e 's#lens_Qwen3-14B#patch_Qwen3-14B#' run_exp2.sh > run_exp3.sh
grep -E "patch|exp3" run_exp3.sh
sbatch run_exp3.sh && squeue -u $USER
#SBATCH -J patch14b
python exp3_backpatch.py --model models/Qwen3-14B \
    --out results/patch_Qwen3-14B
Submitted batch job 678426
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            678426      stud patch14b  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678426      stud patch14b  3355245 PD 2026-10-01T20:36:15      1 gnode02             (Resources)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678357      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
            678424      stud     isv2  3195720 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
(inward) [3355245@slnode01 inward]$ lqutoa
-bash: lqutoa: command not found
(inward) [3355245@slnode01 inward]$ lquota
USED=40.5GB LIMIT=50GB
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678426      stud patch14b  3355245 PD 2026-10-01T20:36:15      1 gnode02             (Resources)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678357      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
            678424      stud     isv2  3195720 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
(inward) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            678429      stud    sim27  3195720 PD 2026-10-01T17:15:45      1 gnode02             (Resources)
            678424      stud     isv2  3195720 PD 2026-10-01T17:31:00      1 gnode02             (Priority)
            678433      stud interact  3310909 PD 2026-10-01T18:31:00      1 gnode02             (Priority)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            678216      stud robust-4  3161355 PD                 N/A      1 (null)              (Dependency)
            678357      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ ls -la exp3_backpatch.py
sacct -j 678426 --format=JobID,State,ExitCode,Elapsed
tail -30 logs/patch14b-678426.out
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 10527 Oct  1 15:36 exp3_backpatch.py
JobID             State ExitCode    Elapsed
------------ ---------- -------- ----------
678426        COMPLETED      0:0   00:09:52
678426.batch  COMPLETED      0:0   00:09:52
300 failed clean items
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 121.58it/s]
source states in 18s
  grid 50/300  (97s)
  grid 100/300  (177s)
  grid 150/300  (256s)
  grid 200/300  (336s)
  grid 250/300  (416s)
  grid 300/300  (496s)

done in 521s
================================================================================
Failed clean items: 300   already correct first token without patch: 96 -> fixable set: 204
                                first-token fix (best of grid)   verified full-answer fix
  REAL back-patch               95  ( 46.6%)              76  ( 37.3%)
  CONTROL (other item's state)   67  ( 32.8%)              49  ( 24.0%)

First-token fix rate per (source s -> target t), REAL / CONTROL, over all items:
  s\t          4          8         12         16         20         24         28         32
   24      0%/0%      36%/37%    36%/36%    36%/37%    35%/38%
   28     12%/17%     37%/35%    37%/34%    36%/33%    35%/31%    38%/23%
   32     41%/20%     37%/10%    37%/10%    37%/10%    37%/10%    35%/10%    32%/9%
   34     40%/12%     38%/9%     38%/8%     35%/8%     37%/7%     36%/7%     34%/6%     33%/6%
   36     40%/8%      40%/7%     38%/7%     38%/6%     38%/6%     37%/5%     37%/5%     38%/4%
   38     39%/4%      37%/4%     37%/4%     37%/4%     35%/4%     37%/3%     38%/3%     35%/3%
```

```
How to read: the headline is the VERIFIED fix rate, REAL vs CONTROL. A real effect means
REAL is clearly above CONTROL. Grid cells show where (which layers) the fix comes from.
(inward) [3355245@slnode01 inward]$ Connection to slogin.hpc.unibocconi.it closed by remote host.
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh slogin@3355245.hpc.unibocconi.it
ssh: Could not resolve hostname 3355245.hpc.unibocconi.it: nodename nor servname provided, or not known
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────────┐
│                 Welcome to  the Bocconi HPC system                          │
│                                                        $$$$$$$$$$$$         │
│                                                        $$$       $$$        │
│                                                        $h         8$        │
│                                  _                     $f         k$        │
│                                 (_)                    $h         8$        │
│                                 | |  _                 $$$       $$$        │
│                                 | | /'_`\              $$$$$$$$$$$$         │
│                                 | |( (_) )             $$$$$$$$$$$$         │
│                                 (_)`\___/'                                  │
│                                                        (Jupiter I)          │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
├── Information ──────────────────────────────────────────────────────────────┤
│ Support Requests  : hpc@unibocconi.it                                       │
│ Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC │
├── Software ─────────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                            │
│ Use 'lquota' to see the available resources in your home                    │
├── Cluster Usage ────────────────────────────────────────────────────────────┤
│ Running computational data processing tasks on the LOGIN node is forbidden. │
│ Please always launch computation using SLURM job submission,               │
│ so that the computation is executed on the COMPUTE nodes.                   │
└─────────────────────────────────────────────────────────────────────────────┘
Last failed login: Thu Oct  1 15:08:01 CEST 2026 from 10.5.165.50 on ssh:notty
There was 1 failed login attempt since the last successful login.
Last login: Thu Oct  1 10:32:35 2026 from 10.5.165.50
(base) [3355245@slnode01 ~]$ cd ~/inward
sed -e 's/lens14b/fixed14b/' -e 's/01:30:00/03:00:00/' -e 's#exp2_lens.py#exp4_fixedpatch.py#' \
    -e 's#lens_Qwen3-14B#fixed_Qwen3-14B --exclude results/patch_Qwen3-14B/patch_items.jsonl#' run_exp2.sh > run_exp4.sh
grep -E "fixed|exp4|03:00" run_exp4.sh
sbatch run_exp4.sh && squeue -u $USER
#SBATCH -J fixed14b
#SBATCH --time=03:00:00
python exp4_fixedpatch.py --model models/Qwen3-14B \
    --out results/fixed_Qwen3-14B --exclude results/patch_Qwen3-14B/patch_items.jsonl
Submitted batch job 679212
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679212      stud fixed14b  3355245 PD       0:00      1 (None)
(base) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)               (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)               (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)               (user env retrieval failed requeued held)
            678854      stud fall2025  3258901 PD                 N/A      1 (null)               (Dependency)
            679131      stud cotd-tra  3343816 PD                 N/A      1 (null)               (Dependency)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679212      stud fixed14b  3355245  R       8:21      1 gnode01
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679212      stud fixed14b  3355245  R      12:21      1 gnode01
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ tail -30 logs/fixed14b-*.out
  450/700  (470s)
  500/700  (515s)
  550/700  (562s)
  600/700  (607s)
  650/700  (656s)
  700/700  (700s)

done in 700s
============================================================================================
Baseline regenerated: 398 wrong, 302 right (population failure share used for net effect: 24.7%)

[36->8]
  fix rate   REAL   56/398 =  14.1%  (95% CI 11.0%-17.8%)   CONTROL   11/398 =   2.8%   REAL vs CONTROL p = 6.1e-09
  break rate REAL   26/302 =   8.6%   CONTROL  257/302 =  85.1%
  net accuracy change on the clean population:  REAL -3.0%   CONTROL -63.4%

[38->4]
  fix rate   REAL   57/398 =  14.3%  (95% CI 11.2%-18.1%)   CONTROL   12/398 =   3.0%   REAL vs CONTROL p = 1.2e-08
  break rate REAL   16/302 =   5.3%   CONTROL  259/302 =  85.8%
  net accuracy change on the clean population:  REAL -0.5%   CONTROL -63.9%

[34->12]
  fix rate   REAL   65/398 =  16.3%  (95% CI 13.0%-20.3%)   CONTROL   12/398 =   3.0%   REAL vs CONTROL p = 1.6e-10
  break rate REAL   22/302 =   7.3%   CONTROL  261/302 =  86.4%
  net accuracy change on the clean population:  REAL -1.5%   CONTROL -64.4%

Any of the 3 fixed pairs fixes: REAL 26.1%   CONTROL 5.3%

How to read: a real effect = REAL fix rate clearly above CONTROL with small p (< 0.01).
A usable fix = positive REAL net accuracy change (fixes outweigh breaks).
(base) [3355245@slnode01 inward]$ cd ~/inward
sed -e 's/lens14b/gated14b/' -e 's#exp2_lens.py#exp5_gated.py#' \
    -e 's#--items results/exp1v2_Qwen3-14B/items.jsonl#--fixed results/fixed_Qwen3-14B#' \
    -e 's#lens_Qwen3-14B#gated_Qwen3-14B#' run_exp2.sh > run_exp5.sh
grep -E "gated|exp5|fixed" run_exp5.sh
sbatch run_exp5.sh && squeue -u $USER
#SBATCH -J gated14b
python exp5_gated.py --model models/Qwen3-14B \
    --fixed results/fixed_Qwen3-14B \
    --out results/gated_Qwen3-14B
Submitted batch job 679230
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679230      stud gated14b  3355245 PD       0:00      1 (None)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679230      stud gated14b  3355245  R       2:24      1 gnode01
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ tail -25 logs/gated14b-*.out
confidence done in 113s; baseline answer differs from exp 4 on 0/700 items (should be ~0)

============================================================================================
Items: 398 baseline-wrong, 302 baseline-right; population failure share 24.7%

How well does LOW confidence flag a wrong answer? (AUROC: 0.5 = useless, 1.0 = perfect)
  p_first    0.652
  mean_logp  0.750
  p_min      0.752

Gated policy, threshold picked on a random half, evaluated on the other half (200 splits).
Ungated = patch everything (experiment 4).  Net = change in overall accuracy on the clean population.
  pair     measure     ungated net  gated net          95% range  fix kept   breaks  % splits >0
  36->8    p_first          -3.02%      -0.34%   -2.89%..+0.63%      3.5%     1.6%          41%
  36->8    mean_logp        -3.02%      +0.38%   -2.32%..+1.42%      4.8%     1.1%          84%
  36->8    p_min            -3.02%      -0.28%   -3.41%..+0.87%      4.2%     1.7%          50%
  38->4    p_first          -0.46%      -0.27%   -2.25%..+0.65%      5.0%     2.0%          46%
  38->4    mean_logp        -0.46%      +0.18%   -1.67%..+1.30%      6.6%     1.9%          64%
  38->4    p_min            -0.46%      -0.18%   -1.88%..+0.98%      6.4%     2.3%          55%
  34->12   p_first          -1.46%      +0.27%   -2.59%..+1.26%      6.1%     1.6%          78%
  34->12   mean_logp        -1.46%      +0.68%   -2.76%..+1.56%      5.9%     1.0%          87%
  34->12   p_min            -1.46%      +0.47%   -1.76%..+1.41%      5.8%     1.3%          76%
```

```
How to read: 'gated net' clearly above 0 with most splits > 0 means confidence-gated back-patching is a
usable improvement. 'fix kept' = share of failures still fixed; 'breaks' = share of correct answers broken.
(base) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=40.5GB LIMIT=50GB
(base) [3355245@slnode01 inward]$ conda activate inward && python -c "import peft; print(peft.__version__)"
0.21.1
(inward) [3355245@slnode01 inward]$ cd ~/inward
for T in self gold; do
sed -e "s/lens14b/inw_$T/" -e 's/01:30:00/03:00:00/' -e 's#exp2_lens.py#exp6_inward.py#' \
    -e "s#lens_Qwen3-14B#inward_${T}_Qwen3-14B --target $T#" run_exp2.sh > run_exp6_$T.sh
sbatch run_exp6_$T.sh
done
grep -h "exp6" run_exp6_*.sh
squeue -u $USER

Submitted batch job 679237
Submitted batch job 679238
python exp6_inward.py --model models/Qwen3-14B \
python exp6_inward.py --model models/Qwen3-14B \
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679238      stud inw_gold  3355245 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            679237      stud inw_self  3355245  R       0:01      1 gnode01
(inward) [3355245@slnode01 inward]$
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679238      stud inw_gold  3355245 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            679237      stud inw_self  3355245  R       1:36      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679238      stud inw_gold  3355245 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            679237      stud inw_self  3355245  R       3:13      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679238      stud inw_gold  3355245 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            679237      stud inw_self  3355245  R       6:09      1 gnode01
(inward) [3355245@slnode01 inward]$ sqeueu -u $USER
-bash: sqeueu: command not found
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ tail -40 logs/inw_self-*.out
tail -40 logs/inw_gold-*.out
train items 864; eval A_newPerson=500, B_newCategory=500, C_bothNew=500
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 123.12it/s]
targets: model's own answer with its own hop-1 answer stated in context (no gold used); 864 examples in 31s; share of targets that happen to be correct: 51.5%
trainable params: 64,225,280 || all params: 14,832,532,480 || trainable%: 0.4330
  epoch 1 step 50/216 loss 0.763 (37s)
  epoch 1 step 100/216 loss 0.404 (75s)
  epoch 2 step 150/216 loss 0.183 (111s)
  epoch 2 step 200/216 loss 0.148 (148s)
training done in 162s

============================================================================================
TARGET = SELF  (model's own answer with its own hop-1 answer stated in context (no gold used))

[A_newPerson]  n=500  (both hops known by base model: 279)
  composed accuracy      base  39.0%  ->  trained  47.4%   (fixed 64, broke 22, McNemar p = 6.5e-06)
  composition gap        base  40.5%  ->  trained  24.4%   (lower is better)
  hop-1 alone accuracy   base  80.7%  ->  trained  81.3%   (should not drop)
  hop-2 alone accuracy   base  64.7%  ->  trained  63.3%   (should not drop)

[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained  52.6%   (fixed 57, broke 17, McNemar p = 3.4e-06)
  composition gap        base  32.4%  ->  trained  17.3%   (lower is better)
  hop-1 alone accuracy   base  79.3%  ->  trained  78.7%   (should not drop)
  hop-2 alone accuracy   base  73.3%  ->  trained  74.7%   (should not drop)

[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained  54.2%   (fixed 37, broke 14, McNemar p = 0.0018)
  composition gap        base  32.3%  ->  trained  25.5%   (lower is better)
  hop-1 alone accuracy   base  84.7%  ->  trained  86.0%   (should not drop)
  hop-2 alone accuracy   base  74.0%  ->  trained  72.7%   (should not drop)

[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
  base    : reaches top-10 in 78% of items, median first layer 33
  trained : reaches top-10 in 80% of items, median first layer 33.0
  per item: earlier after training 16, later 22, same 113 (sign test p = 0.42)

How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
gap down, single-hop accuracy NOT down, and the bridge entity appearing at an EARLIER layer.
train items 864; eval A_newPerson=500, B_newCategory=500, C_bothNew=500
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 126.61it/s]
targets: gold answer from the dataset (external teacher); 864 examples in 0s; share of targets that happen to be correct: 100.0%
trainable params: 64,225,280 || all params: 14,832,532,480 || trainable%: 0.4330
Traceback (most recent call last):
  File "/mnt/beegfsstudents/home/3355245/inward/exp6_inward.py", line 290, in <module>
    main()
  File "/mnt/beegfsstudents/home/3355245/inward/exp6_inward.py", line 169, in main
    ids, att, lab = collate(train_ex[i:i + args.batch])
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/mnt/beegfsstudents/home/3355245/inward/exp6_inward.py", line 150, in collate
    a = tok(t + tok.eos_token, add_special_tokens=False)["input_ids"]
            ~~^~~~~~~~~~~~~~~
TypeError: unsupported operand type(s) for +: 'int' and 'str'
(inward) [3355245@slnode01 inward]$ cd ~/inward
scancel -n lens6b 2>/dev/null
grep -c "hops" exp6_inward.py
mk() { sed -e "s/lens14b/$1/" -e 's/01:30:00/03:00:00/' -e 's#exp2_lens.py#exp6_inward.py#' \
  -e "s#lens_Qwen3-14B#$2 $3#" run_exp2.sh > run_$1.sh; }
mk inw_gold  inward_gold_Qwen3-14B  "--target gold"
mk inw_hops  inward_hops_Qwen3-14B  "--target hops"
mk inw_self1 inward_self_s1_Qwen3-14B "--target self --train-seed 1"
J1=$(sbatch --parsable run_inw_gold.sh)
J2=$(sbatch --parsable --dependency=afterok:$J1 run_inw_hops.sh)
J3=$(sbatch --parsable --dependency=afterok:$J2 run_inw_self1.sh)
sed -e 's/lens14b/lens6b/' -e '/--items/d' \
  -e 's#exp2_lens.py --model models/Qwen3-14B#exp6b_lens.py --model models/Qwen3-14B --eval results/inward_self_Qwen3-14B/eval_C_bothNew.jsonl --adapters self=results/inward_self_Qwen3-14B/adapter gold=resul
ts/inward_gold_Qwen3-14B/adapter hops=results/inward_hops_Qwen3-14B/adapter#' \
  -e 's#--out results/lens_Qwen3-14B#--out results/lens6b_Qwen3-14B#' run_exp2.sh > run_exp6b.sh
sbatch --dependency=afterok:$J3 run_exp6b.sh
squeue -u $USER
4
sbatch: error: QOSMaxSubmitJobPerUserLimit
sbatch: error: Batch job submission failed: Job violates accounting/QOS policy (job submit limit, user's size and/or time limits)
sbatch: error: QOSMaxSubmitJobPerUserLimit
sbatch: error: Batch job submission failed: Job violates accounting/QOS policy (job submit limit, user's size and/or time limits)
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679256      stud inw_hops  3355245 PD       0:00      1 (Dependency)
            679255      stud inw_gold  3355245 PD       0:00      1 (None)
(inward) [3355245@slnode01 inward]$ cd ~/inward
scancel -n inw_hops
cat > run_combo.sh << 'EOF'
#!/bin/bash
#SBATCH -J combo
#SBATCH -p stud
#SBATCH --qos=stud
#SBATCH --gres=gpu:1
#SBATCH --cpus-per-task=8
#SBATCH --mem=64G
#SBATCH --time=05:00:00
#SBATCH -o logs/%x-%j.out
#SBATCH --mail-type=END,FAIL
#SBATCH --mail-user=weiyu.chen2@studbocconi.it

eval "$(conda shell.bash hook)"
conda activate inward
export HF_HUB_OFFLINE=1
cd ~/inward
echo "########## HOPS CONTROL" && \
python exp6_inward.py --model models/Qwen3-14B --items results/exp1v2_Qwen3-14B/items.jsonl \
    --target hops --out results/inward_hops_Qwen3-14B && \
```

```
echo "########## SELF, TRAIN SEED 1" && \
python exp6_inward.py --model models/Qwen3-14B --items results/exp1v2_Qwen3-14B/items.jsonl \
    --target self --train-seed 1 --out results/inward_self_s1_Qwen3-14B && \
echo "########## CORRECTED LENS" && \
python exp6b_lens.py --model models/Qwen3-14B --eval results/inward_self_Qwen3-14B/eval_C_bothNew.jsonl \
    --adapters self=results/inward_self_Qwen3-14B/adapter gold=results/inward_gold_Qwen3-14B/adapter hops=results/inward_hops_Qwen3-14B/adapter \
    --out results/lens6b_Qwen3-14B
EOF
sbatch --dependency=afterok:679255 run_combo.sh
squeue -u $USER
Submitted batch job 679257
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245 PD       0:00      1 (QOSMaxJobsPerUserLimit)
            679255      stud inw_gold  3355245  R       0:38      1 gnode01
(inward) [3355245@slnode01 inward]$ python exp7_audit.py --model models/Qwen3-14B | tee results/audit.txt
================================================================================
(a) bridge/answer first-token overlap in experiment-2 clean items
  birthplace_callingcode           0/77 overlap
  birthplace_capital               2/86 overlap
  birthplace_ccn3                  0/42 overlap
  birthplace_currency             26/66 overlap
  birthplace_currency_short        6/75 overlap
  birthplace_currency_symbol       0/65 overlap
  birthplace_rounded_lat           0/66 overlap
  birthplace_rounded_lng           0/48 overlap
  birthplace_tld                   0/75 overlap

  ALL clean items (as reported before)
    composed, fail     n= 300  bridge ever top-10   85%  median first top-10 layer 33.0  top-10 at layer 36   68%
    composed, success  n= 300  bridge ever top-10   92%  median first top-10 layer 33.0  top-10 at layer 36   82%

  WITHOUT overlapping items (corrected)
    composed, fail     n= 288  bridge ever top-10   84%  median first top-10 layer 33  top-10 at layer 36   68%
    composed, success  n= 278  bridge ever top-10   91%  median first top-10 layer 33.0  top-10 at layer 36   81%

================================================================================
(b) composed prompt vs hop-1-alone prompt (non-overlapping items): first layer where the bridge enters the top-10
  fail     n= 243  median delay +2.0 layers  mean +2.16  later in 79% of items
  success  n= 254  median delay +1.0 layers  mean +1.80  later in 74% of items

================================================================================
(c) scorer spot check, Qwen3-14B. Check by eye that the 'ok' labels are right.

  -- both hops known, composed judged WRONG
  [currency_short] q1='Honduras' | q2='HNL' | composed='USD' | gold=['HNL']
  [tld] q1='Kuwait' | q2='.kw' | composed='.com.qa' | gold=['.kw']
  [currency_short] q1='Turkey' | q2='TRY' | composed='TRL' | gold=['TRY']
  [callingcode] q1='Venezuela' | q2='+58' | composed='+57' | gold=['+58']
  [ccn3] q1='Thailand' | q2='764' | composed='660' | gold=['764']
  [currency_symbol] q1='Ukraine' | q2='₴' | composed='UAH' | gold=['₴']
  [currency_symbol] q1='Pakistan' | q2='Rs.' | composed='PKR' | gold=['₨']
  [currency_short] q1='Romania' | q2='RON' | composed='ROL' | gold=['RON']
  [tld] q1='Sudan' | q2='.sd' | composed='.sc' | gold=['.sd']
  [currency] q1='Honduras' | q2='Honduran Lempira' | composed='Guatemalan quetzal' | gold=['Honduran lempira']
  [currency_symbol] q1='Malaysia' | q2='RM' | composed='$' | gold=['RM']
  [currency_short] q1='Austria' | q2='EUR' | composed='ATS' | gold=['EUR']
  [rounded_lat] q1='Malaysia' | q2='2.5' | composed='1°' | gold=[2]
  [ccn3] q1='Mexico' | q2='484' | composed='52' | gold=['484']
  [currency_symbol] q1='Tina Turner was born in the United States.' | q2='$' | composed='CHF' | gold=['$']

  -- both hops known, composed judged RIGHT
  [callingcode] q1='Zambia' | q2='+260' | composed='+260' | gold=['+260']
  [currency] q1='Australia' | q2='Australian dollar' | composed='Australian dollar' | gold=['Australian dollar']
  [callingcode] q1='Zimbabwe' | q2='+263' | composed='+263' | gold=['+263']
  [tld] q1='Tanzania' | q2='.tz' | composed='.tz' | gold=['.tz']
  [callingcode] q1='Norway' | q2='+47' | composed='+47' | gold=['+47']
  [rounded_lat] q1='Slovakia' | q2='48' | composed='48' | gold=[48]
  [capital] q1='Vietnam' | q2='Hanoi' | composed='Hanoi' | gold=['Hanoi']
  [currency_symbol] q1='Greece' | q2='€' | composed='€' | gold=['€']

================================================================================
(d) fair scaling: same items for every model (clean, both hops known by ALL models)
  common items: 750
  1.7B  gap_clear  73.9%   gap_ctx  63.6%
  8B    gap_clear  27.1%   gap_ctx   4.3%
  14B   gap_clear  15.5%   gap_ctx   0.8%

  same comparison, 8B vs 14B only (more common items):
  common items: 1628
  8B    gap_clear  35.9%   gap_ctx   9.6%
  14B   gap_clear  23.6%   gap_ctx   3.9%
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245 PD       0:00      1 (Dependency)
            679255      stud inw_gold  3355245  R       3:26      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245  R       1:14      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245  R       3:44      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245  R       5:05      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245  R       7:03      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245  R       9:00      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u %USER
squeue: error: Invalid user: %USER

             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245  R      11:32      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            679257      stud    combo  3355245  R      16:36      1 gnode01
(inward) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(inward) [3355245@slnode01 inward]$ cd ~/inward
sacct -j 679255,679257 --format=JobID,JobName,State,ExitCode,Elapsed | grep -v "\."
grep -E "#########|Traceback|Error" logs/combo-679257.out
for f in logs/inw_gold-679255.out logs/combo-679257.out; do grep -E "TARGET =|^\[|composed accuracy|composition gap|hop-1 alone|hop-2 alone|share of targets" $f; done
grep -A 12 "First layer" logs/combo-679257.out
JobID           JobName      State ExitCode    Elapsed
------------ ---------- ---------- -------- ----------
679255         inw_gold  COMPLETED      0:0   00:07:25
679257            combo  COMPLETED      0:0   00:22:45
########### HOPS CONTROL
########### SELF, TRAIN SEED 1
########### CORRECTED LENS
targets: gold answer from the dataset (external teacher); 864 examples in 0s; share of targets that happen to be correct: 100.0%
TARGET = GOLD  (gold answer from the dataset (external teacher))
[A_newPerson]  n=500  (both hops known by base model: 279)
  composed accuracy      base  39.0%  ->  trained  63.0%   (fixed 146, broke 26, McNemar p = 1.8e-21)
  composition gap        base  40.5%  ->  trained  20.8%   (lower is better)
  hop-1 alone accuracy   base  80.7%  ->  trained  79.3%   (should not drop)
  hop-2 alone accuracy   base  64.7%  ->  trained  91.3%   (should not drop)
[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained  62.6%   (fixed 101, broke 11, McNemar p = 2.3e-19)
  composition gap        base  32.4%  ->  trained  15.5%   (lower is better)
  hop-1 alone accuracy   base  79.3%  ->  trained  92.7%   (should not drop)
  hop-2 alone accuracy   base  73.3%  ->  trained  76.0%   (should not drop)
[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained  56.8%   (fixed 63, broke 27, McNemar p = 0.00019)
  composition gap        base  32.3%  ->  trained  22.7%   (lower is better)
  hop-1 alone accuracy   base  84.7%  ->  trained  82.7%   (should not drop)
  hop-2 alone accuracy   base  74.0%  ->  trained  73.3%   (should not drop)
```

```
[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
targets: CONTROL: model's own answers to SINGLE-HOP questions only (no composed questions, no gold); 864 examples in 22s; share of targets that happen to be correct: nan%
TARGET = HOPS  (CONTROL: model's own answers to SINGLE-HOP questions only (no composed questions, no gold))
[A_newPerson]  n=500  (both hops known by base model: 279)
  composed accuracy      base  39.0%  ->  trained  38.6%   (fixed 7, broke 9, McNemar p = 0.8)
  composition gap        base  40.5%  ->  trained  39.8%   (lower is better)
  hop-1 alone accuracy   base  80.7%  ->  trained  82.7%   (should not drop)
  hop-2 alone accuracy   base  64.7%  ->  trained  64.0%   (should not drop)
[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained  43.6%   (fixed 2, broke 7, McNemar p = 0.18)
  composition gap        base  32.4%  ->  trained  33.5%   (lower is better)
  hop-1 alone accuracy   base  79.3%  ->  trained  79.3%   (should not drop)
  hop-2 alone accuracy   base  73.3%  ->  trained  73.3%   (should not drop)
[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained  49.8%   (fixed 3, broke 2, McNemar p = 1)
  composition gap        base  32.3%  ->  trained  32.3%   (lower is better)
  hop-1 alone accuracy   base  84.7%  ->  trained  86.0%   (should not drop)
  hop-2 alone accuracy   base  74.0%  ->  trained  71.3%   (should not drop)
[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
targets: model's own answer with its own hop-1 answer stated in context (no gold used); 864 examples in 30s; share of targets that happen to be correct: 51.5%
TARGET = SELF  (model's own answer with its own hop-1 answer stated in context (no gold used))
[A_newPerson]  n=500  (both hops known by base model: 279)
  composed accuracy      base  39.0%  ->  trained  48.4%   (fixed 65, broke 18, McNemar p = 2.1e-07)
  composition gap        base  40.5%  ->  trained  24.7%   (lower is better)
  hop-1 alone accuracy   base  80.7%  ->  trained  81.3%   (should not drop)
  hop-2 alone accuracy   base  64.7%  ->  trained  65.3%   (should not drop)
[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained  49.2%   (fixed 46, broke 23, McNemar p = 0.0076)
  composition gap        base  32.4%  ->  trained  21.2%   (lower is better)
  hop-1 alone accuracy   base  79.3%  ->  trained  79.3%   (should not drop)
  hop-2 alone accuracy   base  73.3%  ->  trained  74.7%   (should not drop)
[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained  52.2%   (fixed 29, broke 16, McNemar p = 0.072)
  composition gap        base  32.3%  ->  trained  28.0%   (lower is better)
  hop-1 alone accuracy   base  84.7%  ->  trained  86.7%   (should not drop)
  hop-2 alone accuracy   base  74.0%  ->  trained  72.0%   (should not drop)
[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
First layer (of 40) at which the bridge enters the top-10, composed question, n=426
  reference: hop-1 asked directly (base model): hit 97%, median 32
  base   hit   77%  median first layer 33
  self   hit   78%  median first layer 33   vs base: earlier 45, later 70, sign test p = 0.025
         among the 30 items this adapter FIXED: bridge earlier in 5
  gold   hit   75%  median first layer 33   vs base: earlier 105, later 71, sign test p = 0.013
  hops   hit   77%  median first layer 33.0   vs base: earlier 13, later 3, sign test p = 0.021

How to read: a real 'learned to hop earlier' effect = trained median layer clearly below base,
many more 'earlier' than 'later' items with small p. If the layer does not move but accuracy rose,
the adapter improved composition by some other route, and the paper must not claim 'earlier hop'.
(inward) [3355245@slnode01 inward]$ cd ~ && tar czf inward_results_v2.tgz inward/results inward/logs inward/*.py inward/*.sh && ls -lh inward_results_v2.tgz
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 909M Oct  2 10:19 inward_results_v2.tgz
(inward) [3355245@slnode01 ~]$ Connection to slogin.hpc.unibocconi.it closed by remote host.
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────┐
│                 Welcome to  the Bocconi HPC system                      │
│                                                      $$$$$$$$$$$$       │
│                                                      $$$       $$$      │
│                                                      $h         8$      │
│                                  _                   $f         k$      │
│                                 (_)                  $h         8$      │
│                                 | |   _              $$$       $$$      │
│                                 | | /'_`\            $$$$$$$$$$$$       │
│                                 | |( (_) )           $$$$$$$$$$$$       │
│                                 (_)`\___/'                              │
│                                                      (Jupiter I)        │
│                                                                         │
│  2x Dell PowerEdge R750XA:                                              │
│  2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM
│                                                                         │
│  2x Dell PowerEdge R750XA:                                              │
│  2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM
│                                                                         │
├── Information ──────────────────────────────────────────────────────────┤
│ Support Requests  : hpc@unibocconi.it                                   │
│ Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC
├── Software ─────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                        │
│ Use 'lquota' to see the available resources in your home                │
├── Cluster Usage ────────────────────────────────────────────────────────┤
│ Running computational data processing tasks on the LOGIN node is forbidden.
│ Please always launch computation using SLURM job submission,           │
│ so that the computation is executed on the COMPUTE nodes.              │
└─────────────────────────────────────────────────────────────────────────┘
Last login: Fri Oct  2 08:38:09 2026 from 10.5.161.120
(base) [3355245@slnode01 ~]$ ls
BocconiHPC  inward  inward_results.tgz  inward_results_v2.tgz
(base) [3355245@slnode01 ~]$ cd inwards
-bash: cd: inwards: No such file or directory
(base) [3355245@slnode01 ~]$ cd~ inwards
-bash: cd~: command not found
(base) [3355245@slnode01 ~]$ tree
-bash: tree: command not found
(base) [3355245@slnode01 ~]$ cd
(base) [3355245@slnode01 ~]$ tree -a
-bash: tree: command not found
(base) [3355245@slnode01 ~]$ cd ~/inward
tar -xzf exp8_upload.tgz
squeue -u "$USER"
sinfo -p stud -o "%P %l"
tar (child): exp8_upload.tgz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
PARTITION TIMELIMIT
stud 1-00:10:00
(base) [3355245@slnode01 inward]$ cd ~/inward
tar -xzf exp8_upload.tgz
squeue -u "$USER"
sinfo -p stud -o "%P %l"
tar: Ignoring unknown extended header keyword 'LIBARCHIVE.xattr.com.apple.provenance'
tar: Ignoring unknown extended header keyword 'LIBARCHIVE.xattr.com.apple.provenance'
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
PARTITION TIMELIMIT
stud 1-00:10:00
(base) [3355245@slnode01 inward]$ cd ~/inward
ls -lh results/handoff_analysis_20261003/exp8_filtered.py
sbatch --time=24:00:00 results/handoff_analysis_20261003/run_exp8_filtered.sh
squeue -u "$USER"
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 21K Oct  3 14:37 results/handoff_analysis_20261003/exp8_filtered.py
Submitted batch job 680959
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            680959      stud inward-f  3355245 PD       0:00      1 (None)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            680959      stud inward-f  3355245  R       6:03      1 gnode01
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            680959      stud inward-f  3355245  R      15:19      1 gnode01
(base) [3355245@slnode01 inward]$ squeue -u $USEr
squeue: option requires an argument -- 'u'
Try "squeue --help" for more information
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            680959      stud inward-f  3355245  R      27:18      1 gnode01
(base) [3355245@slnode01 inward]$ cd ~/inward
ls -dt results/filtered_*
```

```
ls -lt logs/inward-filtered-*.out
results/filtered_680959
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 74574 Oct  3 16:39 logs/inward-filtered-680959.out
(base) [3355245@slnode01 inward]$ cd ~/inward
sacct -j 680959 --format=JobID,State,ExitCode,Elapsed
tar --exclude='adapter' -czf results/exp8_680959_review.tgz \
  results/filtered_680959 \
  logs/inward-filtered-680959.out
JobID             State ExitCode    Elapsed
------------ ---------- -------- ----------
680959        COMPLETED      0:0   01:52:45
680959.batch  COMPLETED      0:0   01:52:45
(base) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            680321      stud fall2025  3258901 PD                 N/A      1 (null)              (Dependency)
            681038      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(base) [3355245@slnode01 inward]$ cd ~/inward
tar -xzf exp9_upload.tgz
squeue -u "$USER"
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ cd ~/inward
sbatch results/handoff_analysis_20261003/exp9_shuffled/run_exp9_shuffled.sh
squeue -u "$USER"
Submitted batch job 681194
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245 PD       0:00      1 (None)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R       3:26      1 gnode02
(base) [3355245@slnode01 inward]$ squeue -u %USER
squeue: error: Invalid user: %USER

             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ cd ~/inward
ls -dt results/handoff_analysis_20261003/exp9_shuffled/runs_*
ls -lt results/handoff_analysis_20261003/exp9_shuffled/exp9-*.out
results/handoff_analysis_20261003/exp9_shuffled/runs_681194
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 17241 Oct  3 17:40 results/handoff_analysis_20261003/exp9_shuffled/exp9-681194.out
(base) [3355245@slnode01 inward]$
(base) [3355245@slnode01 inward]$ cd ~/inward
sacct -j 681194 --format=JobID,State,ExitCode,Elapsed
tail -n 60 results/handoff_analysis_20261003/exp9_shuffled/exp9-681194.out

tar --exclude='adapter' -czf results/exp9_681194_review.tgz \
  results/handoff_analysis_20261003/exp9_shuffled/runs_681194 \
  results/handoff_analysis_20261003/exp9_shuffled/exp9-681194.out
JobID             State ExitCode    Elapsed
------------ ---------- -------- ----------
681194          RUNNING      0:0   00:21:15
681194.batch    RUNNING      0:0   00:21:15
  composed accuracy      base  39.0%  ->  trained   1.4%   (fixed 6, broke 194, McNemar p = 1.1e-49)
  composition gap        base  40.5%  ->  trained  98.2%   (lower is better)
  hop-1 alone accuracy   base  80.7%  ->  trained  59.3%   (should not drop)
  hop-2 alone accuracy   base  64.7%  ->  trained  64.0%   (should not drop)

[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained   6.4%   (fixed 8, broke 199, McNemar p = 7.4e-49)
  composition gap        base  32.4%  ->  trained  92.1%   (lower is better)
  hop-1 alone accuracy   base  79.3%  ->  trained  54.7%   (should not drop)
  hop-2 alone accuracy   base  73.3%  ->  trained  70.7%   (should not drop)

[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained   6.6%   (fixed 9, broke 224, McNemar p = 7.2e-55)
  composition gap        base  32.3%  ->  trained  90.7%   (lower is better)
  hop-1 alone accuracy   base  84.7%  ->  trained  60.0%   (should not drop)
  hop-2 alone accuracy   base  74.0%  ->  trained  76.7%   (should not drop)

[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
  base    : reaches top-10 in 0.74 of items, median first layer 33.0
  trained : reaches top-10 in 0.02 of items, median first layer 37.0
  per item: earlier after training 0, later 3, same 1 (sign test p = 0.25)

How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
gap down, single-hop accuracy NOT down, mechanism requires separate direct evidence; earlier bridge activation is not required.
train items 864; eval A_newPerson=500, B_newCategory=500, C_bothNew=500
Loading weights: 100%|██████████| 443/443 [00:04<00:00, 101.59it/s]
Allocation diagnostic BEFORE training: {'condition': 'category_confidence50', 'target_cache_sha256': '8e92c717363c58f9aa001b6e7361abc90e9681f57f1bcbdfc1f60b4e3c9fc50d', 'n_pool': 864, 'n_kept': 432, 'target_
changed_n': 0, 'target_changed_share': 0.0, 'assigned_target_correct_n': 226, 'assigned_target_correct_share': 0.5231481481481481, 'original_pool_target_correct_share': 0.5150462962962963, 'original_target_a
uroc': 0.72155748035707907, 'uses_gold_for_allocation': False, 'category_counts': {'birthplace_callingcode': {'available': 216, 'kept': 108, 'target_changed_n': 0, 'assigned_target_correct_n': 84, 'original_t
arget_correct_n': 84}, 'birthplace_capital': {'available': 216, 'kept': 108, 'target_changed_n': 0, 'assigned_target_correct_n': 79, 'original_target_correct_n': 79}, 'birthplace_rounded_lat': {'available':
216, 'kept': 108, 'target_changed_n': 0, 'assigned_target_correct_n': 39, 'original_target_correct_n': 39}, 'birthplace_rounded_lng': {'available': 216, 'kept': 108, 'target_changed_n': 0, 'assigned_target_c
orrect_n': 24, 'original_target_correct_n': 24}}}
trainable params: 64,225,280 || all params: 14,832,532,480 || trainable%: 0.4330
  epoch 1 step 50/108 loss 0.543 (39s)
  epoch 2 step 100/108 loss 0.137 (76s)
training done in 82s

================================================================================================
TARGET = SELF  (category_confidence50: frozen self outputs; gold-free allocation)

[A_newPerson]  n=500  (both hops known by base model: 279)
  composed accuracy      base  39.0%  ->  trained  48.0%   (fixed 66, broke 21, McNemar p = 1.4e-06)
  composition gap        base  40.5%  ->  trained  25.1%   (lower is better)
  hop-1 alone accuracy   base  80.7%  ->  trained  78.7%   (should not drop)
  hop-2 alone accuracy   base  64.7%  ->  trained  64.7%   (should not drop)

[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained  46.6%   (fixed 29, broke 19, McNemar p = 0.19)
  composition gap        base  32.4%  ->  trained  26.6%   (lower is better)
  hop-1 alone accuracy   base  79.3%  ->  trained  80.0%   (should not drop)
  hop-2 alone accuracy   base  73.3%  ->  trained  75.3%   (should not drop)

[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained  51.4%   (fixed 26, broke 17, McNemar p = 0.22)
  composition gap        base  32.3%  ->  trained  28.6%   (lower is better)
  hop-1 alone accuracy   base  84.7%  ->  trained  84.0%   (should not drop)
  hop-2 alone accuracy   base  74.0%  ->  trained  73.3%   (should not drop)

[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
  base    : reaches top-10 in 0.74 of items, median first layer 33.0
  trained : reaches top-10 in 0.74 of items, median first layer 33.0
  per item: earlier after training 9, later 30, same 104 (sign test p = 0.0011)

How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
gap down, single-hop accuracy NOT down, mechanism requires separate direct evidence; earlier bridge activation is not required.
(base) [3355245@slnode01 inward]$ scp 3355245@slogin.hpc.unibocconi.it:~/inward/results/exp9_681194_review.tgz \
  "/Users/danielweiyuchen/Desk
> squeue -j 681194
sacct -j 681194 --format=JobID,State,ExitCode,Elapsed
> ^C
(base) [3355245@slnode01 inward]$ squeue -j 681194
sacct -j 681194 --format=JobID,State,ExitCode,Elapsed
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      22:35      1 gnode02
JobID             State ExitCode    Elapsed
------------ ---------- -------- ----------
681194          RUNNING      0:0   00:22:35
681194.batch    RUNNING      0:0   00:22:35
(base) [3355245@slnode01 inward]$ squeue -u %USER
squeue: error: Invalid user: %USER

             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ sqeue -u %USER
-bash: sqeue: command not found
(base) [3355245@slnode01 inward]$ squeue -u %USER
squeue: error: Invalid user: %USER
```

```
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      24:03      1 gnode02
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      26:11      1 gnode02
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      29:47      1 gnode02
(base) [3355245@slnode01 inward]$ squene -u %USER
-bash: squene: command not found
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      33:28      1 gnode02
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      40:12      1 gnode02
(base) [3355245@slnode01 inward]$ sqeue -u $USER
-bash: sqeue: command not found
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      43:36      1 gnode02
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      47:19      1 gnode02
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      51:54      1 gnode02
(base) [3355245@slnode01 inward]$ squeue -u $SUER
squeue: option requires an argument -- 'u'
Try "squeue --help" for more information
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R      56:21      1 gnode02
(base) [3355245@slnode01 inward]$ tail -n 35 ~/inward/results/handoff_analysis_20261003/exp9_shuffled/exp9-681194.out
================================================================================================
TARGET = SELF  (category_random50: frozen self outputs; gold-free allocation)

[A_newPerson]  n=500  (both hops known by base model: 279)
  composed accuracy      base  39.0%  ->  trained  45.8%   (fixed 61, broke 27, McNemar p = 0.00037)
  composition gap        base  40.5%  ->  trained  27.2%   (lower is better)
  hop-1 alone accuracy   base  80.7%  ->  trained  80.0%   (should not drop)
  hop-2 alone accuracy   base  64.7%  ->  trained  63.3%   (should not drop)

[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained  46.0%   (fixed 28, broke 21, McNemar p = 0.39)
  composition gap        base  32.4%  ->  trained  28.1%   (lower is better)
  hop-1 alone accuracy   base  79.3%  ->  trained  78.7%   (should not drop)
  hop-2 alone accuracy   base  73.3%  ->  trained  75.3%   (should not drop)

[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained  51.6%   (fixed 24, broke 14, McNemar p = 0.14)
  composition gap        base  32.3%  ->  trained  27.6%   (lower is better)
  hop-1 alone accuracy   base  84.7%  ->  trained  85.3%   (should not drop)
  hop-2 alone accuracy   base  74.0%  ->  trained  73.3%   (should not drop)

[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
  base    : reaches top-10 in 0.74 of items, median first layer 33.0
  trained : reaches top-10 in 0.72 of items, median first layer 33.0
  per item: earlier after training 6, later 43, same 92 (sign test p = 5.7e-08)

How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
gap down, single-hop accuracy NOT down, mechanism requires separate direct evidence; earlier bridge activation is not required.
train items 864; eval A_newPerson=500, B_newCategory=500, C_bothNew=500
Loading weights: 100%|██████████| 443/443 [00:04<00:00, 105.40it/s]
Allocation diagnostic BEFORE training: {'condition': 'self_full', 'target_cache_sha256': '8e92c717363c58f9aa001b6e7361abc90e9681f57f1bcbdfc1f60b4e3c9fc50d', 'n_pool': 864, 'n_kept': 864, 'target_changed_n':
0, 'target_changed_share': 0.0, 'assigned_target_correct_n': 445, 'assigned_target_correct_share': 0.5150462962962963, 'original_pool_target_correct_share': 0.5150462962962963, 'original_target_auroc': 0.721
5574803571907, 'uses_gold_for_allocation': False, 'category_counts': {'birthplace_callingcode': {'available': 216, 'kept': 216, 'target_changed_n': 0, 'assigned_target_correct_n': 165, 'original_target_corre
ct_n': 165}, 'birthplace_capital': {'available': 216, 'kept': 216, 'target_changed_n': 0, 'assigned_target_correct_n': 156, 'original_target_correct_n': 156}, 'birthplace_rounded_lat': {'available': 216, 'ke
pt': 216, 'target_changed_n': 0, 'assigned_target_correct_n': 73, 'original_target_correct_n': 73}, 'birthplace_rounded_lng': {'available': 216, 'kept': 216, 'target_changed_n': 0, 'assigned_target_correct_n
': 51, 'original_target_correct_n': 51}}}
trainable params: 64,225,280 || all params: 14,832,532,480 || trainable%: 0.4330
  epoch 1 step 50/216 loss 0.716 (40s)
  epoch 1 step 100/216 loss 0.474 (77s)
  epoch 2 step 150/216 loss 0.210 (114s)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681194      stud inward-e  3355245  R    1:00:09      1 gnode02
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ cd ~/inward
sacct -j 681194 --format=JobID,State,ExitCode,Elapsed
JobID             State ExitCode    Elapsed
------------ ---------- -------- ----------
681194        COMPLETED      0:0   01:22:03
681194.batch  COMPLETED      0:0   01:22:03
(base) [3355245@slnode01 inward]$ tar --exclude='adapter' -czf results/exp9_681194_review.tgz \
  results/handoff_analysis_20261003/exp9_shuffled/runs_681194 \
  results/handoff_analysis_20261003/exp9_shuffled/exp9-681194.out
(base) [3355245@slnode01 inward]$ cd ~/inward
sbatch results/handoff_analysis_20261003/exp10_consistency/run_exp10_consistency.sh
squeue -u 3355245
sbatch: error: Unable to open file results/handoff_analysis_20261003/exp10_consistency/run_exp10_consistency.sh
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ squeue --start -j
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            655043 short_cpu touch_sc   biggio PD 2026-10-06T10:52:00      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            681038      stud cotd-tra  3343816 PD                 N/A      1 (null)              (Dependency)
(base) [3355245@slnode01 inward]$ cd ~/inward
tar --keep-old-files -xzf exp10_upload.tgz &&
sbatch results/handoff_analysis_20261003/exp10_consistency/run_exp10_consistency.sh
squeue -u 3355245
Submitted batch job 681332
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681332      stud inward-e  3355245 PD       0:00      1 (None)
(base) [3355245@slnode01 inward]$ squeue -j 681332
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681332      stud inward-e  3355245  R       0:24      1 gnode01
(base) [3355245@slnode01 inward]$ sqeue -u $USER
-bash: sqeue: command not found
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681332      stud inward-e  3355245  R       7:52      1 gnode01
(base) [3355245@slnode01 inward]$ squene -u $USER
-bash: squene: command not found
(base) [3355245@slnode01 inward]$ sqeue -u $USER
-bash: sqeue: command not found
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681332      stud inward-e  3355245  R      16:57      1 gnode01
(base) [3355245@slnode01 inward]$ cd ~/inward
sacct -j 681332 --format=JobID,State,ExitCode,Elapsed
JobID             State ExitCode    Elapsed
------------ ---------- -------- ----------
681332           FAILED    120:0   00:18:54
681332.batch     FAILED    120:0   00:18:54
(base) [3355245@slnode01 inward]$ tail -n 100 ~/inward/results/handoff_analysis_20261003/exp10_consistency/exp10-681332.out
train items 864; eval A_newPerson=500, B_newCategory=500, C_bothNew=500
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 110.94it/s]
Allocation diagnostic BEFORE training: {'condition': 'self_wrong', 'target_cache_sha256': '8e92c717363c58f9aa001b6e7361abc90e9681f57f1bcbdfc1f60b4e3c9fc50d', 'n_pool': 864, 'n_kept': 228, 'target_changed_n':
 0, 'target_changed_share': 0.0, 'assigned_target_correct_n': 0, 'assigned_target_correct_share': 0.0, 'original_pool_target_correct_share': 0.5069444444444444, 'original_target_auroc': 0.7326033828542028, '
uses_gold_for_allocation': True, 'allocation_scorer': 'scoring_v2_floor', 'subset_seed': 7, 'category_counts': {'birthplace_callingcode': {'available': 216, 'original_correct': 165, 'original_wrong': 51, 'ke
pt': 51, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 0, 'original_target_correct_n': 0}, 'birthplace_capital': {'available': 216, 'original_correct': 156, 'original_wrong': 60, 'kep
t': 60, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 0, 'original_target_correct_n': 0}, 'birthplace_rounded_lat': {'available': 216, 'original_correct': 71, 'original_wrong': 145, '
kept': 71, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 0, 'original_target_correct_n': 0}, 'birthplace_rounded_lng': {'available': 216, 'original_correct': 46, 'original_wrong': 170
```

```
, 'kept': 46, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 0, 'original_target_correct_n': 0}}}
trainable params: 64,225,280 || all params: 14,832,532,480 || trainable%: 0.4330
  epoch 2 step 50/58 loss 0.500 (39s)
training done in 46s

================================================================================
TARGET = SELF  (self_wrong: frozen self outputs; gold-stratified mechanism diagnostic)

[A_newPerson]  n=500  (both hops known by base model: 274)
  composed accuracy      base  38.4%  ->  trained  43.6%   (fixed 47, broke 21, McNemar p = 0.0022)
  composition gap        base  39.8%  ->  trained  28.5%   (lower is better)
  hop-1 alone accuracy   base  80.6%  ->  trained  81.4%   (should not drop)
  hop-2 alone accuracy   base  67.4%  ->  trained  65.0%   (should not drop)

[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained  45.6%   (fixed 28, broke 23, McNemar p = 0.58)
  composition gap        base  32.4%  ->  trained  29.1%   (lower is better)
  hop-1 alone accuracy   base  74.8%  ->  trained  74.8%   (should not drop)
  hop-2 alone accuracy   base  74.2%  ->  trained  76.8%   (should not drop)

[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained  52.0%   (fixed 31, broke 19, McNemar p = 0.12)
  composition gap        base  32.3%  ->  trained  27.3%   (lower is better)
  hop-1 alone accuracy   base  83.8%  ->  trained  85.2%   (should not drop)
  hop-2 alone accuracy   base  78.6%  ->  trained  78.0%   (should not drop)

[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
  base    : reaches top-10 in 0.74 of items, median first layer 33.0
  trained : reaches top-10 in 0.705 of items, median first layer 33
  per item: earlier after training 7, later 45, same 86 (sign test p = 7e-08)

How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
gap down, single-hop accuracy NOT down, mechanism requires separate direct evidence; earlier bridge activation is not required.
train items 864; eval A_newPerson=500, B_newCategory=500, C_bothNew=500
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 115.67it/s]
Allocation diagnostic BEFORE training: {'condition': 'shuffled_wrong', 'target_cache_sha256': '8e92c717363c58f9aa001b6e7361abc90e9681f57f1bcbdfc1f60b4e3c9fc50d', 'n_pool': 864, 'n_kept': 228, 'target_changed
_n': 228, 'target_changed_share': 1.0, 'assigned_target_correct_n': 0, 'assigned_target_correct_share': 0.0, 'original_pool_target_correct_share': 0.5069444444444444, 'original_target_auroc': 0.7326033828542
028, 'uses_gold_for_allocation': True, 'allocation_scorer': 'scoring_v2_floor', 'subset_seed': 7, 'category_counts': {'birthplace_callingcode': {'available': 216, 'original_correct': 165, 'original_wrong': 5
1, 'kept': 51, 'subset_seed': 7, 'target_changed_n': 51, 'assigned_target_correct_n': 0, 'original_target_correct_n': 0}, 'birthplace_capital': {'available': 216, 'original_correct': 156, 'original_wrong': 6
0, 'kept': 60, 'subset_seed': 7, 'target_changed_n': 60, 'assigned_target_correct_n': 0, 'original_target_correct_n': 0}, 'birthplace_rounded_lat': {'available': 216, 'original_correct': 71, 'original_wrong'
: 145, 'kept': 71, 'subset_seed': 7, 'target_changed_n': 71, 'assigned_target_correct_n': 0, 'original_target_correct_n': 0}, 'birthplace_rounded_lng': {'available': 216, 'original_correct': 46, 'original_wr
ong': 170, 'kept': 46, 'subset_seed': 7, 'target_changed_n': 46, 'assigned_target_correct_n': 0, 'original_target_correct_n': 0}}}
trainable params: 64,225,280 || all params: 14,832,532,480 || trainable%: 0.4330
  epoch 2 step 50/58 loss 1.475 (39s)
training done in 45s

================================================================================
TARGET = SELF  (shuffled_wrong: frozen self outputs; gold-stratified mechanism diagnostic)

[A_newPerson]  n=500  (both hops known by base model: 274)
  composed accuracy      base  38.4%  ->  trained   2.6%   (fixed 6, broke 185, McNemar p = 4.1e-47)
  composition gap        base  39.8%  ->  trained  97.4%   (lower is better)
  hop-1 alone accuracy   base  80.6%  ->  trained  54.8%   (should not drop)
  hop-2 alone accuracy   base  67.4%  ->  trained  63.8%   (should not drop)

[B_newCategory]  n=500  (both hops known by base model: 278)
  composed accuracy      base  44.6%  ->  trained   8.6%   (fixed 12, broke 192, McNemar p = 6.5e-43)
  composition gap        base  32.4%  ->  trained  90.3%   (lower is better)
  hop-1 alone accuracy   base  74.8%  ->  trained  50.2%   (should not drop)
  hop-2 alone accuracy   base  74.2%  ->  trained  70.6%   (should not drop)

[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained   9.0%   (fixed 10, broke 213, McNemar p = 1.1e-50)
  composition gap        base  32.3%  ->  trained  88.5%   (lower is better)
  hop-1 alone accuracy   base  83.8%  ->  trained  55.6%   (should not drop)
  hop-2 alone accuracy   base  78.6%  ->  trained  72.0%   (should not drop)

[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
  base    : reaches top-10 in 0.74 of items, median first layer 33.0
  trained : reaches top-10 in 0.015 of items, median first layer 38
  per item: earlier after training 0, later 3, same 0 (sign test p = 0.25)

How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
gap down, single-hop accuracy NOT down, mechanism requires separate direct evidence; earlier bridge activation is not required.
train items 864; eval A_newPerson=500, B_newCategory=500, C_bothNew=500
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 115.37it/s]
Allocation diagnostic BEFORE training: {'condition': 'self_correct', 'target_cache_sha256': '8e92c717363c58f9aa001b6e7361abc90e9681f57f1bcbdfc1f60b4e3c9fc50d', 'n_pool': 864, 'n_kept': 228, 'target_changed_n
': 0, 'target_changed_share': 0.0, 'assigned_target_correct_n': 228, 'assigned_target_correct_share': 1.0, 'original_pool_target_correct_share': 0.5069444444444444, 'original_target_auroc': 0.732603382854202
8, 'uses_gold_for_allocation': True, 'allocation_scorer': 'scoring_v2_floor', 'subset_seed': 7, 'category_counts': {'birthplace_callingcode': {'available': 216, 'original_correct': 165, 'original_wrong': 51,
 'kept': 51, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 51, 'original_target_correct_n': 51}, 'birthplace_capital': {'available': 216, 'original_correct': 156, 'original_wrong': 60
, 'kept': 60, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 60, 'original_target_correct_n': 60}, 'birthplace_rounded_lat': {'available': 216, 'original_correct': 71, 'original_wrong'
: 145, 'kept': 71, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 71, 'original_target_correct_n': 71}, 'birthplace_rounded_lng': {'available': 216, 'original_correct': 46, 'original_w
rong': 170, 'kept': 46, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 46, 'original_target_correct_n': 46}}}
trainable params: 64,225,280 || all params: 14,832,532,480 || trainable%: 0.4330
  epoch 2 step 50/58 loss 0.259 (40s)
training done in 46s
(base) [3355245@slnode01 inward]$ quota -s
df -h ~/inward
tail -n 20 ~/inward/results/handoff_analysis_20261003/exp10_consistency/exp10-681332.out
ls -lh ~/inward/results/handoff_analysis_20261003/exp10_consistency/runs_681332/self_correct_s0/
Filesystem      Size  Used Avail Use% Mounted on
beegfs_nodev    112T   12T  101T  11% /mnt/beegfsstudents

[C_bothNew]  n=500  (both hops known by base model: 322)
  composed accuracy      base  49.6%  ->  trained   9.0%   (fixed 10, broke 213, McNemar p = 1.1e-50)
  composition gap        base  32.3%  ->  trained  88.5%   (lower is better)
  hop-1 alone accuracy   base  83.8%  ->  trained  55.6%   (should not drop)
  hop-2 alone accuracy   base  78.6%  ->  trained  72.0%   (should not drop)

[logit lens, C_bothNew, n=200] bridge entity enters top-10 at layer (of 40):
  base    : reaches top-10 in 0.74 of items, median first layer 33.0
  trained : reaches top-10 in 0.015 of items, median first layer 38
  per item: earlier after training 0, later 3, same 0 (sign test p = 0.25)

How to read: C_bothNew is the main result. Success = composed accuracy up with small p,
gap down, single-hop accuracy NOT down, mechanism requires separate direct evidence; earlier bridge activation is not required.
train items 864; eval A_newPerson=500, B_newCategory=500, C_bothNew=500
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 115.37it/s]
Allocation diagnostic BEFORE training: {'condition': 'self_correct', 'target_cache_sha256': '8e92c717363c58f9aa001b6e7361abc90e9681f57f1bcbdfc1f60b4e3c9fc50d', 'n_pool': 864, 'n_kept': 228, 'target_changed_n
': 0, 'target_changed_share': 0.0, 'assigned_target_correct_n': 228, 'assigned_target_correct_share': 1.0, 'original_pool_target_correct_share': 0.5069444444444444, 'original_target_auroc': 0.732603382854202
8, 'uses_gold_for_allocation': True, 'allocation_scorer': 'scoring_v2_floor', 'subset_seed': 7, 'category_counts': {'birthplace_callingcode': {'available': 216, 'original_correct': 165, 'original_wrong': 51,
 'kept': 51, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 51, 'original_target_correct_n': 51}, 'birthplace_capital': {'available': 216, 'original_correct': 156, 'original_wrong': 60
, 'kept': 60, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 60, 'original_target_correct_n': 60}, 'birthplace_rounded_lat': {'available': 216, 'original_correct': 71, 'original_wrong'
: 145, 'kept': 71, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 71, 'original_target_correct_n': 71}, 'birthplace_rounded_lng': {'available': 216, 'original_correct': 46, 'original_w
rong': 170, 'kept': 46, 'subset_seed': 7, 'target_changed_n': 0, 'assigned_target_correct_n': 46, 'original_target_correct_n': 46}}}
trainable params: 64,225,280 || all params: 14,832,532,480 || trainable%: 0.4330
  epoch 2 step 50/58 loss 0.259 (40s)
training done in 46s
total 342K
drwxr-xr-x 2 3355245 grp_hpcclusterstudentsbocconiusers    3 Oct  3 19:31 adapter
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers  97K Oct  3 19:31 evaluation_manifest.json
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers  114 Oct  3 19:31 initialization.json
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers  632 Oct  3 19:31 run_config.json
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 216K Oct  3 19:31 target_assignment.jsonl
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 1.6K Oct  3 19:31 target_diagnostic.json
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers  25K Oct  3 19:31 train_examples.jsonl
(base) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=50.5GB LIMIT=50GB
(base) [3355245@slnode01 inward]$ du -h --max-depth=1 ~/inward/results | sort -h
du -h --max-depth=1 ~/.cache | sort -h
ls -lh ~/inward/*.tgz ~/inward/results/*.tgz
1.5K    /home/3355245/inward/results/lens6b_Qwen3-14B
70K     /home/3355245/inward/results/patch_Qwen3-14B
287K    /home/3355245/inward/results/fixed_Qwen3-14B
361K    /home/3355245/inward/results/gated_Qwen3-14B
780K    /home/3355245/inward/results/lens_Qwen3-14B
3.0M    /home/3355245/inward/results/exp1_Qwen3-1.7B
3.5M    /home/3355245/inward/results/exp1v2_Qwen3-14B
3.5M    /home/3355245/inward/results/exp1v2_Qwen3-8B
3.6M    /home/3355245/inward/results/exp1_Qwen3-0.6B
```

```
3.8M    /home/3355245/inward/results/exp1v2_Qwen3-1.7B
4.8M    /home/3355245/inward/results/exp1v2_Qwen3-0.6B
246M    /home/3355245/inward/results/inward_gold_Qwen3-14B
246M    /home/3355245/inward/results/inward_hops_Qwen3-14B
246M    /home/3355245/inward/results/inward_self_Qwen3-14B
246M    /home/3355245/inward/results/inward_self_s1_Qwen3-14B
3.6G    /home/3355245/inward/results/filtered_680959
3.7G    /home/3355245/inward/results/handoff_analysis_20261003
8.2G    /home/3355245/inward/results
1.5K    /home/3355245/.cache/conda-anaconda-tos
2.0K    /home/3355245/.cache/conda
2.0M    /home/3355245/.cache/huggingface
5.7M    /home/3355245/.cache/flashinfer
239M    /home/3355245/.cache/vllm
247M    /home/3355245/.cache
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers  73K Oct  3 19:12 /home/3355245/inward/exp10_upload.tgz
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 8.0K Oct  3 14:45 /home/3355245/inward/exp8_upload.tgz
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers  61K Oct  3 17:21 /home/3355245/inward/exp9_upload.tgz
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 914K Oct  3 17:10 /home/3355245/inward/results/exp8_680959_review.tgz
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 1.1M Oct  3 18:45 /home/3355245/inward/results/exp9_681194_review.tgz
(base) [3355245@slnode01 inward]$ cd ~/inward

# 确认要处理的目录及大小
readlink -f results/filtered_680959
du -sh results/filtered_680959

# 查看目录结构
find results/filtered_680959 -maxdepth 2 -type d | sort

# 应有 15 个运行目录、15 个适配器权重文件
find results/filtered_680959 -mindepth 1 -maxdepth 1 -type d | wc -l
find results/filtered_680959 -name adapter_model.safetensors -type f | wc -l

# 确认 exp9/exp10 的依赖文件分别保存在自己的目录
ls -lh results/handoff_analysis_20261003/exp9_shuffled/frozen_targets.jsonl \
       results/handoff_analysis_20261003/exp9_shuffled/reference_exp8/
ls -lh results/handoff_analysis_20261003/exp10_consistency/frozen_targets.jsonl \
       results/handoff_analysis_20261003/exp10_consistency/reference_exp8/
/mnt/beegfsstudents/home/3355245/inward/results/filtered_680959
3.6G    results/filtered_680959
results/filtered_680959
results/filtered_680959/self_confidence_top0.5_s0
results/filtered_680959/self_confidence_top0.5_s0/adapter
results/filtered_680959/self_confidence_top0.5_s1
results/filtered_680959/self_confidence_top0.5_s1/adapter
results/filtered_680959/self_confidence_top0.5_s2
results/filtered_680959/self_confidence_top0.5_s2/adapter
results/filtered_680959/self_confidence_top0.7_s0
results/filtered_680959/self_confidence_top0.7_s0/adapter
results/filtered_680959/self_confidence_top0.7_s1
results/filtered_680959/self_confidence_top0.7_s1/adapter
results/filtered_680959/self_confidence_top0.7_s2
results/filtered_680959/self_confidence_top0.7_s2/adapter
results/filtered_680959/self_confidence_top1.0_s0
results/filtered_680959/self_confidence_top1.0_s0/adapter
results/filtered_680959/self_confidence_top1.0_s1
results/filtered_680959/self_confidence_top1.0_s1/adapter
results/filtered_680959/self_confidence_top1.0_s2
results/filtered_680959/self_confidence_top1.0_s2/adapter
results/filtered_680959/self_random_top0.5_s0
results/filtered_680959/self_random_top0.5_s0/adapter
results/filtered_680959/self_random_top0.5_s1
results/filtered_680959/self_random_top0.5_s1/adapter
results/filtered_680959/self_random_top0.5_s2
results/filtered_680959/self_random_top0.5_s2/adapter
results/filtered_680959/self_random_top0.7_s0
results/filtered_680959/self_random_top0.7_s0/adapter
results/filtered_680959/self_random_top0.7_s1
results/filtered_680959/self_random_top0.7_s1/adapter
results/filtered_680959/self_random_top0.7_s2
results/filtered_680959/self_random_top0.7_s2/adapter
15
15
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 104K Oct  3 17:17 results/handoff_analysis_20261003/exp9_shuffled/frozen_targets.jsonl

results/handoff_analysis_20261003/exp9_shuffled/reference_exp8/:
total 243K
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 80K Oct  3 14:54 eval_A_newPerson.jsonl
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 82K Oct  3 14:54 eval_B_newCategory.jsonl
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 81K Oct  3 14:54 eval_C_bothNew.jsonl
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 104K Oct  3 17:17 results/handoff_analysis_20261003/exp10_consistency/frozen_targets.jsonl

results/handoff_analysis_20261003/exp10_consistency/reference_exp8/:
total 243K
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 80K Oct  3 14:54 eval_A_newPerson.jsonl
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 82K Oct  3 14:54 eval_B_newCategory.jsonl
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 81K Oct  3 14:54 eval_C_bothNew.jsonl
(base) [3355245@slnode01 inward]$ cd ~/inward
python - <<'PY'
from pathlib import Path

root = Path("results")
print(str(root.resolve()) + "/")

def tree(path, prefix="", depth=0):
    if depth >= 3:
        return
    dirs = sorted(p for p in path.iterdir() if p.is_dir())
    for i, p in enumerate(dirs):
        last = i == len(dirs) - 1
        print(prefix + ("└── " if last else "├── ") + p.name + "/")
        tree(p, prefix + ("    " if last else "│   "), depth + 1)

tree(root)
PY
/mnt/beegfsstudents/home/3355245/inward/results/
├── exp1_Qwen3-0.6B/
├── exp1_Qwen3-1.7B/
├── exp1v2_Qwen3-0.6B/
├── exp1v2_Qwen3-1.7B/
├── exp1v2_Qwen3-14B/
├── exp1v2_Qwen3-8B/
├── filtered_680959/
│   ├── self_confidence_top0.5_s0/
│   │   └── adapter/
│   ├── self_confidence_top0.5_s1/
│   │   └── adapter/
│   ├── self_confidence_top0.5_s2/
│   │   └── adapter/
│   ├── self_confidence_top0.7_s0/
│   │   └── adapter/
│   ├── self_confidence_top0.7_s1/
│   │   └── adapter/
│   ├── self_confidence_top0.7_s2/
│   │   └── adapter/
│   ├── self_confidence_top1.0_s0/
│   │   └── adapter/
│   ├── self_confidence_top1.0_s1/
│   │   └── adapter/
│   ├── self_confidence_top1.0_s2/
│   │   └── adapter/
│   ├── self_random_top0.5_s0/
│   │   └── adapter/
│   ├── self_random_top0.5_s1/
│   │   └── adapter/
│   ├── self_random_top0.5_s2/
│   │   └── adapter/
│   ├── self_random_top0.7_s0/
│   │   └── adapter/
│   ├── self_random_top0.7_s1/
│   │   └── adapter/
```

```
│   └── self_random_top0.7_s2/
│       └── adapter/
├── fixed_Qwen3-14B/
├── gated_Qwen3-14B/
├── handoff_analysis_20261003/
│   ├── exp10_consistency/
│   │   ├── __pycache__/
│   │   ├── reference_exp8/
│   │   └── runs_681332/
│   └── exp9_shuffled/
│       ├── __pycache__/
│       ├── reference_exp8/
│       └── runs_681194/
├── inward_gold_Qwen3-14B/
│   └── adapter/
├── inward_hops_Qwen3-14B/
│   └── adapter/
├── inward_self_Qwen3-14B/
│   └── adapter/
├── inward_self_s1_Qwen3-14B/
│   └── adapter/
├── lens6b_Qwen3-14B/
├── lens_Qwen3-14B/
└── patch_Qwen3-14B/
(base) [3355245@slnode01 inward]$ rm -r -- /home/3355245/inward/results/filtered_680959
lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=50.5GB LIMIT=50GB
(base) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=46.6GB LIMIT=50GB
(base) [3355245@slnode01 inward]$ ls -lh ~/inward/results/handoff_analysis_20261003/exp10_consistency/runs_681332/self_correct_s0/adapter/
total 246M
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 1.2K Oct  3 19:31 adapter_config.json
-rw------- 1 3355245 grp_hpcclusterstudentsbocconiusers 246M Oct  3 19:31 adapter_model.safetensors
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 5.1K Oct  3 19:31 README.md
(base) [3355245@slnode01 inward]$ cd ~/inward
tar --keep-old-files -xzf exp10_resume_681332_upload.tgz &&
sbatch results/handoff_analysis_20261003/exp10_consistency/run_exp10_resume_681332.sh
squeue -u 3355245
Submitted batch job 681386
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681386      stud exp10-re  3355245  R       0:00      1 gnode01
(base) [3355245@slnode01 inward]$ sqeueu -u $USER
-bash: sqeueu: command not found
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681386      stud exp10-re  3355245  R       2:53      1 gnode01
(base) [3355245@slnode01 inward]$ # 1) 看有哪些 partition / queue、节点、CPU、内存、GPU
sinfo

# 更详细
sinfo -o "%P %a %l %D %C %m %G"

# 2) 看你账户/用户关联到哪些 account / partition / QOS
sacctmgr show assoc where user=$USER format=User,Account,Partition,QOS,GrpTRES,MaxTRES,MaxJobs,MaxSubmit

# 3) 看有哪些 QoS，以及每个 QoS 的资源上限
sacctmgr show qos format=Name,MaxWall,MaxTRESPerUser,MaxTRESPerJob,GrpTRES,MaxJobsPerUser

# 4) 查看具体 partition 限制
scontrol show partition

# 5) 看 GPU 节点
sinfo -N -o "%N %P %t %c %m %G" | grep -i gpu

# 6) 看你当前 quota / 磁盘占用
quota -s
du -sh ~
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
stud         up 1-00:10:00      2    mix gnode[01-02]
stud         up 1-00:10:00      2   idle gnode[03-04]
PARTITION AVAIL TIMELIMIT NODES CPUS(A/I/O/T) MEMORY GRES
stud up 1-00:10:00 4 30/82/0/112 500000+ gpu:4g.40gb:4,gpu:3g.40gb:4
      User    Account  Partition                QOS       GrpTRES       MaxTRES MaxJobs MaxSubmit
---------- ---------- ---------- -------------------- ------------- ------------- ------- ---------
   3355245    3355245                debug,normal,stud
   3355245    3355245       stud     debug,normal,stud
      Name     MaxWall     MaxTRESPU       MaxTRES       GrpTRES MaxJobsPU
---------- ----------- ------------- ------------- ------------- ---------
    normal             gres/gpu=6,g+                                10
     debug    00:15:00 gres/gpu=2,g+                                 1
  gpu_plus                                        gres/gpu=8
      stud  1-00:00:00 cpu=8,gres/g+    gres/gpu=4                    2
cpu_large+             cpu=2,gres/g+                                100
PartitionName=stud
   AllowGroups=grp_hpcclusterbocconiadmins,grp_hpcclusterstudentsbocconiusers AllowAccounts=ALL AllowQos=stud
   AllocNodes=ALL Default=NO QoS=stud
   DefaultTime=NONE DisableRootJobs=NO ExclusiveUser=NO ExclusiveTopo=NO GraceTime=0 Hidden=NO
   MaxNodes=UNLIMITED MaxTime=1-00:10:00 MinNodes=0 LLN=NO MaxCPUsPerNode=UNLIMITED MaxCPUsPerSocket=UNLIMITED
   Nodes=gnode[01-04]
   PriorityJobFactor=50 PriorityTier=2 RootOnly=NO ReqResv=NO OverSubscribe=NO
   OverTimeLimit=NONE PreemptMode=OFF
   State=UP TotalCPUs=112 TotalNodes=4 SelectTypeParameters=NONE
   JobDefaults=(null)
   DefMemPerCPU=8000 MaxMemPerNode=UNLIMITED
   TRES=cpu=112,mem=2980000M,node=4,billing=240,gres/gpu=32
   TRESBillingWeights=CPU=1.0,GRES/gpu=4.0

gnode01 stud mix 32 990000 gpu:4g.40gb:4,gpu:3g.40gb:4
gnode02 stud mix 24 500000 gpu:4g.40gb:4,gpu:3g.40gb:4
gnode03 stud idle 32 990000 gpu:4g.40gb:4,gpu:3g.40gb:4
gnode04 stud idle 24 500000 gpu:4g.40gb:4,gpu:3g.40gb:4


44G     /home/3355245
(base) [3355245@slnode01 inward]$
(base) [3355245@slnode01 inward]$
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            681386      stud exp10-re  3355245  R      10:27      1 gnode01
(base) [3355245@slnode01 inward]$ Connection to slogin.hpc.unibocconi.it closed by remote host.
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────┐
│                Welcome to  the Bocconi HPC system                       │
│                                                      $$$$$$$$$$$$       │
│                                                      $$$       $$$      │
│                                                      $h         8$      │
│                                  _                   $f         k$      │
│                                 (_)                  $h         8$      │
│                                 | |   _              $$$       $$$      │
│                                 | | /'_`\            $$$       $$$      │
│                                 | |( (_) )           $$$$$$$$$$$$       │
│                                 (_)`\___/'                              │
│                                                      (Jupiter I)        │
│                                                                         │
│ 2x Dell PowerEdge R750XA:                                               │
│ 2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                         │
│ 2x Dell PowerEdge R750XA:                                               │
│ 2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                         │
├── Information ──────────────────────────────────────────────────────────┤
│ Support Requests  : hpc@unibocconi.it                                   │
│ Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC │
├── Software ─────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                        │
│ Use 'lquota' to see the available resources in your home                │
├── Cluster Usage ────────────────────────────────────────────────────────┤
```

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ Running computational data processing tasks on the LOGIN node is forbidden.  │
│ Please always launch computation using SLURM job submission,                 │
│ so that the computation is executed on the COMPUTE nodes.                    │
└──────────────────────────────────────────────────────────────────────────────┘
Last failed login: Sat Oct  3 19:50:55 CEST 2026 from 10.5.105.45 on ssh:notty
There were 3 failed login attempts since the last successful login.
Last login: Sat Oct  3 14:20:02 2026 from 10.5.105.45
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=48.5GB LIMIT=50GB
(base) [3355245@slnode01 ~]$ sacct -u 3355245 -S 2026-10-03 --name=exp10-resume \
  --format=JobID,State,ExitCode,Elapsed
JobID             State ExitCode    Elapsed
------------ ---------- -------- ----------
681386        COMPLETED      0:0   00:54:24
681386.batch  COMPLETED      0:0   00:54:24
(base) [3355245@slnode01 ~]$ cd ~/inward
ls -lt results/handoff_analysis_20261003/exp10_consistency/exp10-resume-*.out
-rw-r--r-- 1 3355245 grp_hpcclusterstudentsbocconiusers 44714 Oct  3 20:56 results/handoff_analysis_20261003/exp10_consistency/exp10-resume-681386.out
(base) [3355245@slnode01 inward]$ cd ~/inward
exp10_dir=results/handoff_analysis_20261003/exp10_consistency
exp10_run=$(ls -dt "$exp10_dir"/resumed_681332_* | head -n 1)
exp10_job=${exp10_run##*_}

tar --exclude='adapter' -czf results/exp10_resumed_review.tgz \
  "$exp10_run" \
  "$exp10_dir/exp10-resume-$exp10_job.out" \
  "$exp10_dir/exp10-681332.out"
(base) [3355245@slnode01 inward]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=48.5GB LIMIT=50GB
(base) [3355245@slnode01 inward]$ Read from remote host slogin.hpc.unibocconi.it: Operation timed out
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────────┐
│                  Welcome to  the Bocconi HPC system                         │
│                                                          $$$$$$$$$$$$       │
│                                                          $$$       $$$      │
│                                                          $h         8$      │
│                                    _                     $f         k$      │
│                                   (_)                    $h         8$      │
│                                   | |  _                 $$$       $$$      │
│                                   | | /'_`\              $$$$$$$$$$$$       │
│                                   | |( (_) )             $$$$$$$$$$$$       │
│                                   (_)`\___/'                                │
│                                                          (Jupiter I)        │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
│  2x Dell PowerEdge R750XA:                                                  │
│  2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
├── Information ──────────────────────────────────────────────────────────────┤
│ Support Requests  : hpc@unibocconi.it                                       │
│ Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC │
├── Software ─────────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                            │
│ Use 'lquota' to see the available resources in your home                    │
├── Cluster Usage ────────────────────────────────────────────────────────────┤
│ Running computational data processing tasks on the LOGIN node is forbidden. │
│ Please always launch computation using SLURM job submission,                │
│ so that the computation is executed on the COMPUTE nodes.                   │
└─────────────────────────────────────────────────────────────────────────────┘
Last login: Sun Oct  4 12:01:50 2026 from 10.5.101.89
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=48.5GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ lquota
USED=48.5GB LIMIT=100GB
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
(base) [3355245@slnode01 ~]$ cd ~/inward
tar --keep-old-files -xzf exp11_upload.tgz &&
(cd results/handoff_analysis_20261003/exp11_iterate && sha256sum -c SHA256SUMS)
lquota
analyze_exp11.py: OK
analyze_exp11_v1.py: OK
exp11_iterate.py: OK
exp11_iterate_v1.py: OK
exp1_gap_v2_v1.py: OK
preflight.py: OK
run_pipeline.py: OK
run_pipeline_v1.py: OK
scoring_v2_v1.py: OK
test_audit.py: OK
run_exp11_iterate.sh: OK
protocol.json: OK
PROTOCOL.txt: OK
HPC_COMMANDS.txt: OK
README.txt: OK
LOCAL_VERIFICATION.json: OK
reference_exp8/eval_A_newPerson.jsonl: OK
reference_exp8/eval_B_newCategory.jsonl: OK
reference_exp8/eval_C_bothNew.jsonl: OK
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=48.5GB LIMIT=100GB
(base) [3355245@slnode01 inward]$ queued=$(squeue -h -u 3355245 | wc -l)
if [ "$queued" -lt 2 ]; then
  sbatch results/handoff_analysis_20261003/exp11_iterate/run_exp11_iterate.sh
else
  echo "已有至少两个任务，暂不提交"
fi
Submitted batch job 685978
(base) [3355245@slnode01 inward]$ squeue =u $USER
squeue: error: Unrecognized option: =u
Usage: squeue [-A account] [--clusters names] [-i seconds] [--jobs jobid]
              [-n name] [-o format] [--only-job-state] [-p partitions]
              [--qos qos] [--reservation reservation] [--sort fields] [--start]
              [--step step_id] [-t states] [-u user_name] [--usage]
              [-L licenses] [-w nodes] [--federation] [--local] [--sibling]
              [--expand-patterns] [--json=data_parser] [--yaml=data_parser]
              [-ahjlrsv]
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ squeue -u 3355245
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
(base) [3355245@slnode01 inward]$ squeue --start
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            685755 short_cpu touch_sc   biggio PD 2026-10-31T10:49:25      1 (null)               (BeginTime)
         660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)               (user env retrieval failed requeued held)
         660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)               (user env retrieval failed requeued held)
            685697      stud cotd-tra  3343816 PD                 N/A      1 (null)               (JobHeldUser)
            685780      stud cotd-tra  3398367 PD                 N/A      1 (null)               (QOSMaxJobsPerUserLimit)
            685906      stud cotd-mig  3343816 PD                 N/A      1 (null)               (QOSMaxJobsPerUserLimit)
            685961      stud cotd-tra  3398367 PD                 N/A      1 (null)               (QOSMaxJobsPerUserLimit)
(base) [3355245@slnode01 inward]$ cat ~/inward/results/handoff_analysis_20261003/exp11_iterate/exp11-685978.out
Traceback (most recent call last):
  File "/mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/preflight.py", line 19, in <module>
    else:raise RuntimeError('lquota unavailable: verify quota explicitly before proceeding')
         ^^^
```

```
fi
Submitted batch job 685985
(base) [3355245@slnode01 inward]$ squeue --start
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            685755 short_cpu touch_sc   biggio PD 2026-10-31T10:49:25      1 (null)              (BeginTime)
          660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
          660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            685697      stud cotd-tra  3343816 PD                 N/A      1 (null)              (JobHeldUser)
            685780      stud cotd-tra  3398367 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
            685906      stud cotd-mig  3343816 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
            685961      stud cotd-tra  3398367 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R       1:53      1 gnode01
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R       9:31      1 gnode01
(base) [3355245@slnode01 inward]$ ssqueu -u $USER
-bash: ssqueu: command not found
(base) [3355245@slnode01 inward]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R      21:04      1 gnode01
(base) [3355245@slnode01 inward]$ Read from remote host slogin.hpc.unibocconi.it: Can't assign requested address
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % squeue --start
zsh: command not found: squeue
(base) danielweiyuchen@Daniels-MacBook-Air ~ % squeue --start
zsh: command not found: squeue
(base) danielweiyuchen@Daniels-MacBook-Air ~ % squeue u $USER
zsh: command not found: squeue
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────────┐
│                Welcome to  the Bocconi HPC system                           │
│                                                          $$$$$$$$$$$$       │
│                                                          $$$       $$$      │
│                                                          $h         8$      │
│                                    _                     $f         k$      │
│                                   (_)                    $h         8$      │
│                                   | |  _                 $$$       $$$      │
│                                   | | /'_`\              $$$       $$$      │
│                                   | |( (_) )             $$$$$$$$$$$$       │
│                                   (_)`\___/'                                │
│                                                          (Jupiter I)        │
│                                                                             │
│ 2x Dell PowerEdge R750XA:                                                   │
│ 2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
│ 2x Dell PowerEdge R750XA:                                                   │
│ 2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM│
├── Information ──────────────────────────────────────────────────────────────┤
│ Support Requests  : hpc@unibocconi.it                                       │
│ Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC │
├── Software ─────────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                            │
│ Use 'lquota' to see the available resources in your home                    │
├── Cluster Usage ────────────────────────────────────────────────────────────┤
│ Running computational data processing tasks on the LOGIN node is forbidden. │
│ Please always launch computation using SLURM job submission,               │
│ so that the computation is executed on the COMPUTE nodes.                   │
└─────────────────────────────────────────────────────────────────────────────┘
Last failed login: Tue Oct  6 13:58:02 CEST 2026 from 10.5.171.227 on ssh:notty
There was 1 failed login attempt since the last successful login.
Last login: Tue Oct  6 13:56:03 2026 from 10.5.171.227
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=49GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R      40:49      1 gnode01
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R      52:07      1 gnode01
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    1:04:20      1 gnode01
(base) [3355245@slnode01 ~]$ qluota
-bash: qluota: command not found
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=49.5GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    1:16:59      1 gnode01
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    1:20:19      1 gnode01
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    1:22:27      1 gnode01
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=49.5GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    1:39:35      1 gnode01
(base) [3355245@slnode01 ~]$ Read from remote host slogin.hpc.unibocconi.it: Can't assign requested address
Connection to slogin.hpc.unibocconi.it closed.
client_loop: send disconnect: Broken pipe
(base) danielweiyuchen@Daniels-MacBook-Air ~ % ssh 3355245@slogin.hpc.unibocconi.it
3355245@slogin.hpc.unibocconi.it's password:
┌── Welcome ──────────────────────────────────────────────────────────────────┐
│                Welcome to  the Bocconi HPC system                           │
│                                                          $$$$$$$$$$$$       │
│                                                          $$$       $$$      │
│                                                          $h         8$      │
│                                    _                     $f         k$      │
│                                   (_)                    $h         8$      │
│                                   | |  _                 $$$       $$$      │
│                                   | | /'_`\              $$$       $$$      │
│                                   | |( (_) )             $$$$$$$$$$$$       │
│                                   (_)`\___/'                                │
│                                                          (Jupiter I)        │
│                                                                             │
│ 2x Dell PowerEdge R750XA:                                                   │
│ 2x Intel Xeon 5317 12 Core, 512GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM │
│                                                                             │
│ 2x Dell PowerEdge R750XA:                                                   │
│ 2x Intel Xeon 6326 16 Core, 1024GB DDR4-3200 RAM, 4 GPU NVIDIA A100 80GB RAM│
├── Information ──────────────────────────────────────────────────────────────┤
│ Support Requests  : hpc@unibocconi.it                                       │
│ Hot to Submit Jobs: https://bocconi.sharepoint.com/sites/BocconiStudentsHPC │
├── Software ─────────────────────────────────────────────────────────────────┤
│ Use 'module avail' to see the installed software                            │
│ Use 'lquota' to see the available resources in your home                    │
├── Cluster Usage ────────────────────────────────────────────────────────────┤
│ Running computational data processing tasks on the LOGIN node is forbidden. │
│ Please always launch computation using SLURM job submission,               │
│ so that the computation is executed on the COMPUTE nodes.                   │
└─────────────────────────────────────────────────────────────────────────────┘
Last login: Tue Oct  6 14:44:51 2026 from 10.5.181.218
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=50.5GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    2:30:31      1 gnode01
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    2:53:10      1 gnode01
```

```
(base) [3355245@slnode01 ~]$ squeue -u 3355245
tail -n 50 "$(ls -t ~/inward/results/handoff_analysis_20261003/exp11_iterate/exp11-*.out | head -n 1)"
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    2:54:14      1 gnode01
frozen_budget round 3 seed 0 step 150 / 648 loss 0.157422736287117
frozen_budget round 3 seed 0 step 200 / 648 loss 0.471100002527236 94
frozen_budget round 3 seed 0 step 250 / 648 loss 0.24925115704536438
frozen_budget round 3 seed 0 step 300 / 648 loss 0.4315410852432251
frozen_budget round 3 seed 0 step 350 / 648 loss 0.0055816336534917355
frozen_budget round 3 seed 0 step 400 / 648 loss 0.011270994320511818
frozen_budget round 3 seed 0 step 450 / 648 loss 0.031248262152075768
frozen_budget round 3 seed 0 step 500 / 648 loss 0.004992285277694464
frozen_budget round 3 seed 0 step 550 / 648 loss 0.0015330903697758913
frozen_budget round 3 seed 0 step 600 / 648 loss 0.00011513981007738039
A_newPerson accuracy 0.488
B_newCategory accuracy 0.52
C_bothNew accuracy 0.532
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage train --out results/handoff_analysis_20261003
/exp11_iterate/runs_685985/shared_r1_s1 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --targets results/handoff_analysis_20261003/exp11_iterat
e/runs_685985/base_targets/targets.jsonl --condition shared --round 1 --train-seed 1 --epochs 2 --batch 8
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 125.77it/s]
shared round 1 seed 1 step 50 / 216 loss 0.363140344619751
shared round 1 seed 1 step 100 / 216 loss 0.4330540597438812
shared round 1 seed 1 step 150 / 216 loss 0.019522586837410927
shared round 1 seed 1 step 200 / 216 loss 0.262781023979187
A_newPerson accuracy 0.486
B_newCategory accuracy 0.504
C_bothNew accuracy 0.514
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage targets --out results/handoff_analysis_202610
03/exp11_iterate/runs_685985/shared_targets_r1_s1 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --adapter results/handoff_analysis_20261003/ex
p11_iterate/runs_685985/shared_r1_s1/adapter --train-seed 1
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 125.04it/s]
targets complete results/handoff_analysis_20261003/exp11_iterate/runs_685985/shared_targets_r1_s1
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage train --out results/handoff_analysis_20261003
/exp11_iterate/runs_685985/reset_r2_s1 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --targets results/handoff_analysis_20261003/exp11_iterate
/runs_685985/shared_targets_r1_s1/targets.jsonl --condition reset --round 2 --train-seed 1 --epochs 2 --batch 8
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 126.36it/s]
reset round 2 seed 1 step 50 / 216 loss 0.4933333396911621
reset round 2 seed 1 step 100 / 216 loss 0.4463973045349121
reset round 2 seed 1 step 150 / 216 loss 0.09513010829687119
reset round 2 seed 1 step 200 / 216 loss 0.05882944166660309
A_newPerson accuracy 0.486
B_newCategory accuracy 0.502
C_bothNew accuracy 0.522
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage targets --out results/handoff_analysis_202610
03/exp11_iterate/runs_685985/reset_targets_r2_s1 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --adapter results/handoff_analysis_20261003/exp
11_iterate/runs_685985/reset_r2_s1/adapter --train-seed 1
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 125.99it/s]
targets complete results/handoff_analysis_20261003/exp11_iterate/runs_685985/reset_targets_r2_s1
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage train --out results/handoff_analysis_20261003
/exp11_iterate/runs_685985/reset_r3_s1 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --targets results/handoff_analysis_20261003/exp11_iterate
/runs_685985/reset_targets_r2_s1/targets.jsonl --condition reset --round 3 --train-seed 1 --epochs 2 --batch 8
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 125.40it/s]
reset round 3 seed 1 step 50 / 216 loss 0.3852313756942749
reset round 3 seed 1 step 100 / 216 loss 0.3807603418827057
reset round 3 seed 1 step 150 / 216 loss 0.1840284913778305
reset round 3 seed 1 step 200 / 216 loss 0.06244531646370888
A_newPerson accuracy 0.464
B_newCategory accuracy 0.514
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=51GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    3:03:16      1 gnode01
(base) [3355245@slnode01 ~]$ sqeue -u $USER
-bash: sqeue: command not found
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    3:09:54      1 gnode01
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=51.5GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    3:16:14      1 gnode01
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    3:48:02      1 gnode01
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    4:15:28      1 gnode01
(base) [3355245@slnode01 ~]$ tail -n 25 ~/inward/results/handoff_analysis_20261003/exp11_iterate/exp11-685985.out
frozen_budget round 3 seed 1 step 100 / 648 loss 0.40375471115112305
frozen_budget round 3 seed 1 step 150 / 648 loss 0.07389669865369797
frozen_budget round 3 seed 1 step 200 / 648 loss 0.3059860169887543
frozen_budget round 3 seed 1 step 250 / 648 loss 0.008306678384542465
frozen_budget round 3 seed 1 step 300 / 648 loss 0.21658073365688324
frozen_budget round 3 seed 1 step 350 / 648 loss 0.0014164053136482835
frozen_budget round 3 seed 1 step 400 / 648 loss 0.020390894263982773
frozen_budget round 3 seed 1 step 450 / 648 loss 0.007392930332571268
frozen_budget round 3 seed 1 step 500 / 648 loss 0.0021514226682484 15
frozen_budget round 3 seed 1 step 550 / 648 loss 0.0002529186604078859
frozen_budget round 3 seed 1 step 600 / 648 loss 7.68685422372073e-05
A_newPerson accuracy 0.476
B_newCategory accuracy 0.5
C_bothNew accuracy 0.512
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage train --out results/handoff_analysis_20261003
/exp11_iterate/runs_685985/shared_r1_s2 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --targets results/handoff_analysis_20261003/exp11_iterat
e/runs_685985/base_targets/targets.jsonl --condition shared --round 1 --train-seed 2 --epochs 2 --batch 8
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 125.37it/s]
shared round 1 seed 2 step 50 / 216 loss 0.312724232673645
shared round 1 seed 2 step 100 / 216 loss 0.8234780430793762
shared round 1 seed 2 step 150 / 216 loss 0.18286536633968353
shared round 1 seed 2 step 200 / 216 loss 0.14577503502368927
A_newPerson accuracy 0.486
B_newCategory accuracy 0.526
C_bothNew accuracy 0.54
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage targets --out results/handoff_analysis_202610
03/exp11_iterate/runs_685985/shared_targets_r1_s2 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --adapter results/handoff_analysis_20261003/ex
p11_iterate/runs_685985/shared_r1_s2/adapter --train-seed 2
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=52.5GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=52.5GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    4:28:31      1 gnode01
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    4:35:37      1 gnode01
(base) [3355245@slnode01 ~]$ squeue --start
             JOBID PARTITION     NAME     USER ST          START_TIME  NODES SCHEDNODES           NODELIST(REASON)
            685755 short_cpu touch_sc   biggio PD 2026-10-31T10:49:25      1 (null)              (BeginTime)
         660314_4 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
         660314_5 gpua100,l gt_extra   seccod PD                 N/A      1 (null)              (user env retrieval failed requeued held)
            685780      stud cotd-tra  3398367 PD                 N/A      1 (null)              (JobHeldUser)
            686006      stud cotd-rea  3343816 PD                 N/A      1 (null)              (JobHeldUser)
            686007      stud cotd-tra  3343816 PD                 N/A      1 (null)              (JobHeldUser)
            686250      stud cotd-mig  3398367 PD                 N/A      1 (null)              (QOSMaxJobsPerUserLimit)
(base) [3355245@slnode01 ~]$ squeue -j 685985
tail -n 35 ~/inward/results/handoff_analysis_20261003/exp11_iterate/exp11-685985.out
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
```

```
            685985      stud inward-e  3355245  R    4:37:20      1 gnode01
frozen_budget round 3 seed 1 step 400 / 648 loss 0.0203908942639827 73
frozen_budget round 3 seed 1 step 450 / 648 loss 0.007392930332571268
frozen_budget round 3 seed 1 step 500 / 648 loss 0.002151422668248415
frozen_budget round 3 seed 1 step 550 / 648 loss 0.0002529186604078859
frozen_budget round 3 seed 1 step 600 / 648 loss 7.68685422372073e-05
A_newPerson accuracy 0.476
B_newCategory accuracy 0.5
C_bothNew accuracy 0.512
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage train --out results/handoff_analysis_20261003
/exp11_iterate/runs_685985/shared_r1_s2 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --targets results/handoff_analysis_20261003/exp11_iterat
e/runs_685985/base_targets/targets.jsonl --condition shared --round 1 --train-seed 2 --epochs 2 --batch 8
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 125.37it/s]
shared round 1 seed 2 step 50 / 216 loss 0.312724232673645
shared round 1 seed 2 step 100 / 216 loss 0.8234780430793762
shared round 1 seed 2 step 150 / 216 loss 0.18286536633968353
shared round 1 seed 2 step 200 / 216 loss 0.14577503502368927
A_newPerson accuracy 0.486
B_newCategory accuracy 0.526
C_bothNew accuracy 0.54
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage targets --out results/handoff_analysis_202610
03/exp11_iterate/runs_685985/shared_targets_r1_s2 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --adapter results/handoff_analysis_20261003/ex
p11_iterate/runs_685985/shared_r1_s2/adapter --train-seed 2
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 127.07it/s]
targets complete results/handoff_analysis_20261003/exp11_iterate/runs_685985/shared_targets_r1_s2
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage train --out results/handoff_analysis_20261003
/exp11_iterate/runs_685985/reset_r2_s2 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --targets results/handoff_analysis_20261003/exp11_iterate
/runs_685985/shared_targets_r1_s2/targets.jsonl --condition reset --round 2 --train-seed 2 --epochs 2 --batch 8
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 125.18it/s]
reset round 2 seed 2 step 50 / 216 loss 0.3299211859703064
reset round 2 seed 2 step 100 / 216 loss 0.6726009845733643
reset round 2 seed 2 step 150 / 216 loss 0.26784375309944153
reset round 2 seed 2 step 200 / 216 loss 0.06952232867479324
A_newPerson accuracy 0.476
B_newCategory accuracy 0.532
C_bothNew accuracy 0.542
RUN: /home/3355245/.conda/envs/inward/bin/python /mnt/beegfsstudents/home/3355245/inward/results/handoff_analysis_20261003/exp11_iterate/exp11_iterate.py --stage targets --out results/handoff_analysis_202610
03/exp11_iterate/runs_685985/reset_targets_r2_s2 --model models/Qwen3-14B --plan results/handoff_analysis_20261003/exp11_iterate/runs_685985/plan --gen-batch 8 --adapter results/handoff_analysis_20261003/exp
11_iterate/runs_685985/reset_r2_s2/adapter --train-seed 2
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|██████████| 443/443 [00:03<00:00, 126.68it/s]
(base) [3355245@slnode01 ~]$ lquota
INFO: Quota usage information is fetched every 30s from the server nodes, thus the displayed values might be slightly out of date.
USED=52.5GB LIMIT=100GB
(base) [3355245@slnode01 ~]$ squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
            685985      stud inward-e  3355245  R    4:54:07      1 gnode01
(base) [3355245@slnode01 ~]$
```